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Rates Vol Lab

Fading vega dislocations

CITI'S TAKE

Intermediate expiry vols have continued to outperform, and we believe it is partly due to hedging demand for MBS floaters with fixed rate caps. Intermediate vol looks high relative to long-dated forward curves, so we recommend shorting 3y10y vol against 10y10y/15y15y rate curve flattener. The 3y10y-1y10y vol slope also looks too steep relative to the level of vol, in our view.

Vol surface — In the US, long-dated vols, such as 10y10y, look the most overvalued based on our macro-vol and PCA relative value models.

Skew — In the US, long-expiry payer skews on all rates have continued to richen. In EUR, payer skews in general have continued to cheapen from their July highs.

CMS curve vol — Implied correlations for USD CMS curve options have declined, reflecting implied curve vols' outperformance vs swaption vols.

SOFR futures convexity adj — Blues and Golds convexity adjustments have been increasing modestly but they are still about 4-5bps too low relative to our model fair values.

Note: Futures trading involves substantial risk of loss.

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See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

Prepared for Christopher Lee

Fading vega dislocations

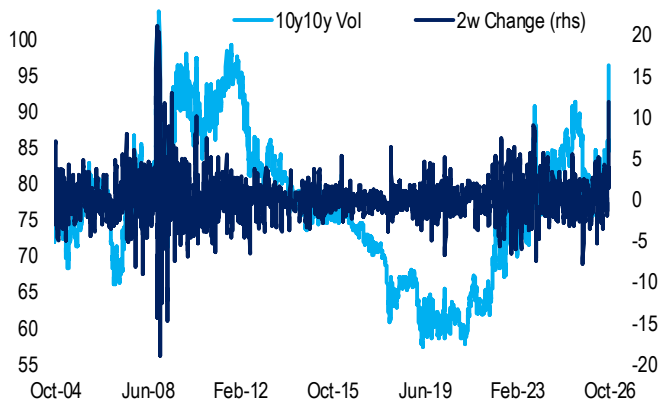
Mike Chang

Intermediate expiry vols have continued to outperform, and we believe it is partly due to hedging demand for MBS floaters with fixed rate caps. Intermediate vol looks high relative to long-dated forward curves, so we recommend shorting 3y10y vol against 10y10y/15y15y rate curve flattener. The 3y10y-1y10y vol slope also looks too steep relative to the level of vol.

Intermediate vega looks high vs long-dated forward curves

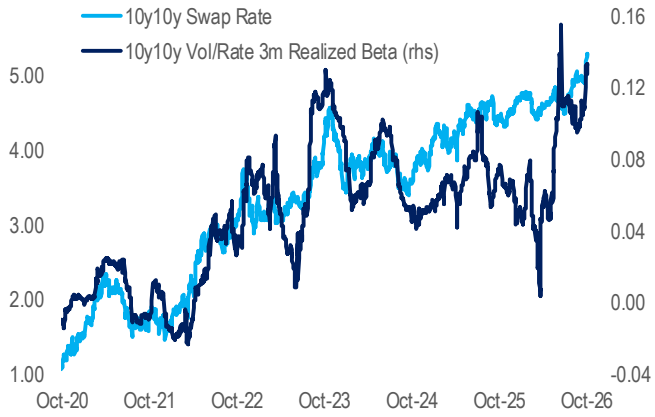
While the increase in overall implied vol has been notable over the past month, the real standout has been the relative outperformance in the intermediate and longer expiries. In long-dated vol for example, the increase in 10y10y over the past 2 weeks has been one of the largest since the GFC, and 10y10y vol is currently trading near its highest level in the past 15 years (Figure 1). The strong performance in long-dated vol has been partly due to the return of its positive directionality with rates amid the recent sell-off. We have previously noted that the directionality (the realized beta) of long-dated vol to its underlying forward rate has been broadly directional with the level of rates throughout most of the post-Covid years (Rates Vol Lab - US and EUR Rates Vol 2025 Outlook). 2025 was the exception, but that dynamic has returned since the start of the Middle East conflict (Figure 2). Given that concerns of long-end rates breaching new highs are unlikely to be alleviated in the near-term, we expect long-dated vol's positively convexed vol/rate directionality to persist and be beneficial for long-dated vol's risk/reward profile, i.e. sharper increase in rate sell-offs versus moderate decrease in rate rallies.

Figure 1. The increase in 10y10y vol over the past 2 weeks has been one of the largest since the GFC



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Source: Citi Research

Figure 2. The beta of long-dated vol to the underlying forward rate has been broadly directional with the level of rates, i.e. a positively convexed vol/rate directionality



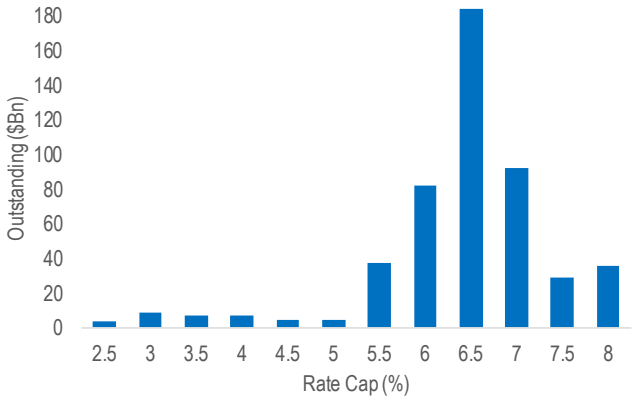
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In intermediate expiries, we believe the recent relative outperformance could have been partly due to the hedging demand from mortgage investors who are holding floaters with high strike fixed rate caps. According to our mortgage strategist (Susan Wang), there are currently $496bn of MBS floaters with fixed caps outstanding, and 72% of them ($358bn) have caps between 6% to 7% (Figure 3). Because of the recent rates sell-off, the forwards are now approaching the cap levels, which would cause the holders of the floaters to take on unexpected duration and short vega risk. We believe some of these investors have either bought intermediate expiry swaptions to hedge the exposure or sold the floaters outright, in which case the dealers likely would have hedged the short vol exposure. We estimate that the $358bn outstanding floaters with 6%-7% caps could have roughly a total maximum short vega of $204mm or 217bn 3y10y swaption straddle

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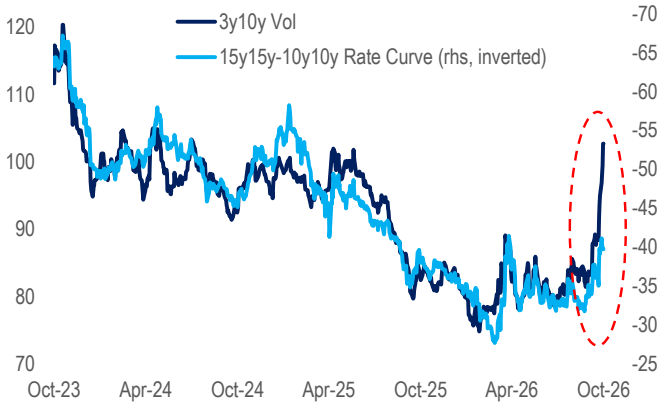
equivalent. While it is difficult to estimate what the incremental demand for vol would be without knowing the proportion that would be hedged (obviously not all), the estimated max short vol exposure suggests that even a modest amount of hedge buying could have caused intermediate expiry vols to outperform.

Figure 3. Our mortgage strategists estimate that there are $496bn of MBS floaters with fixed caps outstanding, and 72% of them ($358bn) have caps between 6% to 7%



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Source: Citi Research, Citi Mortgage Research

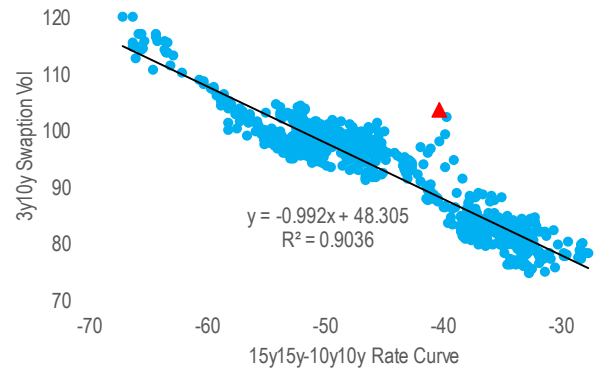
Figure 4. 3y10y vol has increased sharply and outpaced the flattening in the 15y15y-10y10y rate curve



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Looking at the relationship between intermediate vol and long-dated forward rate curves, there is currently clear dislocation that could be attributed to the vol hedging demand (Figure 4). For example, historically the 15y15y-10y10y rate curve tends to be inversely correlated with 3y10y vol, because receiving/(paying) 15y15y swap against 10y10y swap would be net long/(short) convexity. But in recent weeks, 3y10y vol has increased sharply and outpaced the flattening in the 15y15y-10y10y rate curve. Currently, 3y10y vol looks about 14 normals too high relative to the rate curve based on the regression over the past 3 years (Figure 5).

Figure 5. Currently, 3y10y vol looks about 14 normals too high relative to the rate curve based on the regression over the past 3 years



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Figure 6. The 3y10y-1y10y vol slope, which historically tends to move inversely with outright vol, has not flattened as much as expected after the recent spike in vol



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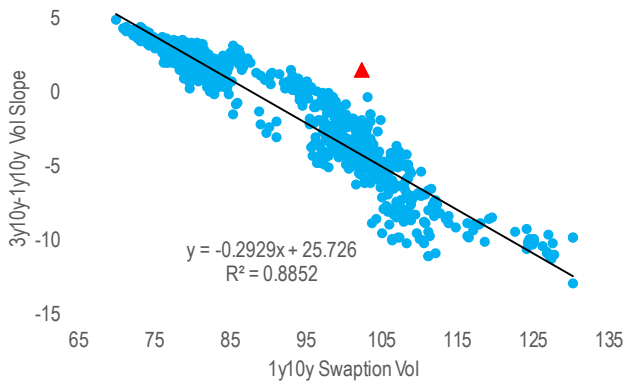
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Rather than being outright short 3y10y vol to monetize its relative richness, we recommend positioning for an eventual normalization in the dislocations between the implied vol and the long-dated forward curve. **Specifically, we recommended selling $30mm 3y10y ATMF straddles at $2.89mm premium against $54mm 10y10y/15y15y curve flattener** (pricing as of 1:30pm 10/2/2026, 3y10y ATMF=5.03%, 10y10y ATMF=5.287%, 15y15y ATMF=4.88%, net curve DV01=$26k, net vega=-$28k; see US Rates Weekly - OAT of the blue). Our target/stop were $400k and -$250k, respectively, and the primary risk to the trade is further outperformance in intermediate expiry vol.

The outperformance of 3y10y vol, in particular, has also been consistent with the relative richness of 3y expiries versus 1y that we have recently highlighted (US Rates Weekly - Neutrals are back in style for fall). Looking at the 3y10y-1y10y vol slope, which historically tends to move inversely with outright vol (3y10y underperforms 1y10y as vol increases), the vol slope currently looks extremely steep after the spike in vol (Figure 6). The 3y10y-1y10y vol slope currently looks about 5.5 normals too steep relative to the level of vol based on regression over the past 3 years (Figure 7).

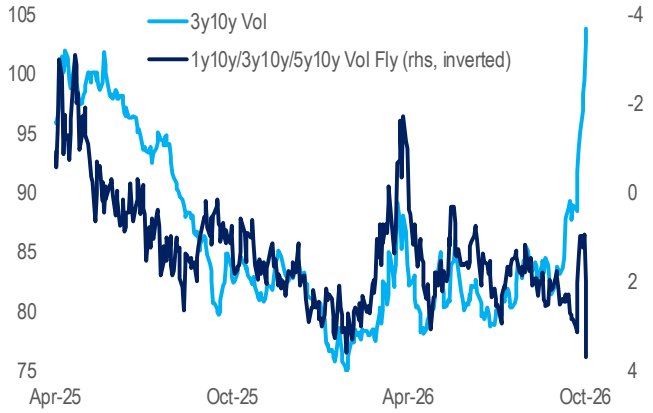
Similarly, we also looked at the 3y10y vs 1y10y and 5y10y on the vol fly, which has typically moved inversely with the outright level of vol, i.e. 3y10y vol tends to underperform 1y10y and 5y10y vol (vol fly tightens) when overall implied vol is increasing (Figure 8). But over the past month, a divergence in the relationship has developed where 3y10y vol has not cheapened relative to 1y10y and 5y10y vols even as implied vols overall richened. The 1y10y/3y10y/5y10y vol fly is currently about 4.5 normals too wide relative to the level of vol based on the relationship over the past 3 years.

Figure 7. The 3y10y-1y10y vol slope currently looks about 5.5 normals too steep relative to the level of vol based on regression over the past 3 years



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Figure 8. The 1y10y/3y10y/5y10y vol fly also looks too high (3y expiry too rich vs 1y and 5y) relative to the outright level of vol



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USD at-the-money swaption vols

USD relative value on the swaption surface

We report levels and z-scores of at-the-money vols, as well as the relative richness/cheapness of vols to rates (based on principal components of the curve).

- Entire swaption vol surface has increased over the last 3 weeks, with the largest moves in upper right vol. (Figure 10).
- Long-dated vols, such as 10y10y, look the most overvalued on the swaption surface based on our macro-vol model (Figure 11) and PCA relative value model (Figure 12).

Figure 9. Implied vols, normals (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	105.8 (2.5)	120.3 (2.7)	119.0 (2.7)	114.0 (2.6)	109.6 (2.7)	102.8 (2.8)	91.8 (2.8)
3m	107.5 (2.7)	118.5 (2.7)	117.7 (2.8)	112.2 (2.9)	107.2 (3.0)	100.6 (3.0)	91.5 (3.0)
6m	116.4 (3.0)	118.4 (2.8)	117.1 (2.9)	111.5 (3.1)	107.8 (3.2)	102.8 (3.2)	93.9 (3.3)
1y	120.8 (2.9)	117.7 (3.0)	115.6 (3.1)	110.7 (3.3)	107.4 (3.3)	103.6 (3.4)	95.9 (3.4)
2y	118.1 (3.2)	115.4 (3.2)	112.7 (3.2)	108.9 (3.3)	106.4 (3.4)	104.0 (3.5)	97.4 (3.5)
3y	114.4 (3.2)	112.3 (3.2)	110.8 (3.3)	107.6 (3.4)	105.9 (3.4)	103.2 (3.5)	96.6 (3.5)
5y	108.3 (3.3)	106.6 (3.4)	105.4 (3.4)	103.3 (3.4)	102.2 (3.4)	100.5 (3.4)	94.5 (3.5)
10y	98.5 (3.5)	98.2 (3.5)	97.7 (3.5)	96.9 (3.4)	96.1 (3.4)	94.7 (3.4)	88.7 (3.5)

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Close 10/2/26. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 10. Change in implied vols since September 11th, normals

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	15.1	22.2	23.1	25.5	27.1	28.5	24.4
3m	8.6	15.1	16.5	20.7	21.5	22.2	20.4
6m	12.1	14.8	15.8	19.1	20.1	21.1	20.1
1y	14.2	14.5	15.4	17.3	18.1	18.8	17.7
2y	15.6	15.7	15.8	16.1	16.9	17.8	16.9
3y	14.8	15.7	16.0	16.4	16.4	16.2	14.6
5y	14.0	13.8	13.8	13.7	13.7	13.7	12.4
10y	10.1	10.1	10.1	10.0	9.9	9.9	8.9

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Note: The table shows changes in ATM implied vols from 10/2/26 to 10/2/26. Source: Citi Research

Figure 11. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	22.7 (0.8)	19.5 (0.6)	17.0 (0.6)	12.3 (0.4)	11.0 (0.4)	7.1 (0.3)	9.7 (0.4)
3m	19.0 (0.7)	15.5 (0.5)	14.0 (0.5)	9.6 (0.4)	7.6 (0.3)	3.8 (0.2)	8.5 (0.4)
6m	19.1 (0.7)	12.6 (0.4)	11.7 (0.4)	8.5 (0.4)	7.8 (0.4)	5.7 (0.3)	10.9 (0.7)
1y	12.8 (0.5)	8.9 (0.3)	8.7 (0.4)	7.7 (0.4)	7.4 (0.4)	6.8 (0.4)	13.7 (1.0)
2y	7.8 (0.4)	7.3 (0.4)	6.8 (0.4)	6.9 (0.4)	7.3 (0.5)	8.2 (0.6)	16.8 (1.4)
3y	6.6 (0.4)	7.0 (0.4)	7.5 (0.5)	7.7 (0.5)	8.7 (0.7)	9.1 (0.7)	18.1 (1.7)
5y	5.8 (0.4)	6.4 (0.5)	6.9 (0.6)	8.1 (0.7)	9.5 (0.9)	11.1 (1.1)	21.2 (2.3)
10y	9.9 (1.2)	11.9 (1.4)	13.3 (1.7)	16.0 (2.0)	17.8 (2.3)	19.6 (2.6)	28.4 (3.3)

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Close 10/2/26. We regress each ATM point on 3 principal components of the swap curve. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We calibrate the model to the entire sample history since 1997. Source: Citi Research

Figure 12. Rich/cheap to PCA of the vol surface, normals (z-score)

exp\tls	1y	2y	3y	5y	7y	10y	30y
1m	8.5 (1.3)	7.2 (1.8)	5.2 (1.3)	5.2 (1.4)	6.2 (1.7)	5.2 (1.3)	3.5 (0.7)
3m	1.7 (0.5)	1.7 (0.8)	1.1 (0.5)	0.8 (0.5)	0.2 (0.2)	-1.2 (-1.0)	-1.1 (-0.4)
6m	0.8 (0.4)	-0.7 (-0.3)	-0.5 (-0.2)	-0.5 (-0.3)	-0.0 (-0.0)	-0.1 (-0.1)	0.0 (0.0)
1y	-2.7 (-0.7)	-2.6 (-0.9)	-1.5 (-0.5)	-1.3 (-0.8)	-0.9 (-0.9)	-0.7 (-0.9)	0.2 (0.1)
2y	-1.6 (-0.5)	-0.9 (-0.3)	-1.0 (-0.4)	-0.8 (-0.5)	-0.3 (-0.4)	0.3 (0.5)	1.8 (1.6)
3y	-0.8 (-0.3)	-0.1 (-0.1)	-0.1 (-0.1)	0.2 (0.2)	0.4 (0.5)	0.3 (0.6)	1.6 (1.8)
5y	-1.0 (-0.6)	-0.6 (-0.5)	-0.4 (-0.4)	-0.4 (-0.6)	-0.4 (-0.7)	-0.4 (-1.2)	0.7 (1.1)
10y	-1.7 (-2.3)	-0.7 (-1.4)	-0.2 (-0.4)	0.2 (0.3)	0.4 (0.8)	0.7 (1.2)	0.9 (1.2)

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Close 10/2/26. We calculate rich/cheap and z-scores in parentheses from PCA on the vol surface based on 3 principal components using 2y of data. By design, this framework only yields relative value. Source: Citi Research

USD implied vs realized volatilities

We calculate the relative richness of short-dated implied vols to 1m and 3m realized vols. We calculate realized vols using both intraday (hourly) and close-to-close sampling.

- Implied vols are broadly trading in-line with both recent intraday (Figure 15) and close-to-close realized vols (Figure 17), aside from the 30y tails which are trading rich.

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Figure 13. 1m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.00 (-0.4)	0.99 (-0.6)	0.98 (-0.8)	1.01 (-0.7)	1.02 (-0.7)	1.02 (-0.7)	1.08 (-0.7)
3m	0.99 (-0.6)	0.99 (-0.7)	0.98 (-0.8)	1.01 (-0.8)	1.02 (-0.7)	1.02 (-0.7)	1.07 (-0.8)
6m	1.00 (-0.5)	0.99 (-0.7)	0.99 (-0.8)	1.02 (-0.7)	1.02 (-0.7)	1.03 (-0.7)	1.07 (-0.7)
1y	1.01 (-0.5)	0.99 (-0.7)	1.01 (-0.8)	1.03 (-0.6)	1.04 (-0.6)	1.03 (-0.6)	1.09 (-0.7)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 14. 3m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.95 (-1.0)	0.98 (-1.0)	0.99 (-1.1)	1.03 (-1.1)	1.04 (-1.1)	1.04 (-1.2)	1.08 (-1.3)
3m	0.96 (-1.0)	0.99 (-1.0)	1.00 (-1.1)	1.03 (-1.1)	1.03 (-1.1)	1.04 (-1.2)	1.07 (-1.3)
6m	0.98 (-0.9)	1.00 (-0.9)	1.00 (-1.0)	1.04 (-1.0)	1.04 (-1.1)	1.04 (-1.2)	1.08 (-1.3)
1y	1.01 (-0.6)	1.01 (-0.8)	1.03 (-0.9)	1.05 (-0.9)	1.05 (-1.0)	1.05 (-1.1)	1.08 (-1.3)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 15. Intraday realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.18 (-0.1)	1.16 (-0.2)	1.15 (-0.2)	1.12 (-0.1)	1.13 (0.1)	1.14 (0.3)	1.24 (1.4)
3m	1.03 (-0.5)	1.07 (-0.5)	1.09 (-0.5)	1.08 (-0.4)	1.09 (-0.4)	1.10 (-0.4)	1.23 (0.9)
6m	0.98 (-0.6)	1.02 (-0.7)	1.05 (-0.7)	1.05 (-0.7)	1.08 (-0.7)	1.11 (-0.6)	1.25 (0.6)
1y	0.97 (-0.7)	1.00 (-0.9)	1.03 (-0.9)	1.04 (-1.0)	1.08 (-1.0)	1.12 (-0.9)	1.27 (0.3)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 16. Intraday realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.47 (1.3)	1.48 (1.5)	1.46 (1.6)	1.41 (1.7)	1.42 (1.9)	1.41 (2.1)	1.45 (3.0)
3m	1.31 (1.4)	1.38 (1.5)	1.39 (1.6)	1.35 (1.7)	1.37 (1.8)	1.36 (1.9)	1.43 (3.2)
6m	1.26 (1.4)	1.31 (1.4)	1.34 (1.5)	1.32 (1.6)	1.35 (1.7)	1.38 (2.0)	1.45 (3.5)
1y	1.26 (1.3)	1.29 (1.2)	1.30 (1.4)	1.31 (1.5)	1.35 (1.7)	1.39 (2.0)	1.47 (3.4)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 17. Close-to-close realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.18 (-0.4)	1.15 (-0.5)	1.13 (-0.6)	1.13 (-0.5)	1.16 (-0.4)	1.16 (-0.4)	1.34 (0.1)
3m	1.02 (-0.8)	1.06 (-0.8)	1.07 (-0.9)	1.09 (-0.8)	1.11 (-0.8)	1.12 (-0.8)	1.32 (-0.3)
6m	0.98 (-0.7)	1.01 (-0.9)	1.04 (-1.0)	1.07 (-0.9)	1.11 (-0.9)	1.14 (-0.9)	1.34 (-0.5)
1y	0.98 (-0.8)	1.00 (-1.1)	1.03 (-1.1)	1.08 (-1.0)	1.12 (-1.0)	1.16 (-1.0)	1.38 (-0.5)

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Note: Close 10/2/26. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 18. Close-to-close realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.40 (0.7)	1.46 (1.1)	1.45 (1.2)	1.44 (1.4)	1.47 (1.5)	1.47 (1.6)	1.56 (2.2)
3m	1.26 (0.7)	1.36 (1.0)	1.39 (1.2)	1.39 (1.2)	1.41 (1.3)	1.42 (1.2)	1.54 (1.7)
6m	1.24 (0.9)	1.31 (1.0)	1.34 (1.1)	1.36 (1.1)	1.41 (1.1)	1.44 (1.1)	1.56 (1.6)
1y	1.28 (1.1)	1.31 (1.0)	1.34 (1.0)	1.37 (1.0)	1.42 (1.0)	1.45 (1.0)	1.59 (1.5)

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Note: Close 10/2/26. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

USD expiry and tail switches

We analyze ATM vol switches across expiries and tails.

- 6m expiries on belly and long-end rates have outperformed the neighboring 3m and 1y expiries (Figure 19). For example, the 6m10y/1y10y vol ratio is at 2.2 sigmas while the 3m10y/6m10y vol ratio is at only 0.9 sigma.
- 2y tails have underperformed 10y tails for all expiries. For example, the 2y2y/2y10y vol ratio is at -2 sigmas (Figure 20).

Figure 19. Short/long expiry ratio (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m/3m	0.98 (1.1)	1.02 (1.3)	1.01 (1.3)	1.02 (0.9)	1.02 (1.4)	1.02 (1.8)	1.00 (1.2)
3m/6m	0.92 (-0.6)	1.00 (1.2)	1.01 (1.3)	1.01 (1.4)	1.00 (1.2)	0.98 (0.9)	0.97 (0.9)
6m/1y	0.96 (1.1)	1.01 (1.4)	1.01 (1.5)	1.01 (1.8)	1.00 (2.1)	0.99 (2.2)	0.98 (2.3)
1y/3y	1.06 (0.9)	1.05 (0.7)	1.04 (0.8)	1.03 (1.6)	1.01 (2.1)	1.00 (2.6)	0.99 (2.5)
3y/5y	1.06 (1.7)	1.05 (1.9)	1.05 (2.3)	1.04 (3.0)	1.04 (3.2)	1.03 (3.3)	1.02 (3.2)
5y/10y	1.10 (2.3)	1.09 (2.3)	1.08 (2.5)	1.07 (2.7)	1.06 (2.8)	1.06 (3.0)	1.07 (3.2)

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Note: Close 10/2/26. The table reports shorter expiry/longer expiry ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 20. Short/long tail ratio (3m z-score)

exp\tails	1y/2y	1y/3y	2y/5y	2y/10y	5y/10y	5y/30y	10y/30y
1m	0.88 (-0.6)	0.89 (-0.6)	1.06 (-0.5)	1.17 (-1.4)	1.11 (-1.9)	1.24 (-0.8)	1.12 (0.2)
3m	0.91 (-1.4)	0.91 (-1.5)	1.06 (-1.6)	1.18 (-1.4)	1.12 (-1.0)	1.23 (-0.8)	1.10 (-0.5)
1y	1.03 (0.5)	1.04 (-0.2)	1.06 (-1.7)	1.14 (-1.9)	1.07 (-1.9)	1.15 (-1.4)	1.08 (-0.7)
2y	1.02 (-0.8)	1.05 (-1.1)	1.06 (-1.3)	1.11 (-2.0)	1.05 (-2.2)	1.12 (-1.5)	1.07 (-0.5)
5y	1.02 (0.2)	1.03 (-1.0)	1.03 (-1.4)	1.06 (-1.3)	1.03 (-0.5)	1.09 (0.1)	1.06 (0.4)
10y	1.00 (-2.0)	1.01 (-1.7)	1.01 (-1.0)	1.04 (-1.5)	1.02 (-1.9)	1.09 (0.7)	1.07 (1.5)

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Note: Close 10/2/26. The table reports shorter tail/longer tail ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

The efficient frontier of USD forward vol trades

We analyze all executable forward vol trades, expressed using gamma-neutral calendar spreads and triangles, and report 10 trades on the efficient frontier based on ex-ante Sharpe (measured by the vol-adjusted roll) and value (measured by 1y-zcore).

- Short-expiry forward vols on front-end rates look to be the most attractive from vol-adjusted carry and entry level perspectives. For example, 3m6m5y fwd vol (short 3m2y2y vs long 9m5y) has positive 0.80 Sharpe carry.

Figure 21. Forward vol trades on the efficient frontier (Sharpe vs z-score)

short expry	long expry	fwd vol	short vol, nv	long vol, nv	fwd vol, nv	3m roll, nv	Shrp	1y ZS	3m ZS
3m2y2y	9m5y	3m6m5y	122.4	111.7	107.2	5.27	0.80	3.5	3.1
6m1y1y	1y2y	6m6m2y	128.6	118.7	108.9	5.22	0.57	2.6	2.7
3m1y1y	9m2y	3m6m2y	132.9	119.5	114.4	5.01	0.56	2.6	2.9
3m1y2y	9m3y	3m6m3y	127.9	117.7	113.6	4.57	0.55	2.7	2.9
1y1y1y	18m2y	1y6m2y	127.7	117.6	95.1	6.06	0.51	1.3	1.3
6m1y2y	1y3y	6m6m3y	123.6	116.7	110.0	4.11	0.50	2.7	2.9
1y1y2y	18m3y	1y6m3y	122.3	115.4	100.4	4.44	0.45	1.8	2.4
3y2y/ 3y3y	5y1y	3y2y1y	108.5	109.1	110.0	1.71	0.40	3.1	3.1
6m2y2y	1y5y	6m6m5y	118.6	111.7	107.0	2.77	0.40	3.7	3.2
1y1y1y	2y1y	1y1y1y	127.7	119.3	110.1	2.89	0.34	2.2	3.0

Note: Close 10/2/26. The table reports 10 best forward vol trades on the efficient frontier based on ex-ante Sharpe ratio and 1y z-score. We analyze both calendar spreads (underlying forwards with matching mid-points) and triangles. The ex-ante Sharpe is calculated as vol-adjusted roll on the surface.

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Source: Citi Research

USD forward volatilities

We calculate forward 1y2y, 1y5y, 1y10y and 1y30y volatilities implied from the spot swaption surface. Forward volatilities reflect market expectations about future short-dated vols and may highlight dislocations on the spot vol surface.

- The term structures for forward vols have flattened over the past month with spot and front-end forward vols increasing more than long-end forward vols outright.

Figure 22. Forward volatilities

	1y2y							1y5y							1y10y							1y30y						
Yrs		1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y	
Fwd	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt	z-scr
spt	118.7	9.4	23.2	28.9	2.8	22.2	0.5	111.7	9.4	23.8	26.5	3.9	17.3	0.6	104.6	9.6	23.2	24.5	4.9	13.9	0.7	96.9	9.2	20.6	21.4	4.9	13.2	1.1
1	116.3	9.0	22.8	26.8	3.6	19.7	0.5	109.2	9.1	20.3	23.3	4.0	14.8	0.5	104.8	10.2	21.1	22.7	5.2	14.1	0.7	98.9	10.6	19.3	20.8	5.6	15.3	1.3
2	113.9	5.3	18.1	23.5	3.2	17.3	0.4	109.3	8.4	20.1	22.9	4.3	14.9	0.5	102.7	7.6	17.4	19.7	4.6	12.0	0.6	93.3	6.9	15.8	15.8	5.2	9.7	0.8
3	108.5	4.0	15.7	19.8	2.9	11.9	0.3	102.8	4.4	15.5	17.1	3.6	8.4	0.3	99.0	4.7	15.0	16.1	4.2	8.3	0.4	91.0	5.4	13.3	14.8	4.7	7.3	0.6
4	101.8	4.1	12.7	16.9	2.7	5.2	0.1	97.9	2.8	12.1	12.7	3.4	3.5	0.1	97.4	3.2	12.9	14.6	3.8	6.7	0.4	88.7	4.1	11.3	13.4	4.3	5.1	0.4
5	102.8	5.7	11.4	14.9	3.6	6.2	0.2	98.3	4.9	12.1	13.4	4.3	3.9	0.1	97.4	4.6	13.3	15.3	4.4	6.7	0.4	85.8	3.6	10.7	12.2	4.2	2.1	0.2
6	100.9	6.1	7.7	10.9	3.4	4.3	0.1	98.7	5.7	9.7	10.5	4.3	4.3	0.2	97.9	5.2	11.6	13.5	4.7	7.2	0.4	85.8	3.6	9.3	9.4	4.8	2.1	0.2
7	98.5	4.6	7.0	9.5	3.5	1.9	0.0	95.3	4.2	7.6	7.8	3.9	0.9	0.0	92.7	3.7	8.8	10.0	4.5	2.0	0.1	82.5	3.2	8.4	9.8	4.5	(1.2)	(0.1)
8	95.0	3.0	6.2	7.4	2.4	(1.5)	(0.0)	92.8	3.1	6.2	7.7	3.8	(1.5)	(0.1)	90.5	2.7	6.6	8.6	3.9	(0.2)	(0.0)	81.9	2.9	7.4	9.9	3.9	(1.8)	(0.2)
9	87.1	2.9	6.3	5.1	0.9	(9.5)	(0.2)	89.5	3.2	6.0	5.8	2.4	(4.9)	(0.2)	86.9	2.8	6.2	6.6	3.1	(3.8)	(0.2)	80.7	3.1	6.8	9.0	4.1	(3.0)	(0.3)

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Note: Close 10/2/26. Forward vols are computed using the standard triangular approximation from spot vols and using 6m realized correlations as a proxy for implied correlations. In addition to levels of forward vols, we calculate 1w and 1m changes, the difference from 1y average (with corresponding z-scores), and the difference from the 10y average of the corresponding 1y expiry spot vol (together with z-scores). Source: Citi Research

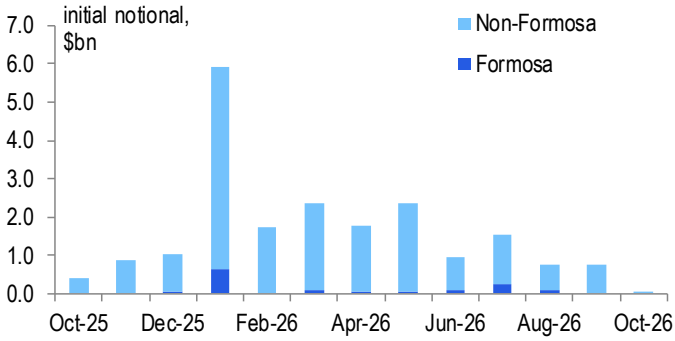
Long-dated callable USD supply

We report supply and redemptions data for long-dated callable bonds.

Prepared for Christopher Lee

- Gross issuance of long-maturity callables in September was one of the smallest YTD and nearly matching August's (Figure 23). We expect issuance for the rest of the year to remain low and broadly in line with last year's, which should continue to provide a tailwind for long-dated vol.

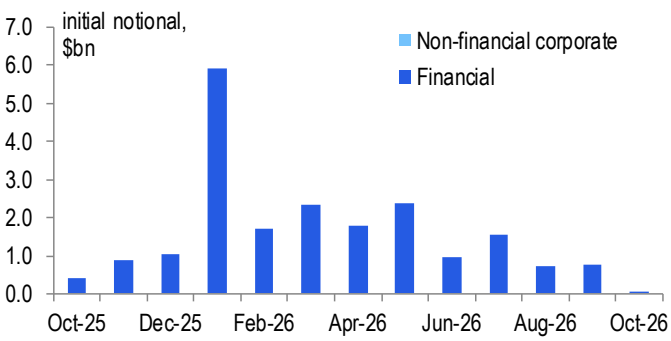
Figure 23. Gross callable supply: Formosa vs Non-Formosa



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Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

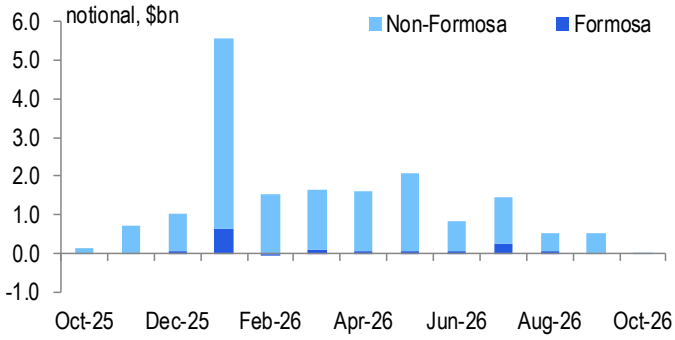
Figure 24. Gross callable supply: Financial vs Non-Financial



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Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

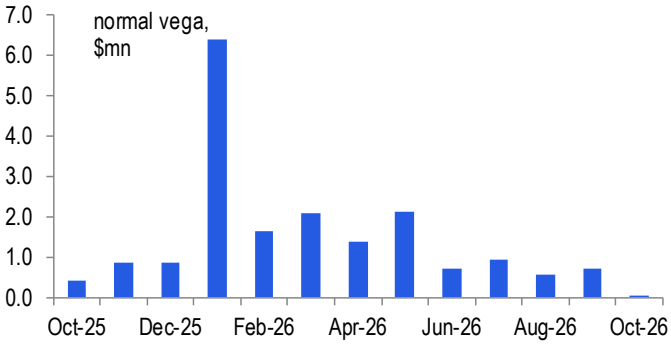
Figure 25. Net callable notional supply



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Source: Citi Research

Figure 26. Vega supply (normal parallel vega)



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Note: We assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

Figure 27. Recent callable supply estimates

	gross issued, initial notional, $bn					redeemed, $bn					net issued, $bn					Vega, $mn
	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	
Oct-26 MTD	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0
Sep-26 (prev month)	0.00	0.75	0.75	0.00	0.75	0.00	0.22	0.22	0.00	0.22	0.00	0.54	0.54	0.00	0.54	0.7
Oct-25 (prev year)	0.00	0.41	0.41	0.00	0.41	0.00	0.26	0.26	0.00	0.26	0.00	0.15	0.15	0.00	0.15	0.4
2026 YTD	1.28	16.86	18.14	0.00	18.14	0.07	2.36	2.42	0.00	2.42	1.21	14.50	15.71	0.00	15.71	16.6
2025 YTD	0.48	11.46	11.94	0.00	11.94	0.00	2.66	2.65	0.02	2.66	0.48	8.80	9.29	-0.02	9.28	12.2
2025 (all year)	0.55	13.33	13.88	0.00	13.88	0.00	2.88	2.87	0.02	2.88	0.55	10.44	11.01	-0.02	10.99	13.9

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Note: To estimate vega, we assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

USD swaption skew

USD skew history and z-scores

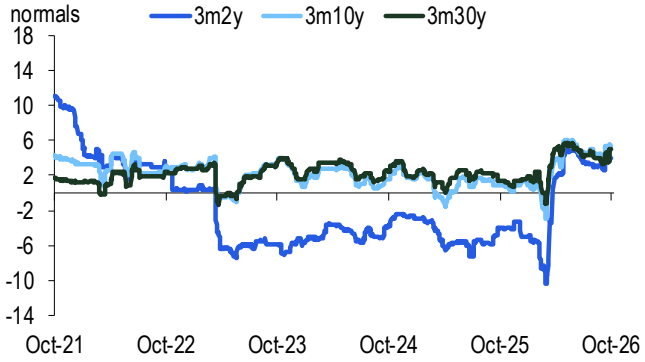
We calculate z-scores of payer and receiver swaption skews across the surface.

- Short-expiry receiver skews on front-end rates have continued to cheapen and are finally trading below ATM vol as we have expected (Figure 30).

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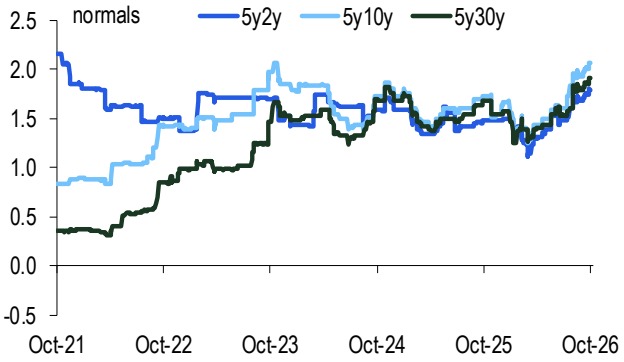
- Long-expiry payer skews on all rates have continued to richen as yields have reached new multi-decade highs (Figure 29). Anecdotally, demand for high strike rate hedges by non-rates investors could have been part of the reason for the richening.

Figure 28. 3m expiry payer skew



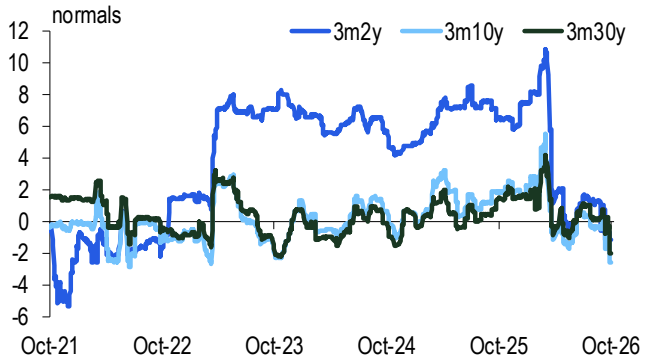
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Note: 25bp-out payer skews for 3m expiry payers. Source: Citi Research

Figure 29. 5y expiry payer skew



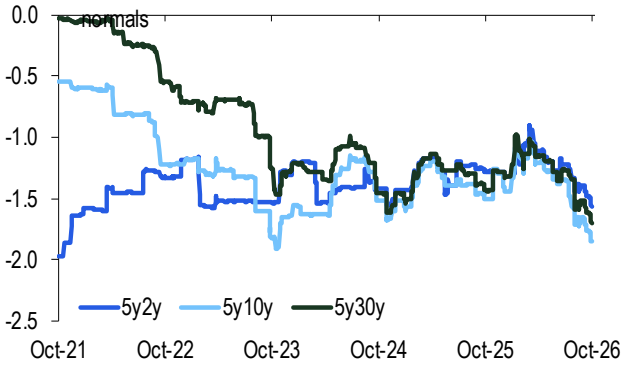
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Note: 25bp-out payer skews for 5y expiry payers. Source: Citi Research

Figure 30. 3m expiry receiver skew



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Note: 25bp-out receiver skews for 3m expiry receivers. Source: Citi Research

Figure 31. 5y expiry receiver skew



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Note: 25bp-out receiver skews for 5y expiry receivers. Source: Citi Research

Figure 32. Skews (1y z-scores) of risk reversals, OTM payer and OTM receivers

	Risk Reversals						Payer skew						Receiver skew					
expr\tail	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y
3m	8.6 (1.6)	5.2 (1.2)	5.9 (1.1)	6.9 (1.3)	7.9 (1.6)	7.0 (1.7)	5.8 (1.3)	4.0 (1.0)	4.4 (0.9)	4.8 (1.0)	5.3 (1.2)	4.9 (1.1)	-2.8 (-2.0)	-1.2 (-1.5)	-1.6 (-1.3)	-2.1 (-1.6)	-2.6 (-2.0)	-2.0 (-2.6)
6m	3.1 (1.2)	4.3 (1.4)	5.1 (1.4)	5.1 (1.5)	5.2 (1.6)	4.1 (1.4)	2.6 (1.1)	3.1 (1.3)	3.4 (1.2)	3.4 (1.3)	3.5 (1.4)	3.0 (1.2)	-0.5 (-1.4)	-1.2 (-1.6)	-1.7 (-1.5)	-1.7 (-1.6)	-1.7 (-1.8)	-1.1 (-1.6)
1y	2.2 (1.3)	2.9 (1.5)	3.8 (2.0)	4.1 (2.0)	4.4 (2.0)	3.9 (1.8)	1.8 (1.1)	2.1 (1.5)	2.3 (1.7)	2.5 (1.8)	2.7 (1.8)	2.4 (1.6)	-0.5 (-1.4)	-0.8 (-1.6)	-1.4 (-2.2)	-1.6 (-2.2)	-1.7 (-2.3)	-1.5 (-1.9)
2y	1.8 (0.4)	2.7 (1.3)	3.8 (2.7)	4.0 (2.8)	4.2 (2.6)	3.8 (2.4)	1.2 (0.4)	1.6 (1.2)	2.1 (2.6)	2.2 (2.6)	2.3 (2.5)	2.1 (2.2)	-0.6 (-0.5)	-1.1 (-1.4)	-1.7 (-2.9)	-1.8 (-2.9)	-1.9 (-2.7)	-1.7 (-2.5)
3y	3.1 (1.7)	3.1 (1.8)	3.5 (2.2)	3.8 (2.4)	4.3 (2.5)	3.7 (2.0)	1.8 (1.6)	1.7 (1.7)	1.9 (2.1)	2.1 (2.2)	2.3 (2.4)	2.0 (1.9)	-1.4 (-1.7)	-1.3 (-1.9)	-1.6 (-2.4)	-1.7 (-2.5)	-2.0 (-2.7)	-1.7 (-2.1)
5y	3.3 (1.7)	3.4 (2.3)	3.4 (1.9)	3.6 (2.2)	3.9 (2.3)	3.6 (2.3)	1.8 (1.5)	1.8 (2.2)	1.8 (1.8)	1.9 (2.0)	2.1 (2.2)	1.9 (2.2)	-1.6 (-1.9)	-1.6 (-2.5)	-1.6 (-2.0)	-1.7 (-2.3)	-1.9 (-2.4)	-1.7 (-2.5)
10y	3.1 (1.7)	3.1 (1.8)	3.3 (1.9)	3.4 (1.9)	3.6 (1.9)	3.5 (1.6)	1.6 (1.6)	1.6 (1.7)	1.7 (1.7)	1.8 (1.8)	1.9 (1.8)	1.8 (1.4)	-1.5 (-1.8)	-1.5 (-2.0)	-1.6 (-2.1)	-1.6 (-2.1)	-1.8 (-2.1)	-1.7 (-1.7)

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Note: Close 10/2/26. The table reports skews and 1y z-scores in parentheses for risk reversals, payer and receiver skews, defined in normal vol terms. We use 25bp OTM. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

USD implied vs realized skew

We compare implied swaption skews with realized skews. We calculate realized skews as the model value of OTM options from the empirical SABR model calibrated to the recent (3m) realized dynamics of ATM volatilities and rates.

- Long-dated risk reversal skews are still trading rich to our fair value model but less so than before (Figure 35).

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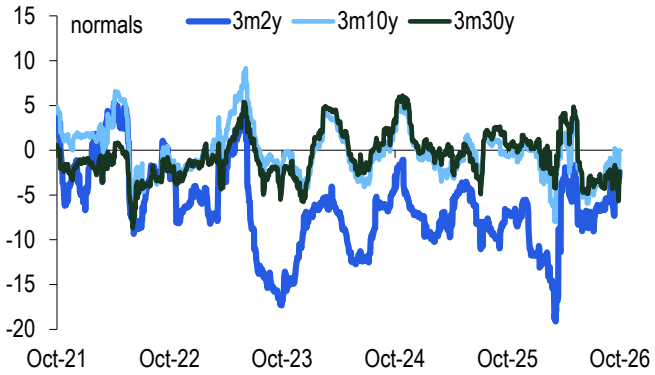
Figure 33. Implied vs realized skews (spread to the model for risk reversals, price/model ratio for payers and receivers)

	Risk Reversals (1Y z-score)					OTM payers (1Y z-score)					OTM receivers (1Y z-score)				
expr\tail	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y
3m	1.9 (2.6)	-2.4 (1.5)	-1.8 (1.0)	0.0 (0.7)	-2.4 (-0.6)	1.02 (1.7)	1.00 (1.2)	1.00 (0.8)	1.01 (0.3)	1.00 (-1.0)	1.00 (-2.6)	1.02 (-1.6)	1.02 (-1.3)	1.01 (-0.9)	1.03 (0.0)
6m	-1.3 (2.0)	-0.5 (2.0)	-0.3 (1.7)	-0.1 (0.5)	-2.1 (-1.0)	1.00 (1.6)	1.01 (1.7)	1.01 (1.4)	1.01 (0.2)	1.00 (-1.0)	1.02 (-1.9)	1.01 (-1.9)	1.01 (-1.8)	1.01 (-0.6)	1.02 (0.6)
1y	-1.5 (1.5)	-1.3 (1.4)	-0.6 (0.8)	0.4 (0.0)	-0.8 (-1.1)	1.00 (1.6)	1.00 (1.5)	1.00 (0.7)	1.01 (-0.2)	1.00 (-1.2)	1.01 (-1.8)	1.01 (-1.6)	1.01 (-1.0)	1.00 (-0.1)	1.01 (0.9)
2y	-1.5 (-0.5)	-0.6 (0.2)	0.5 (0.4)	1.4 (0.5)	0.7 (-0.5)	1.00 (-0.4)	1.00 (0.2)	1.00 (0.2)	1.01 (0.1)	1.01 (-0.7)	1.01 (0.3)	1.01 (-0.4)	1.00 (-0.4)	0.99 (-0.4)	1.00 (0.4)
3y	0.2 (0.5)	0.2 (0.2)	0.8 (0.0)	1.4 (0.3)	1.1 (-0.6)	1.00 (0.4)	1.00 (0.1)	1.01 (-0.1)	1.01 (0.0)	1.01 (-0.9)	1.00 (-0.5)	1.00 (-0.2)	1.00 (0.1)	0.99 (-0.1)	1.00 (0.7)
5y	0.9 (-0.5)	1.0 (-0.2)	0.9 (-0.6)	1.5 (-0.1)	1.4 (-0.9)	1.01 (-0.6)	1.01 (-0.4)	1.01 (-0.7)	1.01 (-0.5)	1.01 (-1.5)	1.00 (0.6)	1.00 (0.4)	1.00 (0.7)	0.99 (0.4)	0.99 (1.2)
10y	1.2 (-1.4)	1.2 (-1.3)	1.4 (-1.2)	1.8 (-1.3)	1.8 (-1.6)	1.01 (-1.6)	1.01 (-1.5)	1.01 (-1.6)	1.01 (-1.8)	1.01 (-2.1)	0.99 (1.6)	0.99 (1.5)	0.99 (1.5)	0.99 (1.7)	0.99 (1.9)

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Note: Close 10/2/26. We calibrate an empirical SABR model to the recent observed dynamics of ATM vols and rates. We use a 2y rolling window to estimate a backbone parameter, and a 3m window to estimate correlation and vol-of-vol parameters. The empirical model matches ATM implied vols exactly, so it's different from the market only in skew parameters. We calculate "realized skew" as the theoretical values of OTM payers, receivers and risk reversals from this empirical model. The table shows market-"realized" spreads of risk reversals, and market/"realized" ratios of OTM payers and receivers together with 1y z-scores of these spreads and ratios in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

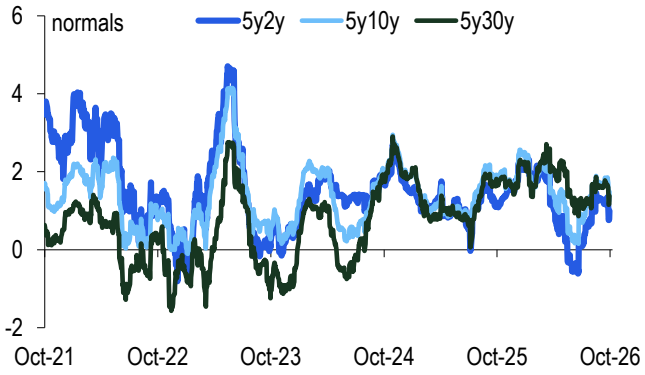
Figure 34. 3m expiry risk reversals/model spread



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Note: The chart shows model-"realized" spreads for 25bp-out risk reversals in 3m expiries using the same methodology as in Figure 33. Source: Citi Research.

Figure 35. 5y expiry risk reversals/model spread



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Note: The chart shows model-"realized" spreads for 100bp-out risk reversals in 5y expiries using the same methodology as in Figure 33. Source: Citi Research.

USD Board/swaption relative value

We evaluate the relative pricing of Board and swaption volatilities.

- Board vols, especially for US contracts, are trading rich to swaption vols (Figure 36), and the implied board/swaption vol ratios are currently higher than the recent realized board/swaption vol ratios.
- US board put skew is trading well above board call skews and in-line with swaption payer skews (Figure 39).

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Figure 36. Board vols vs swaptions

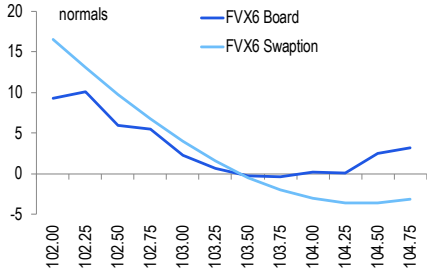
option	FVX6	FVZ6	TYX6	TYZ6	USX6	USZ6
Board vol	127.1	129.9	122.0	122.4	105.5	109.0
1w chg	11.6	13.4	9.9	9.0	13.2	14.9
Board/ 1m RV	1.36	1.39	1.41	1.41	1.56	1.62
Board/ 3m RV	1.57	1.60	1.57	1.58	1.67	1.73
CTD-matched swpn vol	123.2	125.6	115.3	116.0	98.3	100.6
1w chg	8.5	12.7	7.6	10.1	7.9	10.4
swpn/ 1m RV	1.31	1.34	1.34	1.34	1.47	1.51
swpn/ 3m RV	1.57	1.61	1.60	1.61	1.69	1.73
Board/swpn	1.03	1.03	1.06	1.05	1.07	1.08
1w ago	1.01	1.03	1.04	1.07	1.02	1.04
ftrs 1m RV/swap 1m RV	0.99	0.99	1.01	1.01	1.01	1.01
ftrs 3m RV/swap 3m RV	1.04	1.04	1.08	1.08	1.08	1.08

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Note: Close 10/2/26. The table shows implied ATM (interpolated) Board vols vs CTD-matched swaption vols with corresponding realized volatilities

Source: Citi Research

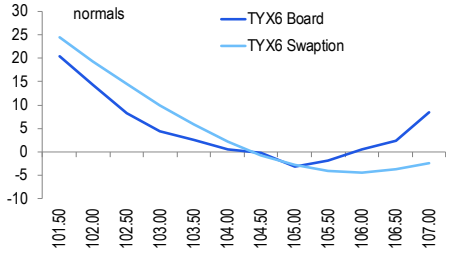
Figure 37. FV Board vs swaption skew



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Note: Close 10/2/26. Swaption strike = Board strike yield+invoice spread. Source: Citi Research

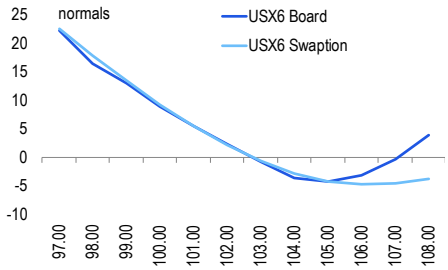
Figure 38. TY Board vs swaption skew



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Note: Close 10/2/26. Swaption strike = Board strike yield+invoice spread. Source: Citi Research

Figure 39. US Board vs swaption skew



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Note: Close 10/2/26. Swaption strike = Board strike yield+invoice spread. Source: Citi Research

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Figure 40. Conditional spread trades

							puts/payers								calls/receivers							
optn	expiry	futures price	fut DV01	swap DV01	swpn ntl, $mn	invc sprd, bp	put strike	strike yield, %	costless payer strike,	strike spread, bp	bp pick-up	put vol	payer vol	vol ratio	call strike	strike yield, %	costless receiver strike, %	strike spread, bp	bp pick-up	call vol	rcvr vol	vol ratio
FVX6	10/23/26	103-14	3.92	3.68	106.4	-31.0	103.00	5.23	4.89	-33.7	2.7	129.4	126.1	1.03	103.75	5.04	4.70	-33.7	2.6	126.7	120.7	1.05
FVX6	10/23/26	103-14	3.92	3.68	106.4	-31.0	102.75	5.29	4.95	-34.1	3.0	132.6	128.6	1.03	104.00	4.97	4.64	-33.0	1.9	127.4	119.9	1.06
FVX6	10/23/26	103-14	3.92	3.68	106.4	-31.0	102.50	5.36	5.02	-33.4	2.3	133.1	131.8	1.01	104.25	4.91	4.58	-32.6	1.5	127.2	119.5	1.06
FVX6	10/23/26	103-14	3.92	3.68	106.4	-31.0	102.25	5.42	5.08	-33.8	2.8	137.2	134.8	1.02	104.50	4.85	4.53	-31.4	0.4	129.7	119.6	1.08
FVX6	10/23/26	103-14	3.92	3.68	106.4	-31.0	102.00	5.49	5.17	-31.9	0.9	136.5	139.3	0.98	104.75	4.78	4.47	-30.9	-0.1	130.3	120.0	1.09
FVZ6	11/20/26	103-14	3.92	3.68	106.4	-31.0	103.25	5.16	4.84	-32.5	1.5	130.6	126.2	1.03	103.50	5.10	4.77	-32.9	1.9	129.6	125.0	1.04
FVZ6	11/20/26	103-14	3.92	3.68	106.4	-31.0	103.00	5.23	4.90	-32.6	1.5	132.0	127.7	1.03	103.75	5.04	4.71	-33.1	2.0	127.7	124.1	1.03
FVZ6	11/20/26	103-14	3.92	3.68	106.4	-31.0	102.75	5.29	4.97	-32.2	1.1	132.4	129.6	1.02	104.00	4.97	4.65	-32.8	1.7	127.4	123.6	1.03
FVZ6	11/20/26	103-14	3.92	3.68	106.4	-31.0	102.50	5.36	5.03	-32.5	1.5	135.2	131.5	1.03	104.25	4.91	4.59	-32.4	1.4	127.7	123.3	1.04
FVZ6	11/20/26	103-14	3.92	3.68	106.4	-31.0	102.25	5.42	5.10	-31.7	0.7	135.7	134.1	1.01	104.50	4.85	4.53	-31.9	0.8	128.7	123.4	1.04
FVZ6	11/20/26	103-14	3.92	3.68	106.4	-31.0	102.00	5.49	5.17	-31.4	0.3	137.6	136.6	1.01	104.75	4.78	4.47	-31.7	0.6	129.1	123.8	1.04
TYX6	10/23/26	104-11+	5.91	5.60	105.5	-38.4	104.00	5.27	4.86	-41.0	2.6	122.6	116.5	1.05	104.50	5.19	4.79	-39.4	1.0	121.8	114.2	1.07
TYX6	10/23/26	104-11+	5.91	5.60	105.5	-38.4	103.50	5.36	4.95	-40.9	2.5	124.7	119.9	1.04	105.00	5.10	4.71	-39.2	0.8	118.8	112.2	1.06
TYX6	10/23/26	104-11+	5.91	5.60	105.5	-38.4	103.00	5.44	5.04	-40.4	2.0	126.4	124.2	1.02	105.50	5.02	4.64	-38.0	-0.3	120.2	111.2	1.08
TYX6	10/23/26	104-11+	5.91	5.60	105.5	-38.4	102.50	5.53	5.13	-40.3	1.9	130.3	128.7	1.01	106.00	4.94	4.57	-36.6	-1.7	122.5	110.9	1.10
TYX6	10/23/26	104-11+	5.91	5.60	105.5	-38.4	102.00	5.62	5.21	-41.0	2.7	136.3	133.1	1.02	106.50	4.85	4.50	-35.4	-3.0	124.4	111.2	1.12
TYX6	10/23/26	104-11+	5.91	5.60	105.5	-38.4	101.50	5.70	5.28	-41.8	3.5	142.4	137.6	1.03	107.00	4.77	4.45	-32.4	-5.9	130.5	111.8	1.17
TYZ6	11/20/26	104-11+	5.91	5.60	105.5	-38.4	104.00	5.27	4.86	-40.8	2.5	124.5	116.8	1.07	104.50	5.19	4.80	-38.6	0.2	121.5	115.5	1.05
TYZ6	11/20/26	104-11+	5.91	5.60	105.5	-38.4	103.50	5.36	4.95	-41.2	2.8	127.0	118.9	1.07	105.00	5.10	4.72	-37.9	-0.5	121.4	114.3	1.06
TYZ6	11/20/26	104-11+	5.91	5.60	105.5	-38.4	103.00	5.44	5.03	-41.3	3.0	129.2	121.6	1.06	105.50	5.02	4.64	-37.5	-0.9	120.6	113.6	1.06
TYZ6	11/20/26	104-11+	5.91	5.60	105.5	-38.4	102.50	5.53	5.12	-41.3	2.9	131.6	124.6	1.06	106.00	4.94	4.57	-36.8	-1.6	121.3	113.5	1.07
TYZ6	11/20/26	104-11+	5.91	5.60	105.5	-38.4	102.00	5.62	5.20	-41.7	3.3	135.1	127.9	1.06	106.50	4.85	4.49	-36.0	-2.3	122.4	113.8	1.08
TYZ6	11/20/26	104-11+	5.91	5.60	105.5	-38.4	101.50	5.70	5.29	-41.3	3.0	137.8	131.7	1.05	107.00	4.77	4.42	-34.8	-3.5	124.6	114.5	1.09
USX6	10/23/26	102-24	13.44	11.94	112.5	-72.7	102.00	5.83	5.08	-75.4	2.7	107.9	99.4	1.09	103.00	5.76	5.03	-72.8	0.1	104.7	97.6	1.07
USX6	10/23/26	102-24	13.44	11.94	112.5	-72.7	101.00	5.91	5.15	-75.8	3.1	111.0	102.3	1.08	104.00	5.68	4.96	-72.6	-0.1	102.0	95.6	1.07
USX6	10/23/26	102-24	13.44	11.94	112.5	-72.7	100.00	5.98	5.22	-76.2	3.5	114.4	105.8	1.08	105.00	5.61	4.89	-72.0	-0.7	101.4	94.3	1.08
USX6	10/23/26	102-24	13.44	11.94	112.5	-72.7	99.00	6.06	5.29	-76.8	4.1	118.5	109.5	1.08	106.00	5.54	4.83	-71.0	-1.7	102.4	93.6	1.09
USX6	10/23/26	102-24	13.44	11.94	112.5	-72.7	98.00	6.14	5.37	-76.9	4.2	122.0	113.8	1.07	107.00	5.47	4.77	-69.5	-3.2	105.2	93.6	1.12
USX6	10/23/26	102-24	13.44	11.94	112.5	-72.7	97.00	6.22	5.44	-78.1	5.4	127.7	117.7	1.09	108.00	5.40	4.73	-67.3	-5.4	109.4	93.8	1.17
USZ6	11/20/26	102-24	13.44	11.94	112.5	-72.7	102.00	5.83	5.08	-75.6	2.9	110.5	101.2	1.09	103.00	5.76	5.04	-71.6	-1.1	108.5	100.4	1.08
USZ6	11/20/26	102-24	13.44	11.94	112.5	-72.7	101.00	5.91	5.14	-76.3	3.6	113.7	103.0	1.10	104.00	5.68	4.97	-71.6	-1.1	105.8	99.1	1.07
USZ6	11/20/26	102-24	13.44	11.94	112.5	-72.7	100.00	5.98	5.21	-77.1	4.4	116.9	105.1	1.11	105.00	5.61	4.90	-71.6	-1.1	104.0	98.3	1.06
USZ6	11/20/26	102-24	13.44	11.94	112.5	-72.7	99.00	6.06	5.29	-77.6	4.8	119.5	107.7	1.11	106.00	5.54	4.82	-71.5	-1.3	103.5	98.0	1.06
USZ6	11/20/26	102-24	13.44	11.94	112.5	-72.7	98.00	6.14	5.36	-77.8	5.1	122.0	110.6	1.10	107.00	5.47	4.75	-71.6	-1.1	102.8	98.2	1.05
USZ6	11/20/26	102-24	13.44	11.94	112.5	-72.7	97.00	6.22	5.43	-79.0	6.3	126.2	113.4	1.11	108.00	5.40	4.69	-71.1	-1.6	104.0	98.7	1.05

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Note: Close 10/2/26. Swaption notional is for 1000 Board contracts. The costless swaption strike makes the DV01-neutral trade costless. The BP pick-up is the spread between the costless strike spread and invoice spread. Futures trading involves substantial risk of loss. Source: Citi Research

USD midcurve swaptions

We report levels and z-scores of at-the-money vols, implied correlations, and implied/realized ratios of correlations. We also compute the relative richness/cheapness of midcurve vols to principal components of the plain vanilla surface.

- Short-dated midcurve vols on 5y5y look rich to swaption vols based on the implied correlations (Figure 42) and the multi-variable regressions (Figure 44).

Figure 41. Implied vols, normals (3m z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	141.6 (3.0)	134.4 (3.1)	129.7 (3.0)	136.4 (3.0)	130.4 (3.0)	125.3 (3.0)	108.0 (2.9)	96.6 (2.7)
3m	132.9 (2.9)	126.0 (3.1)	121.6 (3.1)	127.9 (3.0)	122.4 (3.1)	117.9 (3.0)	102.4 (2.9)	95.8 (2.9)
6m	128.6 (3.0)	120.8 (3.2)	118.8 (3.2)	123.6 (3.1)	118.6 (3.2)	114.5 (3.2)	105.1 (3.3)	97.5 (3.1)
1y	127.7 (3.0)	118.7 (3.1)	118.3 (3.1)	122.3 (3.0)	117.9 (3.1)	113.5 (3.1)	104.5 (3.1)	99.3 (3.2)
2y	113.9 (3.1)	113.6 (3.2)	106.2 (3.3)	113.2 (3.1)	109.0 (3.2)	104.9 (3.2)	108.3 (3.6)	99.2 (3.5)

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Close 10/2/26. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 42. Implied correlations, % (3y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	97.6 (0.0)	97.8 (-0.6)	99.3 (0.2)	88.6 (-0.2)	95.3 (-0.3)	96.6 (-0.3)	94.7 (-0.9)	97.0 (-0.7)
3m	99.1 (0.6)	98.9 (0.1)	99.3 (-0.4)	95.7 (0.5)	96.7 (-0.3)	97.5 (-0.4)	95.6 (-1.1)	97.5 (-0.3)
6m	97.1 (-0.1)	99.4 (0.6)	99.2 (-0.4)	94.8 (0.4)	97.2 (0.2)	97.9 (0.0)	96.4 (-0.4)	97.6 (-0.2)
1y	95.2 (-1.5)	99.2 (0.9)	99.0 (-0.9)	91.6 (-0.2)	96.1 (0.0)	98.1 (0.3)	97.4 (1.8)	97.8 (0.1)
2y	99.8 (0.0)	99.3 (-0.6)	99.9 (-0.1)	97.5 (-0.3)	99.1 (-0.3)	99.8 (-0.1)	97.1 (0.3)	98.6 (1.1)

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Close 10/2/26. The table shows implied correlations (3y z-scores in parentheses). Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

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Figure 43. Implied/realized correlation ratios (1y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	1.00 (0.0)	0.99 (-0.7)	1.00 (-0.4)	0.95 (-0.2)	0.98 (-0.9)	0.99 (-0.1)	0.99 (0.6)	1.01 (0.1)
3m	1.01 (0.4)	1.00 (-0.1)	1.00 (0.0)	1.01 (0.1)	1.00 (-0.5)	1.00 (0.4)	1.00 (0.4)	1.01 (0.8)
6m	0.99 (0.0)	1.00 (0.6)	1.00 (0.5)	0.99 (0.2)	1.00 (0.6)	1.00 (0.5)	1.01 (0.8)	1.01 (0.4)
1y	0.96 (-2.3)	1.00 (0.4)	0.99 (-0.3)	0.94 (-1.1)	0.98 (0.1)	1.00 (0.3)	1.01 (1.7)	1.01 (0.5)
2y	1.01 (-0.4)	1.00 (-0.9)	1.00 (-1.4)	1.00 (-0.6)	1.01 (-1.3)	1.01 (-1.2)	1.00 (0.5)	1.01 (0.7)

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Close 10/2/26. The table shows the ratios of implied to 3m realized correlations (in daily changes). The 1y z-scores of ratios are shown in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 44. Rich/cheap to plain vanilla swaptions (z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	-1.1 (-0.2)	6.2 (0.7)	-4.2 (-0.5)	4.2 (0.8)	0.5 (0.1)	2.1 (0.4)	6.5 (2.2)	5.6 (1.6)
3m	-2.8 (-0.7)	-2.4 (-0.4)	2.9 (0.6)	-1.9 (-0.5)	0.7 (0.2)	1.2 (0.3)	4.6 (1.9)	3.6 (1.2)
6m	1.0 (0.3)	-4.9 (-1.1)	0.6 (0.1)	-0.6 (-0.2)	-1.5 (-0.4)	-2.6 (-0.7)	1.5 (1.1)	2.8 (1.0)
1y	3.6 (1.4)	-6.1 (-1.9)	1.1 (0.3)	-1.1 (-0.5)	-1.1 (-0.3)	-3.0 (-0.8)	-1.9 (-1.9)	1.8 (0.6)
2y	-0.6 (-0.3)	0.8 (0.3)	-2.4 (-0.7)	0.1 (0.1)	-1.0 (-0.3)	-2.3 (-0.6)	0.2 (0.2)	-1.1 (-0.4)

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Close 10/2/26. We regress midcurve vols on two corresponding plain vanilla vols using last 3y of data. For example, 1y2y1y is regressed against 1y3y and 1y2y vols. We calculate rich/cheap and z-scores in brackets. Source: Citi Research

USD curve (spread) CMS options

We report levels and z-scores of at-the-money vols, rich/cheap on PCA, implied correlations, and implied/realized vol ratios for curve options.

- The implied correlations for short and longer expiry CMS curve options have declined in recent weeks, reflecting implied curve vols' outperformance relative to swaption vols (Figure 47).

Figure 45. Implied vols, normals (3m z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	39.6 (2.4)	71.4 (2.6)	94.6 (2.6)	32.9 (2.2)	69.2 (2.3)	40.1 (2.8)
3m	40.6 (2.4)	70.6 (2.6)	94.1 (2.6)	32.5 (2.6)	68.9 (2.6)	38.8 (2.9)
6m	40.7 (2.5)	71.0 (2.6)	94.7 (2.7)	32.5 (2.9)	69.5 (2.8)	39.0 (3.0)
1y	40.7 (2.6)	70.5 (2.7)	92.3 (2.9)	32.3 (3.1)	69.2 (3.0)	39.1 (3.2)
2y	40.0 (3.0)	63.9 (3.1)	84.4 (3.1)	30.3 (3.2)	63.3 (3.3)	37.5 (3.2)
3y	38.0 (3.1)	57.6 (3.1)	79.5 (3.1)	30.3 (3.2)	58.3 (3.2)	33.4 (3.3)
5y	31.8 (3.2)	47.7 (3.2)	64.8 (3.2)	28.1 (3.2)	49.4 (3.2)	30.7 (3.2)

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Close 10/2/26. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

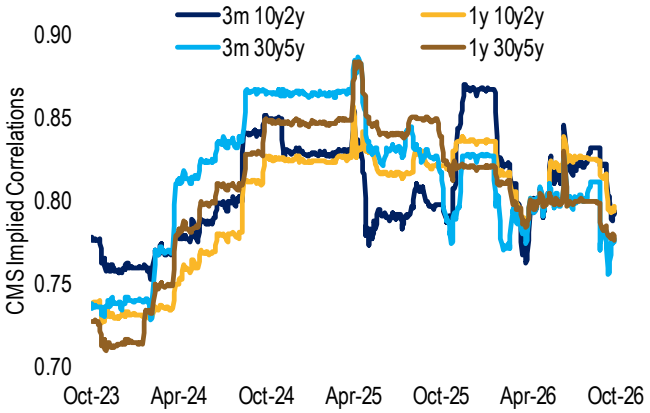
Figure 46. Rich/cheap to PCA, normals (z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	-2.6 (-0.9)	5.9 (1.9)	8.8 (2.5)	0.6 (0.3)	6.4 (1.7)	2.5 (0.9)
3m	-2.6 (-1.2)	3.6 (1.6)	5.6 (2.0)	-0.1 (0.0)	4.1 (1.3)	0.0 (0.0)
6m	-4.0 (-2.4)	0.3 (0.2)	1.7 (0.7)	-1.7 (-1.8)	2.5 (0.8)	-0.9 (-0.7)
1y	-5.6 (-3.8)	-5.1 (-2.5)	-5.5 (-1.5)	-2.7 (-2.6)	0.1 (0.0)	-0.5 (-0.6)
2y	-3.3 (-2.7)	-5.4 (-3.8)	-4.0 (-2.0)	-1.8 (-1.4)	-1.7 (-0.9)	-0.7 (-0.7)
3y	-2.4 (-1.9)	-4.6 (-4.3)	-0.2 (-0.2)	1.4 (1.7)	-0.6 (-0.3)	-1.5 (-1.3)
5y	3.7 (3.7)	-5.5 (-3.9)	-0.2 (-0.2)	4.2 (3.9)	0.0 (0.0)	1.5 (1.4)

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Close 10/2/26. We calculate rich/cheap and z-scores in parentheses from PCA on curve vols using two principal components and 3y of data. By design, this framework only yields relative value. Source: Citi Research

Figure 47. Implied correlations of CMS curve options



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Note: The chart shows implied correlation of options on the 2s10s curve with different expirations. Source: Citi Research

Figure 48. Implied/realized ratios

US Implied/1m Realized Vol Ratio (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.40 (-0.2)	1.48 (0.2)	1.29 (0.0)	1.28 (-0.4)	1.25 (0.0)	1.22 (0.0)
3m	1.39 (-0.5)	1.39 (-0.1)	1.21 (-0.3)	1.24 (-0.6)	1.22 (-0.3)	1.17 (-0.3)
6m	1.37 (-0.6)	1.36 (-0.2)	1.19 (-0.4)	1.24 (-0.7)	1.24 (-0.4)	1.19 (-0.5)
1y	1.42 (-0.7)	1.38 (-0.4)	1.19 (-0.6)	1.30 (-0.8)	1.29 (-0.5)	1.24 (-0.6)

implied/3m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.52 (0.3)	1.59 (1.2)	1.43 (1.0)	1.43 (0.6)	1.44 (1.3)	1.48 (1.6)
3m	1.55 (0.3)	1.54 (1.1)	1.38 (0.7)	1.40 (0.4)	1.42 (1.0)	1.43 (1.1)
6m	1.58 (0.3)	1.54 (1.0)	1.38 (0.7)	1.42 (0.5)	1.45 (0.9)	1.46 (0.9)
1y	1.69 (0.3)	1.61 (0.8)	1.41 (0.5)	1.51 (0.6)	1.54 (0.9)	1.55 (0.8)

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Close 10/2/26. Close-to-close realized vol is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

SOFR futures convexity adjustments

We calculate SOFR/FRA convexity adjustment for 1y SOFR packs and compare them to model values calibrated to cap/floor volatilities.

- Convexity adjustments for Blues (Figure 50) and Golds (Figure 51) have been drifting modestly higher but they are well below their model fair values. Our model fair values have rebounded meaningfully along with the higher rates vol and are suggesting that Gold convexity adjustments are the most undervalued at about -5bps relative to our model.
- Note: Futures trading involves substantial risk of loss.

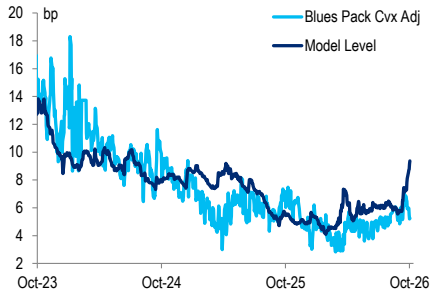
Figure 49. Convexity adjustments for 1y SOFR futures packs (vs CME swaps)

	convexity adjustment					vs the model				implied vs realized vol			
ED Pack	Cvx Adj (bp)	1 Week Chg (bp)	3m Z-Score	1Y Z-Score	Model	Vs Model	3m Z-Score	1Y Z-Score	3m Roll (short cvx,	Implied Vol	Realized Vol	Implied/ Realized	Cap vol Impl/Rlzd
Z7-U8	1.50	0.03	2.20	3.75	2.30	-0.81	-2.57	-1.67	0.45	107.3	87.1	1.2	1.4
H8-Z8	1.98	0.03	2.20	3.75	3.03	-1.06	-2.86	-1.77	0.48	106.8	85.0	1.3	1.5
M8-H9	2.50	0.06	2.25	**3.79**	3.81	-1.30	-3.01	-1.86	0.53	105.9	83.1	1.3	1.5
U8-M9	3.07	0.12	**2.36**	**3.91**	4.60	-1.53	-3.06	-1.93	0.57	105.0	81.4	**1.3**	1.5
Z8-U9	3.69	0.21	**2.51**	**4.15**	5.42	-1.73	-3.03	-2.00	0.62	104.1	79.8	**1.3**	1.5
H9-Z9	4.23	0.25	**2.33**	3.53	6.29	-2.07	-3.19	-2.56	0.54	101.8	77.9	**1.3**	1.5
M9-H0	4.65	0.18	1.43	2.05	7.24	-2.60	-3.33	-2.94	0.42	98.2	76.6	1.3	1.5
U9-M0	4.96	0.00	0.31	0.94	8.27	-3.30	-3.37	-3.00	0.32	94.0	75.5	1.2	1.5
Z9-U0	5.24	-0.24	-0.43	0.34	9.38	-4.13	-3.33	-2.95	0.28	89.9	74.8	1.2	1.6
H0-Z0	5.69	-0.40	-0.46	0.28	10.53	-4.83	-3.40	-2.86	0.45	87.6	74.3	1.2	1.6
M0-H1	6.50	-0.33	0.08	0.60	11.70	-5.20	-3.39	-2.69	**0.81**	88.0	73.9	1.2	1.5
U0-M1	7.63	-0.14	0.82	1.20	12.91	-5.28	-3.37	-2.51	**1.12**	89.8	73.6	1.2	1.5
Z0-U1	8.96	0.08	1.37	1.96	14.12	-5.16	-3.31	-2.34	**1.33**	92.1	73.3	1.3	1.5

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Note: Close 10/2/26. Convexity adjustments for 1y SOFR packs are computed as the spread between the pack's rate (the average of 4 SOFR rates in the pack) and matched-maturity forward 1y CME swap rate. The model for convexity adjustment is the Ho-Lee model calibrated to cap/floor vols. Implied vol is calculated by matching the model to the observed convexity adjustment. Realized vol is 3m realized vol of the corresponding pack. For each valuation metric, we mark three best short convexity trades in **bold**. Note that these are the most attractive in the chart but not necessarily trades that we would recommend at this point in time. Source: Citi Research

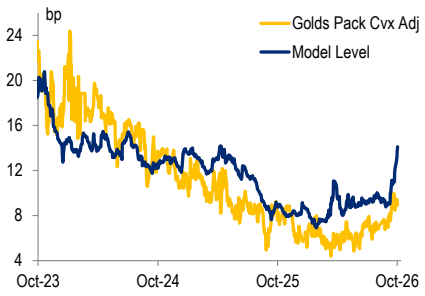
Figure 50. Blues pack (rolling) vs cap/floor-based model



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The chart shows the convexity adjustment of the Blues pack vs the (Ho-Lee) model. Source: Citi Research

Figure 51. Golds pack (rolling) vs cap/floor-based model



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The chart shows the convexity adjustment of the Golds pack vs the (Ho-Lee) model. Source: Citi Research

Prepared for Christopher Lee

USD conditional curves

We analyze conditional curves based on the entry point, implied/realized betas, and carry.

Figure 52. Conditional curves: bear trades

		spt and fwd crvs			costless				imp vs rlzd beta			bear steepener				bear flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/51.1	20.0	10.2	2.6	15.1	12.6	2.6	44	0.10	0.01	-0.09	9.8	-4.9	4.9	0.0	-9.8	4.9	-4.9	0.0
3m 1y3y	100/34.9	25.4	11.7	2.6	16.3	15.2	2.7	33	0.09	-0.02	-0.11	13.6	-4.6	9.1	0.0	-13.6	4.6	-9.1	0.0
3m 1y5y	100/22.0	28.5	12.7	2.4	14.8	14.9	2.5	24	0.04	-0.09	-0.13	15.8	-2.1	13.7	0.0	-15.8	2.1	-13.7	0.0
3m 2y5y	100/43.0	8.5	2.6	1.1	-0.2	2.4	1.1	15	-0.06	-0.09	-0.04	6.0	2.8	8.7	0.0	-6.0	-2.8	-8.7	0.0
3m 2y10y	100/24.1	19.1	12.0	0.2	3.9	-2.9	0.1	10	-0.18	-0.25	-0.08	7.1	8.2	15.2	0.0	-7.1	-8.2	-15.2	0.0
3m 2y30y	100/12.2	27.6	18.5	-0.4	5.7	-14.3	-0.4	9	-0.33	-0.52	-0.20	9.1	12.8	21.9	0.0	-9.1	-12.8	-21.9	0.0
3m 3y5y	100/62.9	3.2	1.0	0.5	-1.4	-0.2	0.5	10	-0.05	-0.06	-0.01	2.1	2.4	4.6	0.0	-2.1	-2.4	-4.6	0.0
3m 3y10y	100/35.3	13.7	10.5	-0.2	2.7	-5.6	-0.2	8	-0.17	-0.22	-0.05	3.3	7.8	11.1	0.0	-3.3	-7.8	-11.1	0.0
3m 3y30y	100/17.9	22.2	16.9	-0.7	4.5	-16.9	-0.6	9	-0.30	-0.47	-0.16	5.3	12.4	17.7	0.0	-5.3	-12.4	-17.7	0.0
3m 5y10y	100/56.0	10.6	9.5	-0.6	4.3	-5.3	-0.6	8	-0.11	-0.15	-0.04	1.1	5.2	6.3	0.0	-1.1	-5.2	-6.3	0.0
3m 5y30y	100/28.5	19.0	15.9	-0.9	6.3	-16.5	-0.9	10	-0.23	-0.37	-0.14	3.1	9.6	12.8	0.0	-3.1	-9.6	-12.8	0.0
3m 10y30y	100/50.8	8.4	6.5	-1.2	2.3	-11.1	-1.1	12	-0.10	-0.20	-0.10	2.0	4.2	6.1	0.0	-2.0	-4.2	-6.1	0.0
6m 1y2y	100/51.2	20.0	3.4	2.5	4.6	7.4	2.4	29	0.02	-0.04	-0.06	16.6	-1.2	15.4	4.4	-16.6	1.2	-15.4	-4.4
6m 1y3y	100/34.9	25.4	2.7	2.1	3.2	8.3	2.2	23	0.01	-0.08	-0.09	22.6	-0.4	22.2	5.5	-22.6	0.4	-22.2	-5.5
6m 1y5y	100/22.0	28.5	2.6	1.4	-0.3	6.9	1.4	18	-0.04	-0.16	-0.12	26.0	2.9	28.9	6.1	-26.0	-2.9	-28.9	-6.1
6m 2y5y	100/42.9	8.5	-0.8	0.3	-5.0	-0.6	0.4	11	-0.06	-0.12	-0.06	9.4	4.2	13.6	2.0	-9.4	-4.2	-13.6	-2.0
6m 2y10y	100/24.0	19.1	8.1	-0.2	-1.7	-5.9	-0.1	8	-0.15	-0.28	-0.13	11.0	9.8	20.8	2.0	-11.0	-9.8	-20.8	-2.0
6m 2y30y	100/12.2	27.6	13.0	-0.7	-3.1	-17.4	-0.6	8	-0.29	-0.54	-0.26	14.6	16.1	30.7	3.0	-14.6	-16.1	-30.7	-3.0
6m 3y5y	100/63.0	3.2	-0.2	-0.1	-3.6	-1.5	0.0	8	-0.05	-0.07	-0.02	3.3	3.4	6.7	0.9	-3.3	-3.4	-6.7	-0.9
6m 3y10y	100/35.3	13.7	8.8	-0.5	-0.2	-6.8	-0.4	7	-0.14	-0.23	-0.09	4.9	9.0	14.0	1.1	-4.9	-9.0	-14.0	-1.1
6m 3y30y	100/17.9	22.2	13.6	-0.8	-1.6	-18.3	-0.7	8	-0.26	-0.48	-0.21	8.5	15.2	23.8	2.2	-8.5	-15.2	-23.8	-2.2
6m 5y10y	100/56.0	10.6	9.0	-0.7	3.5	-5.3	-0.6	8	-0.08	-0.15	-0.07	1.6	5.4	7.0	0.2	-1.6	-5.4	-7.0	-0.2
6m 5y30y	100/28.5	19.0	13.8	-1.0	2.4	-16.8	-0.9	9	-0.19	-0.37	-0.18	5.2	11.4	16.6	1.5	-5.2	-11.4	-16.6	-1.5
6m 10y30y	100/50.8	8.4	4.8	-1.2	-0.7	-11.3	-1.2	10	-0.09	-0.19	-0.10	3.6	5.6	9.2	1.3	-3.6	-5.6	-9.2	-1.3
1y 1y2y	100/51.0	20.0	-1.7	1.0	-4.3	1.0	0.4	19	-0.02	-0.05	-0.03	21.7	2.6	24.3	1.5	-21.7	-2.6	-24.3	-1.5
1y 1y3y	100/34.9	25.4	-3.6	0.5	-8.0	0.1	0.2	17	-0.04	-0.10	-0.06	29.0	4.4	33.4	1.8	-29.0	-4.4	-33.4	-1.8
1y 1y5y	100/22.0	28.5	-3.5	0.0	-12.3	-2.3	-0.1	14	-0.09	-0.18	-0.09	32.0	8.8	40.8	1.5	-32.0	-8.8	-40.8	-1.5
1y 2y5y	100/43.1	8.5	-1.8	-0.5	-7.9	-3.5	-0.3	9	-0.06	-0.12	-0.06	10.4	6.1	16.5	0.0	-10.4	-6.1	-16.5	0.0
1y 2y10y	100/24.1	19.1	7.9	-0.7	-4.7	-9.6	-0.5	7	-0.14	-0.27	-0.14	11.2	12.6	23.8	0.2	-11.2	-12.6	-23.8	-0.2
1y 2y30y	100/12.3	27.6	10.4	-1.0	-9.7	-21.2	-0.8	8	-0.25	-0.51	-0.26	17.1	20.2	37.3	0.8	-17.1	-20.2	-37.3	-0.8
1y 3y5y	100/62.9	3.2	0.1	-0.7	-4.2	-2.5	-0.5	8	-0.04	-0.07	-0.03	3.1	4.3	7.4	-0.3	-3.1	-4.3	-7.4	0.3
1y 3y10y	100/35.3	13.7	9.8	-0.8	-0.9	-8.7	-0.7	7	-0.11	-0.22	-0.11	3.9	10.7	14.6	-0.1	-3.9	-10.7	-14.6	0.1
1y 3y30y	100/17.9	22.2	12.3	-1.1	-5.8	-20.2	-0.9	8	-0.22	-0.44	-0.22	9.9	18.2	28.0	0.6	-9.9	-18.2	-28.0	-0.6
1y 5y10y	100/56.1	10.6	9.7	-0.9	3.5	-6.2	-0.8	8	-0.07	-0.14	-0.07	0.8	6.2	7.1	0.2	-0.8	-6.2	-7.1	-0.2
1y 5y30y	100/28.5	19.0	12.2	-1.1	-1.2	-17.8	-1.1	8	-0.16	-0.34	-0.18	6.8	13.5	20.2	0.8	-6.8	-13.5	-20.2	-0.8
1y 10y30y	100/50.8	8.4	2.5	-1.3	-4.4	-11.5	-1.3	9	-0.08	-0.18	-0.10	5.9	6.9	12.9	0.7	-5.9	-6.9	-12.9	-0.7
2y 1y2y	100/51.5	20.0	-2.0	-0.7	-4.4	-1.1	-0.1	19	-0.02	-0.04	-0.02	22.0	2.4	24.4	-0.4	-22.0	-2.4	-24.4	0.4
2y 1y3y	100/35.2	25.4	-2.1	-0.8	-7.4	-2.9	-0.4	18	-0.05	-0.07	-0.02	27.5	5.2	32.7	-0.6	-27.5	-5.2	-32.7	0.6
2y 1y5y	100/22.1	28.5	0.8	-0.9	-9.3	-6.8	-0.7	15	-0.08	-0.14	-0.05	27.7	10.1	37.8	-0.9	-27.7	-10.1	-37.8	0.9
2y 2y5y	100/43.0	8.5	2.9	-1.0	-4.9	-5.8	-0.9	11	-0.06	-0.09	-0.04	5.7	7.8	13.4	-0.5	-5.7	-7.8	-13.4	0.5
2y 2y10y	100/24.1	19.1	15.3	-1.0	1.2	-11.3	-0.9	9	-0.11	-0.22	-0.11	3.8	14.1	18.0	-0.9	-3.8	-14.1	-18.0	0.9
2y 2y30y	100/12.3	27.6	13.4	-1.2	-9.2	-22.7	-1.1	8	-0.19	-0.39	-0.20	14.1	22.6	36.8	-0.2	-14.1	-22.6	-36.8	0.2
2y 3y5y	100/62.9	3.2	3.0	-1.0	-1.9	-4.0	-1.0	9	-0.03	-0.07	-0.04	0.2	4.9	5.0	-0.3	-0.2	-4.9	-5.0	0.3
2y 3y10y	100/35.3	13.7	15.4	-1.1	4.3	-9.5	-0.9	9	-0.08	-0.19	-0.11	-1.7	11.1	9.5	-0.7	1.7	-11.1	-9.5	0.7
2y 3y30y	100/17.9	22.2	13.5	-1.2	-6.0	-20.9	-1.2	8	-0.16	-0.36	-0.20	8.7	19.5	28.2	0.0	-8.7	-19.5	-28.2	0.0
2y 5y10y	100/56.1	10.6	12.4	-1.1	6.2	-5.6	-0.9	10	-0.05	-0.12	-0.07	-1.8	6.2	4.4	-0.4	1.8	-6.2	-4.4	0.4
2y 5y30y	100/28.5	19.0	10.5	-1.3	-4.0	-16.9	-1.2	8	-0.12	-0.27	-0.15	8.5	14.5	23.0	0.3	-8.5	-14.5	-23.0	-0.3
2y 10y30y	100/50.8	8.4	-1.9	-1.5	-9.9	-11.3	-1.5	7	-0.06	-0.14	-0.07	10.3	8.0	18.4	0.7	-10.3	-8.0	-18.4	-0.7
3y 1y2y	100/51.2	20.0	0.9	-1.0	-1.7	-1.9	-0.8	22	-0.02	-0.01	0.01	19.1	2.6	21.7	-0.3	-19.1	-2.6	-21.7	0.3
3y 1y3y	100/34.9	25.4	2.8	-1.1	-3.0	-3.5	-1.1	19	-0.03	-0.05	-0.02	22.5	5.8	28.3	-0.5	-22.5	-5.8	-28.3	0.5
3y 1y5y	100/22.0	28.5	7.8	-1.1	-2.6	-6.0	-0.9	17	-0.06	-0.12	-0.06	20.7	10.5	31.2	-0.8	-20.7	-10.5	-31.2	0.8
3y 2y5y	100/42.9	8.5	6.9	-1.1	-0.8	-4.2	-0.9	15	-0.04	-0.10	-0.06	1.6	7.7	9.3	-0.5	-1.6	-7.7	-9.3	0.5
3y 2y10y	100/24.1	19.1	21.1	-1.1	6.3	-10.2	-1.0	11	-0.09	-0.20	-0.11	-2.0	14.8	12.8	-0.5	2.0	-14.8	-12.8	0.5
3y 2y30y	100/12.3	27.6	13.8	-1.3	-11.4	-20.8	-1.3	7	-0.16	-0.33	-0.17	13.7	25.3	39.0	0.3	-13.7	-25.3	-39.0	-0.3
3y 3y5y	100/63.0	3.2	5.0	-1.1	0.5	-2.5	-0.6	12	-0.03	-0.06	-0.03	-1.8	4.5	2.7	-0.2	1.8	-4.5	-2.7	0.2
3y 3y10y	100/35.4	13.7	19.2	-1.1	7.6	-8.5	-0.9	10	-0.07	-0.15	-0.08	-5.5	11.6	6.1	-0.3	5.5	-11.6	-6.1	0.3
3y 3y30y	100/18.0	22.2	11.9	-1.3	-10.1	-19.2	-1.3	7	-0.15	-0.28	-0.14	10.3	22.0	32.3	0.5	-10.3	-22.0	-32.3	-0.5
3y 5y10y	100/56.2	10.6	14.2	-1.2	7.3	-6.0	-1.1	11	-0.04	-0.09	-0.05	-3.6	6.9	3.3	-0.1	3.6	-6.9	-3.3	0.1
3y 5y30y	100/28.6	19.0	6.9	-1.4	-10.2	-16.6	-1.4	7	-0.11	-0.21	-0.10	12.1	17.2	29.3	0.8	-12.1	-17.2	-29.3	-0.8
3y 10y30y	100/50.8	8.4	-7.3	-1.6	-17.4	-10.6	-1.7	5	-0.06	-0.11	-0.05	15.7	10.1	25.8	0.8	-15.7	-10.1	-25.8	-0.8

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Note: Close 10/2/26. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m carry is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bear steepeners in red, and three best bear flatteners in green. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

Figure 53. Conditional curves: bull trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bull steepener				bull flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/51.1	20.0	10.2	2.6	5.7	6.7	2.5	31	0.10	0.01	-0.09	9.8	4.5	14.3	14.5	-9.8	-4.5	-14.3	-14.5
3m 1y3y	100/34.9	25.4	11.7	2.6	7.6	9.1	2.4	25	0.09	-0.02	-0.11	13.6	4.2	17.8	18.2	-13.6	-4.2	-17.8	-18.2
3m 1y5y	100/22.0	28.5	12.7	2.4	10.9	9.1	2.1	23	0.04	-0.09	-0.13	15.8	1.9	17.7	18.1	-15.8	-1.9	-17.7	-18.1
3m 2y5y	100/43.0	8.5	2.6	1.1	5.1	2.2	1.0	20	-0.06	-0.09	-0.04	6.0	-2.5	3.4	3.7	-6.0	2.5	-3.4	-3.7
3m 2y10y	100/24.1	19.1	12.0	0.2	19.4	-0.6	0.2	20	-0.18	-0.25	-0.08	7.1	-7.4	-0.3	-0.1	-7.1	7.4	0.3	0.1
3m 2y30y	100/12.2	27.6	18.5	-0.4	30.0	-7.0	-0.4	19	-0.33	-0.52	-0.20	9.1	-11.5	-2.5	-1.1	-9.1	11.5	2.5	1.1
3m 3y5y	100/62.9	3.2	1.0	0.5	3.2	-0.2	0.3	19	-0.05	-0.06	-0.01	2.1	-2.2	-0.1	0.0	-2.1	2.2	0.1	0.0
3m 3y10y	100/35.3	13.7	10.5	-0.2	17.5	-3.0	-0.2	21	-0.17	-0.22	-0.05	3.3	-7.0	-3.8	-3.2	-3.3	7.0	3.8	3.2
3m 3y30y	100/17.9	22.2	16.9	-0.7	28.1	-9.4	-0.7	18	-0.30	-0.47	-0.16	5.3	-11.2	-5.9	-1.1	-5.3	11.2	5.9	1.1
3m 5y10y	100/56.0	10.6	9.5	-0.6	14.1	-2.8	-0.5	22	-0.11	-0.15	-0.04	1.1	-4.7	-3.5	-3.2	-1.1	4.7	3.5	3.2
3m 5y30y	100/28.5	19.0	15.9	-0.9	24.5	-9.2	-0.8	19	-0.23	-0.37	-0.14	3.1	-8.6	-5.5	-1.1	-3.1	8.6	5.5	1.1
3m 10y30y	100/50.8	8.4	6.5	-1.2	10.1	-6.7	-1.1	17	-0.10	-0.20	-0.10	2.0	-3.6	-1.6	-1.1	-2.0	3.6	1.6	1.1
6m 1y2y	100/51.2	20.0	3.4	2.5	2.3	5.1	2.5	25	0.02	-0.04	-0.06	16.6	1.1	17.7	2.5	-16.6	-1.1	-17.7	-2.5
6m 1y3y	100/34.9	25.4	2.7	2.1	2.3	5.9	2.1	22	0.01	-0.08	-0.09	22.6	0.4	23.0	3.7	-22.6	-0.4	-23.0	-3.7
6m 1y5y	100/22.0	28.5	2.6	1.4	5.2	4.3	1.3	20	-0.04	-0.16	-0.12	26.0	-2.7	23.3	3.5	-26.0	2.7	-23.3	-3.5
6m 2y5y	100/42.9	8.5	-0.8	0.3	2.9	-0.7	0.2	18	-0.06	-0.12	-0.06	9.4	-3.8	5.6	1.2	-9.4	3.8	-5.6	-1.2
6m 2y10y	100/24.0	19.1	8.1	-0.2	16.9	-5.2	-0.4	19	-0.15	-0.28	-0.13	11.0	-8.8	2.2	1.1	-11.0	8.8	-2.2	-1.1
6m 2y30y	100/12.2	27.6	13.0	-0.7	27.3	-12.2	-0.7	18	-0.29	-0.54	-0.26	14.6	-14.3	0.3	1.2	-14.6	14.3	-0.3	-1.2
6m 3y5y	100/63.0	3.2	-0.2	-0.1	2.9	-1.5	-0.3	18	-0.05	-0.07	-0.02	3.3	-3.1	0.3	0.1	-3.3	3.1	-0.3	-0.1
6m 3y10y	100/35.3	13.7	8.8	-0.5	16.9	-6.0	-0.6	20	-0.14	-0.23	-0.09	4.9	-8.1	-3.1	0.1	-4.9	8.1	3.1	-0.1
6m 3y30y	100/17.9	22.2	13.6	-0.8	27.1	-13.0	-0.9	18	-0.26	-0.48	-0.21	8.5	-13.5	-5.0	0.3	-8.5	13.5	5.0	-0.3
6m 5y10y	100/56.0	10.6	9.0	-0.7	13.8	-4.4	-0.8	21	-0.08	-0.15	-0.07	1.6	-4.8	-3.2	0.1	-1.6	4.8	3.2	-0.1
6m 5y30y	100/28.5	19.0	13.8	-1.0	23.8	-11.4	-1.0	18	-0.19	-0.37	-0.18	5.2	-10.0	-4.8	0.3	-5.2	10.0	4.8	-0.3
6m 10y30y	100/50.8	8.4	4.8	-1.2	9.8	-7.1	-1.1	17	-0.09	-0.19	-0.10	3.6	-4.9	-1.3	0.1	-3.6	4.9	1.3	-0.1
1y 1y2y	100/51.0	20.0	-1.7	1.0	0.7	1.6	1.4	24	-0.02	-0.05	-0.03	21.7	-2.4	19.3	0.1	-21.7	2.4	-19.3	-0.1
1y 1y3y	100/34.9	25.4	-3.6	0.5	0.5	0.3	0.7	21	-0.04	-0.10	-0.06	29.0	-4.1	24.9	0.0	-29.0	4.1	-24.9	0.0
1y 1y5y	100/22.0	28.5	-3.5	0.0	4.6	-3.0	-0.1	20	-0.09	-0.18	-0.09	32.0	-8.1	23.9	-0.2	-32.0	8.1	-23.9	0.2
1y 2y5y	100/43.1	8.5	-1.8	-0.5	3.7	-4.6	-0.6	19	-0.06	-0.12	-0.06	10.4	-5.5	4.8	-0.3	-10.4	5.5	-4.8	0.3
1y 2y10y	100/24.1	19.1	7.9	-0.7	19.3	-9.8	-0.8	20	-0.14	-0.27	-0.14	11.2	-11.4	-0.2	-1.1	-11.2	11.4	0.2	1.1
1y 2y30y	100/12.3	27.6	10.4	-1.0	28.5	-16.9	-1.0	18	-0.25	-0.51	-0.26	17.1	-18.1	-0.9	-0.7	-17.1	18.1	0.9	0.7
1y 3y5y	100/62.9	3.2	0.1	-0.7	3.9	-3.3	-0.9	20	-0.04	-0.07	-0.03	3.1	-3.8	-0.7	-0.2	-3.1	3.8	0.7	0.2
1y 3y10y	100/35.3	13.7	9.8	-0.8	19.4	-8.5	-0.9	22	-0.11	-0.22	-0.11	3.9	-9.6	-5.6	-0.9	-3.9	9.6	5.6	0.9
1y 3y30y	100/17.9	22.2	12.3	-1.1	28.4	-15.6	-1.1	18	-0.22	-0.44	-0.22	9.9	-16.1	-6.3	-0.5	-9.9	16.1	6.3	0.5
1y 5y10y	100/56.1	10.6	9.7	-0.9	15.2	-5.1	-0.9	24	-0.07	-0.14	-0.07	0.8	-5.5	-4.7	-0.8	-0.8	5.5	4.7	0.8
1y 5y30y	100/28.5	19.0	12.2	-1.1	24.0	-12.2	-1.1	18	-0.16	-0.34	-0.18	6.8	-11.8	-5.0	-0.4	-6.8	11.8	5.0	0.4
1y 10y30y	100/50.8	8.4	2.5	-1.3	8.4	-7.2	-1.2	16	-0.08	-0.18	-0.10	5.9	-5.9	0.0	0.3	-5.9	5.9	0.0	-0.3
2y 1y2y	100/51.5	20.0	-2.0	-0.7	0.2	-3.6	-1.0	23	-0.02	-0.04	-0.02	22.0	-2.2	19.8	-0.2	-22.0	2.2	-19.8	0.2
2y 1y3y	100/35.2	25.4	-2.1	-0.8	2.7	-5.9	-1.0	23	-0.05	-0.07	-0.02	27.5	-4.9	22.6	-0.5	-27.5	4.9	-22.6	0.5
2y 1y5y	100/22.1	28.5	0.8	-0.9	10.1	-8.0	-1.0	22	-0.08	-0.14	-0.05	27.7	-9.3	18.4	-0.9	-27.7	9.3	-18.4	0.9
2y 2y5y	100/43.0	8.5	2.9	-1.0	9.7	-4.5	-1.0	24	-0.06	-0.09	-0.04	5.7	-6.8	-1.2	-0.7	-5.7	6.8	1.2	0.7
2y 2y10y	100/24.1	19.1	15.3	-1.0	27.6	-9.9	-1.1	25	-0.11	-0.22	-0.11	3.8	-12.3	-8.5	-0.9	-3.8	12.3	8.5	0.9
2y 2y30y	100/12.3	27.6	13.4	-1.2	33.1	-15.8	-1.2	21	-0.19	-0.39	-0.20	14.1	-19.6	-5.5	-0.4	-14.1	19.6	5.5	0.4
2y 3y5y	100/62.9	3.2	3.0	-1.0	7.1	-2.2	-1.0	26	-0.03	-0.07	-0.04	0.2	-4.1	-4.0	-0.4	-0.2	4.1	4.0	0.4
2y 3y10y	100/35.3	13.7	15.4	-1.1	24.9	-7.6	-1.1	26	-0.08	-0.19	-0.11	-1.7	-9.5	-11.2	-0.6	1.7	9.5	11.2	0.6
2y 3y30y	100/17.9	22.2	13.5	-1.2	30.2	-13.6	-1.2	19	-0.16	-0.36	-0.20	8.7	-16.7	-8.0	-0.1	-8.7	16.7	8.0	0.1
2y 5y10y	100/56.1	10.6	12.4	-1.1	17.6	-5.4	-1.2	27	-0.05	-0.12	-0.07	-1.8	-5.2	-7.0	-0.2	1.8	5.2	7.0	0.2
2y 5y30y	100/28.5	19.0	10.5	-1.3	22.6	-11.5	-1.3	17	-0.12	-0.27	-0.15	8.5	-12.0	-3.6	0.2	-8.5	12.0	3.6	-0.2
2y 10y30y	100/50.8	8.4	-1.9	-1.5	4.7	-6.1	-1.3	13	-0.06	-0.14	-0.07	10.3	-6.6	3.7	0.4	-10.3	6.6	-3.7	-0.4
3y 1y2y	100/51.2	20.0	0.9	-1.0	3.1	-2.2	-1.1	27	-0.02	-0.01	0.01	19.1	-2.2	16.9	-0.2	-19.1	2.2	-16.9	0.2
3y 1y3y	100/34.9	25.4	2.8	-1.1	7.5	-3.4	-1.0	25	-0.03	-0.05	-0.02	22.5	-4.7	17.9	-0.3	-22.5	4.7	-17.9	0.3
3y 1y5y	100/22.0	28.5	7.8	-1.1	16.4	-6.0	-1.2	25	-0.06	-0.12	-0.06	20.7	-8.6	12.1	-0.4	-20.7	8.6	-12.1	0.4
3y 2y5y	100/42.9	8.5	6.9	-1.1	13.2	-3.8	-1.2	27	-0.04	-0.10	-0.06	1.6	-6.3	-4.7	-0.3	-1.6	6.3	4.7	0.3
3y 2y10y	100/24.1	19.1	21.1	-1.1	33.2	-7.1	-1.2	28	-0.09	-0.20	-0.11	-2.0	-12.1	-14.1	-0.5	2.0	12.1	14.1	0.5
3y 2y30y	100/12.3	27.6	13.8	-1.3	34.7	-11.9	-1.2	22	-0.16	-0.33	-0.17	13.7	-20.9	-7.1	0.2	-13.7	20.9	7.1	-0.2
3y 3y5y	100/63.0	3.2	5.0	-1.1	8.8	-2.6	-1.4	28	-0.03	-0.06	-0.03	-1.8	-3.8	-5.6	-0.1	1.8	3.8	5.6	0.1
3y 3y10y	100/35.4	13.7	19.2	-1.1	28.7	-5.8	-1.2	28	-0.07	-0.15	-0.08	-5.5	-9.6	-15.0	-0.4	5.5	9.6	15.0	0.4
3y 3y30y	100/18.0	22.2	11.9	-1.3	30.1	-10.7	-1.2	19	-0.15	-0.28	-0.14	10.3	-18.2	-7.9	0.3	-10.3	18.2	7.9	-0.3
3y 5y10y	100/56.2	10.6	14.2	-1.2	19.8	-3.3	-1.1	29	-0.04	-0.09	-0.05	-3.6	-5.6	-9.2	-0.2	3.6	5.6	9.2	0.2
3y 5y30y	100/28.6	19.0	6.9	-1.4	20.9	-8.2	-1.2	16	-0.11	-0.21	-0.10	12.1	-14.0	-1.9	0.4	-12.1	14.0	1.9	-0.4
3y 10y30y	100/50.8	8.4	-7.3	-1.6	0.7	-5.1	-1.3	11	-0.06	-0.11	-0.05	15.7	-8.0	7.7	0.6	-15.7	8.0	-7.7	-0.6

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Note: Close 10/2/26. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **bull steepeners in red**, and three **best bull flatteners in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

USD conditional flies

We analyze conditional fly trades based on the entry point, directionality with rates and carry.

Figure 54. Conditional payer flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long payer in the belly				short payer in the belly			
curve	notionals, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, ZC	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	143.2/100/31.4	11.1	5.4	2.2	43	91	1.5	0.9	3.5	8.9	7.3	2.2	0	5.7	-3.5	2.3	0.0	-5.7	3.5	-2.3	0.0
3m 1y3y7y	143.2/100/23.5	9.3	3.9	1.8	63	97	-0.6	-0.4	4.6	8.5	8.5	1.8	85	5.4	-4.6	0.8	0.0	-5.4	4.6	-0.8	0.0
3m 2y3y5y	73.1/100/31.4	1.1	0.3	1.0	49	96	-0.9	-1.9	1.0	1.3	1.4	0.9	95	0.8	-1.0	-0.2	0.0	-0.8	1.0	0.2	0.0
3m 2y5y7y	116.3/100/37.4	2.4	-0.2	2.2	13	-44	4.5	2.1	-0.3	-0.5	1.9	2.3	20	2.6	0.3	2.9	0.0	-2.6	-0.3	-2.9	0.0
3m 2y5y10y	116.3/100/28.0	-1.0	-3.4	2.2	54	63	2.2	1.6	1.1	-2.3	3.8	2.5	35	2.4	-1.1	1.3	0.0	-2.4	1.1	-1.3	0.0
3m 2y10y30y	207.4/100/25.4	5.3	2.8	2.2	39	-61	9.0	2.5	-1.9	0.9	3.8	2.2	17	2.5	1.9	4.4	0.0	-2.5	-1.9	-4.4	0.0
3m 3y5y7y	79.5/100/37.4	-0.3	-1.0	2.5	23	-37	1.7	2.1	-0.1	-1.1	0.9	2.8	23	0.7	0.1	0.9	0.0	-0.7	-0.1	-0.9	0.0
3m 3y5y10y	79.5/100/28.0	-3.7	-4.2	1.3	69	96	-0.6	-1.1	1.3	-2.9	2.8	1.6	77	0.5	-1.3	-0.8	0.0	-0.5	1.3	0.8	0.0
3m 3y7y10y	106.3/100/37.5	0.0	-1.2	0.8	5	-74	4.0	2.2	-0.8	-2.1	0.3	0.9	0	1.2	0.8	2.0	0.0	-1.2	-0.8	-2.0	0.0
3m 3y7y15y	106.3/100/28.0	-5.7	-6.6	2.5	61	5	2.3	2.1	0.0	-6.6	3.0	2.6	18	0.9	0.0	0.9	0.0	-0.9	0.0	-0.9	0.0
3m 3y10y30y	141.8/100/25.4	2.6	2.0	2.3	44	-70	6.5	2.8	-1.7	0.3	2.8	2.1	0	0.6	1.7	2.4	0.0	-0.6	-1.7	-2.4	0.0
3m 5y10y15y	89.2/100/37.3	-0.4	-0.6	0.1	-4	-84	3.2	2.5	-1.6	-2.3	-0.6	-0.1	0	0.3	1.6	1.9	0.0	-0.3	-1.6	-1.9	0.0
3m 5y10y30y	89.2/100/25.4	1.1	1.5	2.2	64	-17	2.5	2.7	-0.5	1.0	3.2	2.0	28	-0.4	0.5	0.1	0.0	0.4	-0.5	-0.1	0.0
3m 10y20y30y	80.1/100/40.7	11.5	11.3	-1.0	-23	-82	1.8	1.3	-0.9	10.4	-2.5	-1.0	42	0.2	0.9	1.1	0.0	-0.2	-0.9	-1.1	0.0
6m 1y3y5y	143.3/100/31.5	11.1	1.4	2.4	20	40	3.7	1.7	1.9	3.4	4.4	2.1	0	9.7	-1.9	7.7	2.3	-9.7	1.9	-7.7	-2.3
6m 1y3y7y	143.3/100/23.5	9.3	0.1	1.9	50	86	1.8	1.0	3.1	3.2	5.8	1.7	75	9.2	-3.1	6.1	2.2	-9.2	3.1	-6.1	-2.2
6m 2y3y5y	73.4/100/31.5	1.1	-0.3	1.1	50	94	-0.5	-1.1	1.3	1.1	1.1	0.8	94	1.4	-1.3	0.1	0.1	-1.4	1.3	-0.1	-0.1
6m 2y5y7y	116.6/100/37.4	2.4	-1.8	1.2	2	-69	4.8	2.2	-0.9	-2.7	0.6	1.2	16	4.2	0.9	5.1	0.9	-4.2	-0.9	-5.1	-0.9
6m 2y5y10y	116.6/100/28.0	-1.0	-4.9	2.2	47	11	2.6	1.8	0.6	-4.3	2.5	2.5	29	3.9	-0.6	3.3	0.9	-3.9	0.6	-3.3	-0.9
6m 2y10y30y	208.0/100/25.4	5.3	1.6	1.4	30	-74	9.7	2.5	-2.0	-0.3	2.6	1.6	16	3.7	2.0	5.7	0.5	-3.7	-2.0	-5.7	-0.5
6m 3y5y7y	79.4/100/37.4	-0.3	-1.4	1.8	17	-63	1.6	2.1	-0.5	-2.0	0.4	1.6	17	1.2	0.5	1.7	0.4	-1.2	-0.5	-1.7	-0.4
6m 3y5y10y	79.4/100/28.0	-3.7	-4.6	1.3	69	95	-0.6	-1.1	0.9	-3.6	2.3	1.4	71	0.9	-0.9	-0.1	0.3	-0.9	0.9	0.1	-0.3
6m 3y7y10y	106.2/100/37.5	0.0	-1.9	0.1	-3	-82	4.0	2.3	-1.3	-3.2	-0.4	0.3	8	1.9	1.3	3.2	0.5	-1.9	-1.3	-3.2	-0.5
6m 3y7y15y	106.2/100/28.0	-5.7	-7.1	2.5	56	-39	2.4	2.2	-0.2	-7.3	2.3	2.7	16	1.4	0.2	1.6	0.3	-1.4	-0.2	-1.6	-0.3
6m 3y10y30y	141.7/100/25.4	2.6	2.0	1.6	37	-77	6.6	2.8	-1.6	0.4	2.4	1.8	16	0.7	1.6	2.3	-0.1	-0.7	-1.6	-2.3	0.1
6m 5y10y15y	89.2/100/37.3	-0.4	-0.7	-0.1	-9	-86	3.3	2.5	-1.5	-2.2	-0.5	0.0	7	0.3	1.5	1.8	-0.1	-0.3	-1.5	-1.8	0.1
6m 5y10y30y	89.2/100/25.4	1.1	2.1	2.2	61	-37	2.6	2.8	0.1	2.1	3.3	2.3	35	-1.0	-0.1	-1.1	-0.5	1.0	0.1	1.1	0.5
6m 10y20y30y	80.0/100/40.7	11.5	11.2	-1.0	-25	-82	1.6	1.3	-1.3	9.9	-2.6	-1.1	38	0.3	1.3	1.6	0.2	-0.3	-1.3	-1.6	-0.2
1y 1y3y5y	143.2/100/31.4	11.1	-1.8	2.2	7	-49	4.2	2.0	-0.1	-1.9	1.2	1.6	36	12.9	0.1	13.0	1.0	-12.9	-0.1	-13.0	-1.0
1y 1y3y7y	143.2/100/23.5	9.3	-3.4	2.0	50	46	2.4	1.4	1.3	-2.0	2.8	1.6	61	12.6	-1.3	11.3	1.0	-12.6	1.3	-11.3	-1.0
1y 2y3y5y	73.0/100/31.4	1.1	-1.0	1.3	54	92	-0.2	-0.4	1.2	0.2	1.0	1.0	91	2.1	-1.2	0.9	0.3	-2.1	1.2	-0.9	-0.3
1y 2y5y7y	116.1/100/37.4	2.4	-2.4	-0.1	-5	-82	4.0	2.2	-1.6	-4.0	-0.7	0.1	14	4.9	1.6	6.5	0.0	-4.9	-1.6	-6.5	0.0
1y 2y5y10y	116.1/100/28.0	-1.0	-5.8	2.1	45	-42	2.1	1.8	0.0	-5.8	1.3	2.2	27	4.8	0.0	4.7	-0.1	-4.8	0.0	-4.7	0.1
1y 2y10y30y	207.1/100/25.4	5.3	2.7	0.4	25	-81	8.9	2.6	-2.7	0.0	0.6	0.6	16	2.6	2.7	5.4	-0.3	-2.6	-2.7	-5.4	0.3
1y 3y5y7y	79.5/100/37.4	-0.3	-1.5	0.1	15	-77	1.1	1.8	-0.7	-2.2	0.0	0.4	16	1.2	0.7	1.9	-0.2	-1.2	-0.7	-1.9	0.2
1y 3y5y10y	79.5/100/28.0	-3.7	-4.8	1.1	73	93	-0.8	-1.5	0.9	-3.9	2.0	1.2	68	1.1	-0.9	0.2	-0.2	-1.1	0.9	-0.2	0.2
1y 3y7y10y	106.3/100/37.5	0.0	-1.8	-0.5	-11	-88	3.3	2.3	-1.9	-3.7	-1.3	-0.4	8	1.7	1.9	3.7	-0.1	-1.7	-1.9	-3.7	0.1
1y 3y7y15y	106.3/100/28.0	-5.7	-6.8	2.1	53	-67	2.1	2.4	-0.6	-7.4	1.5	2.2	16	1.1	0.6	1.7	-0.2	-1.1	-0.6	-1.7	0.2
1y 3y10y30y	141.7/100/25.4	2.6	3.7	0.7	33	-82	6.0	2.7	-1.8	1.8	1.3	0.9	20	-1.0	1.8	0.8	-0.4	1.0	-1.8	-0.8	0.4
1y 5y10y15y	89.2/100/37.3	-0.4	-0.1	-0.4	-11	-87	3.1	2.5	-1.7	-1.9	-0.9	-0.3	7	-0.2	1.7	1.5	0.0	0.2	-1.7	-1.5	0.0
1y 5y10y30y	89.2/100/25.4	1.1	3.6	2.2	59	-51	2.6	2.8	0.3	3.9	2.9	2.5	45	-2.6	-0.3	-2.8	-0.3	2.6	0.3	2.8	0.3
1y 10y20y30y	80.0/100/40.7	11.5	11.2	-1.2	-25	-81	1.3	1.1	-1.8	9.3	-2.7	-1.3	34	0.4	1.8	2.2	0.1	-0.4	-1.8	-2.2	-0.1

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Note: Close 10/2/26. Conditional flies are structured by buying (selling) payers in the belly vs selling (buying) payers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best long belly payer trades in red, and three best short belly payer trades in green. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

Figure 55. Conditional receiver flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long receiver in the belly				short receiver in the belly			
curve	ntl, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, zs	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	143.2/100/31.4	11.1	5.4	2.2	43	91	1.5	0.9	3.2	2.2	-58.0	-2.1	0	-5.7	-3.2	-8.9	-5.9	5.7	3.2	8.9	5.9
3m 1y3y7y	143.2/100/23.5	9.3	3.9	1.8	63	97	-0.6	-0.4	4.2	-0.3	-32.4	-2.3	65	-5.4	-4.2	-9.6	-5.5	5.4	4.2	9.6	5.5
3m 2y3y5y	73.1/100/31.4	1.1	0.3	1.0	49	96	-0.9	-1.9	0.9	-0.7	-62.3	-2.1	85	-0.8	-0.9	-1.8	-0.9	0.8	0.9	1.8	0.9
3m 2y5y7y	116.3/100/37.4	2.4	-0.2	2.2	13	-44	4.5	2.1	-0.3	0.1	-36.5	-2.1	0	-2.6	0.3	-2.4	-2.8	2.6	-0.3	2.4	2.8
3m 2y5y10y	116.3/100/28.0	-1.0	-3.4	2.2	54	63	2.2	1.6	1.0	-4.5	-31.7	-2.2	29	-2.4	-1.0	-3.4	-2.5	2.4	1.0	3.4	2.5
3m 2y10y30y	207.4/100/25.4	5.3	2.8	2.2	39	-61	9.0	2.5	-1.7	4.4	-52.8	-2.3	23	-2.5	1.7	-0.9	-2.7	2.5	-1.7	0.9	2.7
3m 3y5y7y	79.5/100/37.4	-0.3	-1.0	2.5	23	-37	1.7	2.1	-0.1	-0.9	-37.4	-2.2	24	-0.7	0.1	-0.6	-0.8	0.7	-0.1	0.6	0.8
3m 3y5y10y	79.5/100/28.0	-3.7	-4.2	1.3	69	96	-0.6	-1.1	1.2	-5.4	-32.7	-2.3	53	-0.5	-1.2	-1.7	-0.5	0.5	1.2	1.7	0.5
3m 3y7y10y	106.3/100/37.5	0.0	-1.2	0.8	5	-74	4.0	2.2	-0.7	-0.5	-34.6	-2.1	15	-1.2	0.7	-0.4	-1.2	1.2	-0.7	0.4	1.2
3m 3y7y15y	106.3/100/28.0	-5.7	-6.6	2.5	61	5	2.3	2.1	0.0	-6.6	-37.1	-2.3	17	-0.9	0.0	-0.9	-0.9	0.9	0.0	0.9	0.9
3m 3y10y30y	141.8/100/25.4	2.6	2.0	2.3	44	-70	6.5	2.8	-1.5	3.5	-53.7	-2.3	24	-0.6	1.5	0.9	-0.7	0.6	-1.5	-0.9	0.7
3m 5y10y15y	89.2/100/37.3	-0.4	-0.6	0.1	-4	-84	3.2	2.5	-1.4	0.8	-48.8	-2.0	14	-0.3	1.4	1.2	-0.3	0.3	-1.4	-1.2	0.3
3m 5y10y30y	89.2/100/25.4	1.1	1.5	2.2	64	-17	2.5	2.7	-0.5	2.0	-62.4	-2.2	34	0.4	0.5	0.9	0.5	-0.4	-0.5	-0.9	-0.5
3m 10y20y30y	80.1/100/40.7	11.5	11.3	-1.0	-23	-82	1.8	1.3	-0.8	12.1	-92.4	-2.3	50	-0.2	0.8	0.6	-0.2	0.2	-0.8	-0.6	0.2
6m 1y3y5y	143.3/100/31.5	11.1	1.4	2.4	20	40	3.7	1.7	1.7	-0.3	-64.2	-2.1	0	-9.7	-1.7	-11.4	-0.8	9.7	1.7	11.4	0.8
6m 1y3y7y	143.3/100/23.5	9.3	0.1	1.9	50	86	1.8	1.0	2.8	-2.7	-40.2	-2.3	59	-9.2	-2.8	-11.9	-0.2	9.2	2.8	11.9	0.2
6m 2y3y5y	73.4/100/31.5	1.1	-0.3	1.1	50	94	-0.5	-1.1	1.2	-1.4	-66.4	-2.1	78	-1.4	-1.2	-2.5	0.1	1.4	1.2	2.5	-0.1
6m 2y5y7y	116.6/100/37.4	2.4	-1.8	1.2	2	-69	4.8	2.2	-0.8	-0.9	-43.8	-2.2	19	-4.2	0.8	-3.4	-1.1	4.2	-0.8	3.4	1.1
6m 2y5y10y	116.6/100/28.0	-1.0	-4.9	2.2	47	11	2.6	1.8	0.5	-5.4	-35.4	-2.2	27	-3.9	-0.5	-4.4	-0.3	3.9	0.5	4.4	0.3
6m 2y10y30y	208.0/100/25.4	5.3	1.6	1.4	30	-74	9.7	2.5	-1.8	3.4	-55.0	-2.3	21	-3.7	1.8	-1.9	-1.6	3.7	-1.8	1.9	1.6
6m 3y5y7y	79.4/100/37.4	-0.3	-1.4	1.8	17	-63	1.6	2.1	-0.5	-0.9	-44.0	-2.2	24	-1.2	0.5	-0.7	-0.3	1.2	-0.5	0.7	0.3
6m 3y5y10y	79.4/100/28.0	-3.7	-4.6	1.3	69	95	-0.6	-1.1	0.8	-5.4	-35.6	-2.3	53	-0.9	-0.8	-1.7	0.4	0.9	0.8	1.7	-0.4
6m 3y7y10y	106.2/100/37.5	0.0	-1.9	0.1	-3	-82	4.0	2.3	-1.2	-0.7	-37.8	-2.1	14	-1.9	1.2	-0.7	-0.8	1.9	-1.2	0.7	0.8
6m 3y7y15y	106.2/100/28.0	-5.7	-7.1	2.5	56	-39	2.4	2.2	-0.2	-6.9	-38.6	-2.3	17	-1.4	0.2	-1.2	-0.2	1.4	-0.2	1.2	0.2
6m 3y10y30y	141.7/100/25.4	2.6	2.0	1.6	37	-77	6.6	2.8	-1.4	3.4	-55.2	-2.3	24	-0.7	1.4	0.8	-0.8	0.7	-1.4	-0.8	0.8
6m 5y10y15y	89.2/100/37.3	-0.4	-0.7	-0.1	-9	-86	3.3	2.5	-1.3	0.6	-49.4	-2.1	13	-0.3	1.3	1.0	-0.8	0.3	-1.3	-1.0	0.8
6m 5y10y30y	89.2/100/25.4	1.1	2.1	2.2	61	-37	2.6	2.8	0.0	2.0	-62.6	-2.2	34	1.0	0.0	1.0	0.1	-1.0	0.0	-1.0	-0.1
6m 10y20y30y	80.0/100/40.7	11.5	11.2	-1.0	-25	-82	1.6	1.3	-1.2	12.4	-93.1	-2.3	51	-0.3	1.2	0.9	-0.5	0.3	-1.2	-0.9	0.5
1y 1y3y5y	143.2/100/31.4	11.1	-1.8	2.2	7	-49	4.2	2.0	-0.1	-1.7	-71.4	-2.1	36	-12.9	0.1	-12.8	-0.1	12.9	-0.1	12.8	0.1
1y 1y3y7y	143.2/100/23.5	9.3	-3.4	2.0	50	46	2.4	1.4	1.2	-4.5	-51.0	-2.3	54	-12.6	-1.2	-13.8	0.3	12.6	1.2	13.8	-0.3
1y 2y3y5y	73.0/100/31.4	1.1	-1.0	1.3	54	92	-0.2	-0.4	1.1	-2.1	-70.5	-2.2	72	-2.1	-1.1	-3.2	0.6	2.1	1.1	3.2	-0.6
1y 2y5y7y	116.1/100/37.4	2.4	-2.4	-0.1	-5	-82	4.0	2.2	-1.4	-1.0	-53.0	-2.3	19	-4.9	1.4	-3.5	-0.8	4.9	-1.4	3.5	0.8
1y 2y5y10y	116.1/100/28.0	-1.0	-5.8	2.1	45	-42	2.1	1.8	0.0	-5.8	-41.4	-2.3	27	-4.8	0.0	-4.8	-0.2	4.8	0.0	4.8	0.2
1y 2y10y30y	207.1/100/25.4	5.3	2.7	0.4	25	-81	8.9	2.6	-2.4	5.1	-57.5	-2.3	24	-2.6	2.4	-0.2	-0.6	2.6	-2.4	0.2	0.6
1y 3y5y7y	79.5/100/37.4	-0.3	-1.5	0.1	15	-77	1.1	1.8	-0.6	-0.9	-52.3	-2.3	24	-1.2	0.6	-0.6	-0.5	1.2	-0.6	0.6	0.5
1y 3y5y10y	79.5/100/28.0	-3.7	-4.8	1.1	73	93	-0.8	-1.5	0.8	-5.7	-40.9	-2.4	51	-1.1	-0.8	-1.9	0.1	1.1	0.8	1.9	-0.1
1y 3y7y10y	106.3/100/37.5	0.0	-1.8	-0.5	-11	-88	3.3	2.3	-1.7	-0.1	-43.5	-2.3	17	-1.7	1.7	-0.1	-0.8	1.7	-1.7	0.1	0.8
1y 3y7y15y	106.3/100/28.0	-5.7	-6.8	2.1	53	-67	2.1	2.4	-0.5	-6.2	-41.7	-2.3	18	-1.1	0.5	-0.6	-0.2	1.1	-0.5	0.6	0.2
1y 3y10y30y	141.7/100/25.4	2.6	3.7	0.7	33	-82	6.0	2.7	-1.6	5.3	-57.0	-2.3	29	1.0	1.6	2.6	-0.2	-1.0	-1.6	-2.6	0.2
1y 5y10y15y	89.2/100/37.3	-0.4	-0.1	-0.4	-11	-87	3.1	2.5	-1.5	1.4	-50.6	-2.1	19	0.2	1.5	1.7	-0.4	-0.2	-1.5	-1.7	0.4
1y 5y10y30y	89.2/100/25.4	1.1	3.6	2.2	59	-51	2.6	2.8	0.2	3.4	-62.5	-2.3	42	2.6	-0.2	2.3	0.7	-2.6	0.2	-2.3	-0.7
1y 10y20y30y	80.0/100/40.7	11.5	11.2	-1.2	-25	-81	1.3	1.1	-1.6	12.8	-93.0	-2.3	52	-0.4	1.6	1.2	-0.8	0.4	-1.6	-1.2	0.8

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Note: Close 10/2/26. Conditional flies are structured by buying (selling) receivers in the belly vs selling (buying) receivers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best long belly receiver trades in red, and three best short belly receiver trades in green. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

EUR at-the-money swaption vols

EUR relative value on the swaption surface

We report levels and z-scores of at-the-money vols, as well as the relative richness/cheapness of vols to rates (based on principal components of the curve).

- Implied vols of short-term expiries have continued to outperform longer-end expiries across most tails (Figure 57). Our PCA model is still flagging 1m expiries of tails in the belly as rich (Figure 59).

Figure 56. Implied vols, normals (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	110.3 (2.5)	128.5 (2.6)	125.1 (2.7)	117.9 (2.8)	110.7 (3.0)	101.6 (3.2)	81.1 (3.9)
3m	107.9 (2.3)	118.6 (2.3)	115.4 (2.4)	109.3 (2.6)	104.3 (2.8)	97.6 (3.1)	82.0 (3.6)
6m	111.1 (2.3)	116.5 (2.4)	113.2 (2.5)	107.4 (2.6)	103.0 (2.8)	96.9 (3.0)	83.0 (3.5)
1y	115.0 (2.4)	114.2 (2.4)	110.4 (2.5)	104.0 (2.6)	99.6 (2.7)	93.5 (2.8)	83.0 (3.2)
2y	109.4 (2.4)	106.5 (2.4)	103.9 (2.5)	99.9 (2.7)	96.6 (2.7)	92.1 (2.8)	82.9 (3.1)
3y	103.8 (2.4)	102.2 (2.4)	99.7 (2.5)	96.7 (2.6)	94.2 (2.7)	90.9 (2.9)	82.0 (3.2)
5y	96.0 (2.7)	95.3 (2.7)	93.6 (2.7)	91.0 (2.8)	89.2 (2.9)	87.0 (3.0)	79.0 (3.4)
10y	83.3 (2.6)	83.4 (2.6)	82.6 (2.7)	80.8 (2.8)	79.6 (2.8)	78.1 (2.9)	71.7 (3.0)

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Close 10/2/26. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 57. Change in implied vols since September 11th, normals

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	23.1	29.2	30.3	32.6	32.9	33.4	25.4
3m	17.1	19.9	20.2	21.1	21.7	22.1	20.0
6m	16.9	18.3	18.8	19.6	19.4	19.0	17.0
1y	17.8	16.2	16.3	16.0	15.3	14.0	12.0
2y	14.3	12.2	12.4	13.0	12.3	11.1	9.7
3y	11.9	10.7	10.9	11.4	11.2	10.6	9.1
5y	9.9	9.6	9.5	9.4	9.4	9.4	8.2
10y	5.4	5.4	5.5	5.7	5.8	6.0	5.2

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Note: The table shows changes in ATM implied vols from 10/2/26 to 10/2/26. Source: Citi Research

Figure 58. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	21.9 (1.1)	19.7 (0.9)	15.3 (0.7)	10.3 (0.5)	7.0 (0.4)	3.8 (0.2)	-15.1 (-0.6)
3m	15.2 (0.8)	10.5 (0.5)	7.3 (0.4)	4.1 (0.2)	2.9 (0.2)	1.1 (0.1)	-12.9 (-0.6)
6m	12.2 (0.7)	8.8 (0.5)	6.7 (0.4)	4.7 (0.3)	3.9 (0.3)	2.1 (0.1)	-9.5 (-0.5)
1y	9.9 (0.6)	8.0 (0.5)	6.7 (0.4)	5.0 (0.4)	3.8 (0.3)	1.2 (0.1)	-6.0 (-0.4)
2y	7.1 (0.5)	6.4 (0.4)	6.5 (0.5)	6.6 (0.5)	5.6 (0.4)	3.8 (0.3)	-1.5 (-0.1)
3y	7.2 (0.5)	7.7 (0.6)	7.6 (0.6)	8.0 (0.6)	7.4 (0.6)	6.3 (0.5)	1.6 (0.1)
5y	9.0 (0.7)	9.6 (0.8)	9.6 (0.8)	9.7 (0.9)	9.4 (0.9)	8.7 (0.8)	5.2 (0.5)
10y	8.1 (0.9)	8.4 (0.9)	9.0 (1.0)	9.3 (1.1)	9.5 (1.2)	9.6 (1.2)	8.5 (1.3)

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Close 10/2/26. We regress each ATM point on 3 principal components of the swap curve. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We calibrate the model to the entire sample history since 2002. Source: Citi Research

Figure 59. Rich/cheap to PCA of the vol surface, normals (z-score)

exp\tls	1y	2y	3y	5y	7y	10y	30y
1m	5.5 (1.0)	11.1 (2.5)	10.5 (3.2)	8.0 (3.1)	5.8 (2.5)	3.6 (1.4)	-3.7 (-1.4)
3m	-0.5 (-0.1)	2.4 (0.7)	1.9 (1.0)	2.1 (1.7)	1.5 (1.4)	0.9 (0.6)	-1.8 (-1.2)
6m	-1.9 (-0.5)	0.3 (0.2)	0.6 (0.5)	1.0 (1.1)	0.5 (0.5)	0.5 (0.4)	-1.8 (-1.7)
1y	-0.5 (-0.3)	-0.5 (-0.5)	-0.3 (-0.4)	-0.3 (-0.4)	-0.8 (-0.9)	-1.4 (-1.4)	-2.0 (-2.0)
2y	-2.1 (-2.3)	-3.2 (-3.0)	-2.5 (-2.5)	-0.8 (-0.9)	-0.3 (-0.4)	-0.7 (-0.8)	-1.5 (-1.5)
3y	-2.5 (-2.4)	-2.6 (-2.4)	-2.3 (-2.3)	-0.5 (-0.7)	0.2 (0.2)	0.3 (0.4)	-0.6 (-0.7)
5y	-1.3 (-1.2)	-1.0 (-1.0)	-0.2 (-0.3)	0.2 (0.4)	0.6 (1.5)	0.8 (2.0)	0.9 (1.0)
10y	-1.9 (-2.7)	-1.4 (-2.1)	-0.4 (-0.8)	-0.1 (-0.4)	0.5 (1.5)	0.9 (1.8)	2.9 (2.6)

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Close 10/2/26. We calculate rich/cheap and z-scores in parentheses from PCA on the vol surface based on 3 principal components using 2y of data. By design, this framework only yields relative value. Source: Citi Research

EUR implied vs realized volatilities

We calculate the relative richness of short-dated implied vols to 1m and 3m realized vols. We calculate realized vols using both intraday (hourly) and close-to-close sampling.

- 1m expiry implied vols are trading well above both intraday realized vols (Figure 62) and close-to-close realized vols (Figure 64), especially when compared to the longer expiries.

Figure 60. 1m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.93 (-1.1)	0.96 (-0.9)	0.98 (-0.8)	1.00 (-0.9)	1.00 (-0.9)	1.02 (-0.8)	1.16 (-0.6)
3m	0.93 (-1.1)	0.97 (-0.9)	0.99 (-0.8)	1.01 (-0.9)	1.01 (-0.9)	1.02 (-0.8)	1.16 (-0.6)
6m	0.95 (-0.9)	0.98 (-0.8)	0.99 (-0.8)	1.01 (-0.9)	1.01 (-0.8)	1.02 (-0.8)	1.15 (-0.6)
1y	0.99 (-0.8)	1.01 (-0.7)	1.00 (-0.8)	1.01 (-0.9)	1.01 (-0.9)	1.02 (-0.9)	1.14 (-0.7)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 61. 3m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.97 (-1.3)	0.99 (-1.0)	1.00 (-1.0)	1.02 (-1.8)	1.02 (-1.7)	1.03 (-1.5)	1.17 (-0.7)
3m	0.97 (-1.3)	0.99 (-1.0)	1.00 (-1.1)	1.02 (-1.7)	1.02 (-1.7)	1.03 (-1.5)	1.17 (-0.7)
6m	0.98 (-1.0)	1.00 (-1.0)	1.00 (-1.2)	1.02 (-1.8)	1.02 (-1.7)	1.03 (-1.5)	1.17 (-0.7)
1y	1.00 (-1.0)	1.01 (-1.0)	1.01 (-1.6)	1.02 (-1.8)	1.02 (-1.7)	1.04 (-1.5)	1.16 (-0.7)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Figure 62. Intraday realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.38 (0.6)	1.29 (0.6)	1.25 (0.7)	1.29 (1.3)	1.31 (1.1)	1.27 (1.1)	1.10 (0.3)
3m	1.09 (0.1)	1.09 (0.1)	1.11 (0.2)	1.17 (0.6)	1.20 (0.5)	1.20 (0.4)	1.10 (-0.0)
6m	0.95 (-0.0)	1.01 (-0.0)	1.06 (0.1)	1.13 (0.4)	1.18 (0.4)	1.18 (0.2)	1.10 (-0.1)
1y	0.92 (-0.1)	1.00 (-0.1)	1.06 (0.1)	1.12 (0.2)	1.15 (0.1)	1.15 (-0.1)	1.10 (-0.3)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 63. Intraday realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.72 (1.6)	1.63 (1.7)	1.60 (1.9)	1.61 (2.4)	1.62 (2.2)	1.57 (2.2)	1.39 (1.6)
3m	1.38 (1.1)	1.39 (1.2)	1.41 (1.3)	1.46 (1.8)	1.49 (1.7)	1.49 (1.7)	1.39 (1.3)
6m	1.21 (0.9)	1.29 (1.0)	1.34 (1.2)	1.41 (1.7)	1.46 (1.6)	1.46 (1.5)	1.40 (1.2)
1y	1.18 (0.9)	1.28 (1.0)	1.34 (1.3)	1.40 (1.5)	1.43 (1.4)	1.42 (1.3)	1.39 (1.1)

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Note: Close 10/2/26. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 64. Close-to-close realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.29 (0.1)	1.24 (0.2)	1.23 (0.3)	1.29 (0.3)	1.31 (0.3)	1.30 (0.3)	1.27 (-0.2)
3m	1.02 (-0.3)	1.06 (-0.2)	1.09 (-0.2)	1.17 (-0.1)	1.21 (-0.1)	1.23 (-0.1)	1.27 (-0.5)
6m	0.90 (-0.3)	0.99 (-0.3)	1.05 (-0.3)	1.14 (-0.3)	1.18 (-0.3)	1.21 (-0.3)	1.27 (-0.7)
1y	0.91 (-0.4)	1.00 (-0.4)	1.06 (-0.4)	1.13 (-0.4)	1.16 (-0.5)	1.17 (-0.6)	1.25 (-1.1)

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Note: Close 10/2/26. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 65. Close-to-close realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.67 (1.2)	1.61 (1.6)	1.59 (1.7)	1.64 (1.7)	1.65 (1.6)	1.62 (1.5)	1.62 (1.3)
3m	1.33 (0.8)	1.37 (1.0)	1.40 (1.1)	1.49 (1.1)	1.53 (1.1)	1.54 (1.0)	1.63 (1.1)
6m	1.18 (0.8)	1.28 (0.9)	1.34 (1.0)	1.44 (1.0)	1.49 (0.9)	1.51 (0.8)	1.63 (1.1)
1y	1.18 (0.7)	1.29 (0.8)	1.36 (0.9)	1.43 (0.8)	1.47 (0.7)	1.47 (0.6)	1.61 (1.0)

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Note: Close 10/2/26. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

EUR expiry and tail switches

We analyze ATM vol switches across expiries and tails.

- Short expiries have outperformed longer expiries for 30y tails, for example the 6m30y/1y30y ratio is at +3.7sigmas (Figure 66).
- 1y tails have generally outperformed 2y tails, particularly for longer expiries. For example, the 10y1y/10y3y vol ratio is at +3.3sigmas (Figure 67).

Figure 66. Short/long expiry ratio (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m/3m	1.02 (2.4)	1.08 (2.8)	1.08 (2.9)	1.08 (2.6)	1.06 (2.7)	1.04 (2.8)	0.99 (2.1)
3m/6m	0.97 (1.3)	1.02 (1.4)	1.02 (1.5)	1.02 (1.9)	1.01 (2.3)	1.01 (2.6)	0.99 (3.2)
6m/1y	0.97 (0.8)	1.02 (1.3)	1.03 (1.6)	1.03 (2.0)	1.03 (2.5)	1.04 (3.3)	1.00 (3.7)
1y/3y	1.11 (1.8)	1.12 (2.0)	1.11 (2.1)	1.08 (2.2)	1.06 (2.3)	1.03 (2.4)	1.01 (2.7)
3y/5y	1.08 (1.4)	1.07 (1.3)	1.07 (1.5)	1.06 (1.8)	1.06 (1.9)	1.05 (1.9)	1.04 (1.8)
5y/10y	1.15 (2.5)	1.14 (2.5)	1.13 (2.5)	1.13 (2.5)	1.12 (2.7)	1.11 (2.9)	1.10 (3.5)

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Note: Close 10/2/26. The table reports shorter expiry/longer expiry ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

Figure 67. Short/long tail ratio (3m z-score)

exp\tails	1y/2y	1y/3y	2y/5y	2y/10y	5y/10y	5y/30y	10y/30y
1m	0.86 (0.1)	0.88 (0.0)	1.09 (-0.7)	1.27 (-0.5)	1.16 (-0.2)	1.45 (0.4)	1.25 (0.9)
3m	0.91 (0.2)	0.94 (-0.0)	1.09 (-0.6)	1.21 (-0.4)	1.12 (-0.1)	1.33 (0.0)	1.19 (0.2)
1y	1.01 (1.2)	1.04 (0.9)	1.10 (0.7)	1.22 (0.8)	1.11 (0.9)	1.25 (1.0)	1.13 (1.1)
2y	1.03 (1.8)	1.05 (1.2)	1.07 (0.3)	1.16 (0.8)	1.08 (1.2)	1.21 (1.3)	1.11 (1.3)
5y	1.01 (1.5)	1.03 (1.3)	1.05 (1.0)	1.10 (1.0)	1.05 (0.9)	1.15 (1.2)	1.10 (1.4)
10y	1.00 (3.3)	1.01 (0.3)	1.03 (0.2)	1.07 (0.6)	1.03 (0.8)	1.13 (1.5)	1.09 (1.7)

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Note: Close 10/2/26. The table reports shorter tail/longer tail ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

EUR forward volatilities

We calculate forward 1y2y, 1y5y, 1y10y and 1y30y volatilities implied from the spot swaption surface. Forward volatilities reflect market expectations about future short-dated vols and may highlight dislocations on the spot vol surface.

- Over the last month, forward vols have generally moved higher and the vol term structures have inverted (Figure 68).

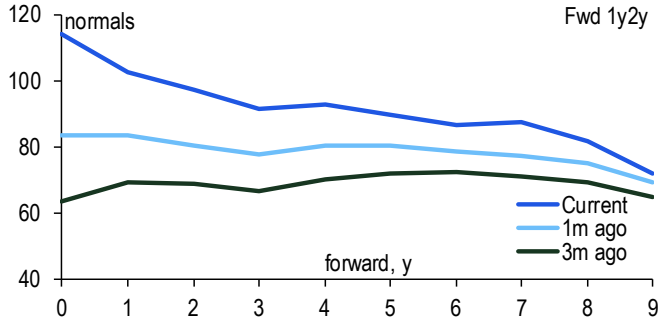
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Figure 68. Forward volatilities

	1y2y							1y5y							1y10y							1y30y						
Yrs		1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y	
Fwd	Vol	chg	chg	avg	z-scr	spt	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt	z-scr
spt	114.2	4.3	30.7	45.9	2.5	56.0	1.5	104.0	5.1	26.5	37.4	2.9	40.7	1.3	93.5	5.8	21.4	29.2	3.5	27.6	1.1	83.0	6.5	17.1	20.2	3.9	16.9	0.9
1	102.6	1.8	19.3	34.3	2.7	44.4	1.2	98.3	4.3	18.5	30.1	3.0	35.0	1.1	91.2	4.1	15.0	24.2	3.3	25.3	1.0	80.5	3.5	10.4	15.5	3.7	14.3	0.7
2	97.2	(1.7)	16.6	28.4	2.9	39.0	1.0	93.0	0.1	15.6	24.6	3.2	29.7	1.0	89.0	3.4	14.1	21.6	3.7	23.0	0.9	75.5	3.4	8.8	12.1	4.1	9.4	0.5
3	91.5	0.4	13.8	22.3	3.7	33.3	0.9	87.5	0.3	13.0	19.3	3.6	24.2	0.8	85.1	3.4	12.1	17.7	4.0	19.1	0.8	71.2	3.8	7.5	9.1	4.4	5.1	0.3
4	92.7	5.2	12.3	22.8	3.7	34.5	0.9	84.7	1.5	9.0	16.1	3.4	21.4	0.7	81.2	2.7	8.2	14.0	3.6	15.3	0.6	66.8	3.1	4.8	6.5	2.9	0.7	0.0
5	89.8	0.3	9.2	17.3	3.4	31.6	0.8	82.7	(0.9)	6.9	12.6	3.3	19.4	0.6	79.0	1.1	5.5	11.5	3.6	13.1	0.5	60.4	0.9	0.3	2.2	1.2	(5.7)	(0.3)
6	86.5	(0.5)	7.8	12.7	3.8	28.2	0.7	78.8	(1.9)	4.2	7.7	2.8	15.5	0.5	75.8	0.3	3.2	7.8	3.2	9.8	0.4	56.2	(0.1)	(2.0)	(0.9)	(0.3)	(9.9)	(0.5)
7	87.6	(1.3)	10.1	14.3	3.7	29.4	0.8	79.2	(2.4)	6.3	10.0	3.2	15.9	0.5	72.3	(2.1)	3.5	7.0	2.8	6.4	0.3	55.6	(2.5)	0.5	1.2	0.4	(10.5)	(0.5)
8	81.6	(2.9)	6.5	9.6	2.8	23.4	0.6	74.2	(3.5)	3.6	6.4	2.4	11.0	0.4	66.7	(3.7)	0.7	3.1	1.0	0.8	0.0	52.0	(4.2)	(0.9)	(0.7)	(0.2)	(14.1)	(0.7)
9	72.2	(5.1)	3.1	3.3	0.7	13.9	0.4	68.2	(4.4)	0.7	1.5	0.6	5.0	0.2	60.5	(5.3)	(1.6)	(1.8)	(0.4)	(5.5)	(0.2)	48.7	(5.6)	(2.0)	(2.9)	(0.6)	(17.5)	(0.9)

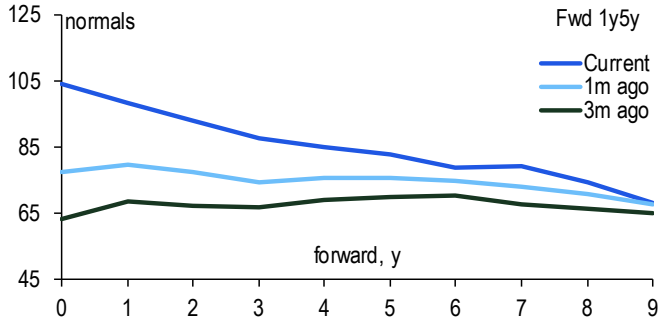
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Note: Close 10/2/26. Forward vols are computed using the standard triangular approximation from spot vols and using 6m realized correlations as a proxy for implied correlations. In addition to levels of forward vols, we calculate 1w and 1m changes, the difference from 1y average (with corresponding z-scores), and the difference from the 10y average of the corresponding 1y expiry spot vol (together with z-scores). Source: Citi Research

Figure 69. Forward vol of 1y2y



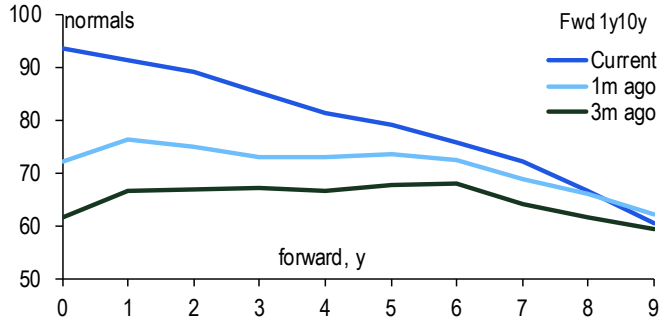
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Note: Close 10/2/26. Source: Citi Research

Figure 70. Forward vol of 1y5y



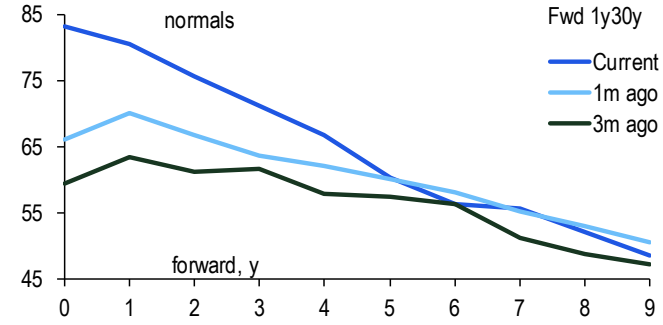
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Figure 71. Forward vol of 1y10y



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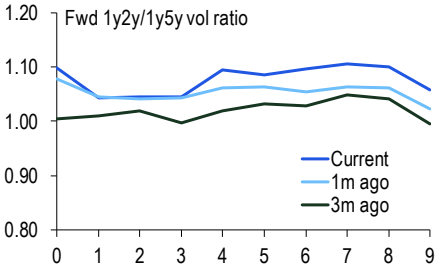
Figure 72. Forward vol of 1y30y



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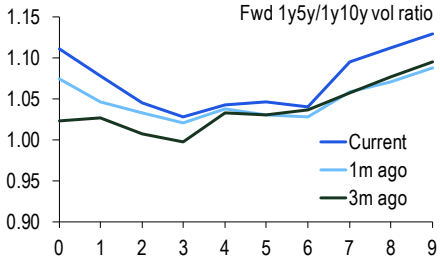
Prepared for Christopher Lee

Figure 73. Fwd 1y2y/1y5y vol ratio



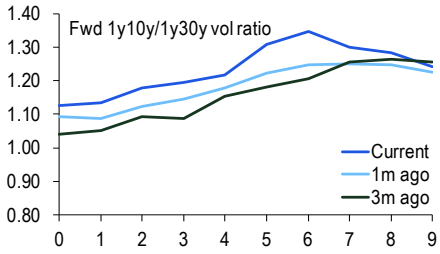
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Note: Close 10/2/26. Source: Citi Research

Figure 74. Fwd 1y5y/1y10y vol ratio



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Figure 75. Fwd 1y10y/1y30y vol ratio



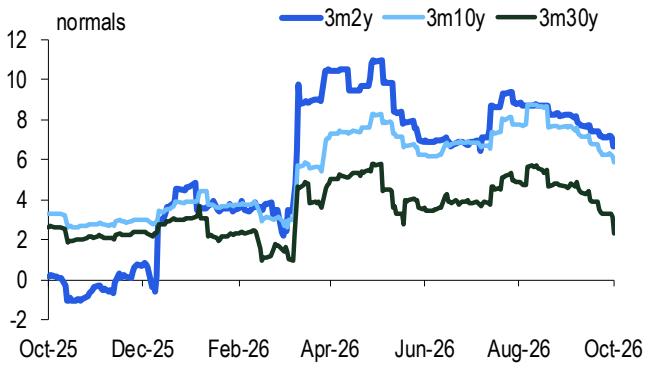
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Note: Close 10/2/26. Source: Citi Research

EUR swaption skew

We calculate z-scores of payer and receiver swaption skews across the surface.

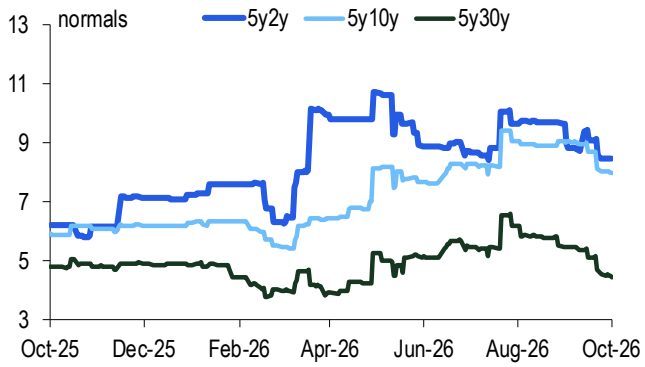
- Payer skews with 3m expiries have continued moving lower from their early July highs (Figure 76), while 3m expiry receiver skews have generally moved sideways over the past month (Figure 78).
- Risk reversals on 10y tails look rich across most expiries, reflecting a lower receiver skew over the past year (Figure 80).

Figure 76. 3m expiry payer skew



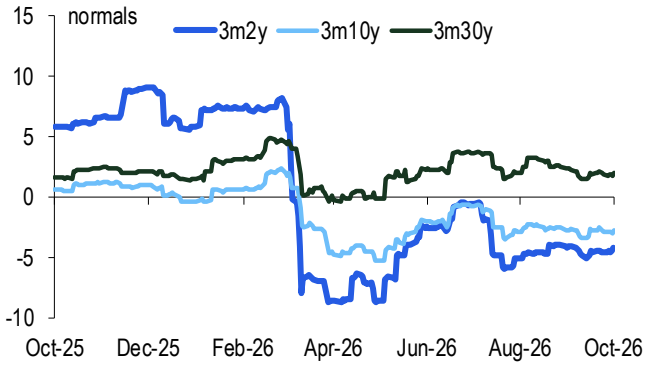
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Note: 25bp-out payer skews for 3m expiry payers. Source: Citi Research

Figure 77. 5y expiry payer skew



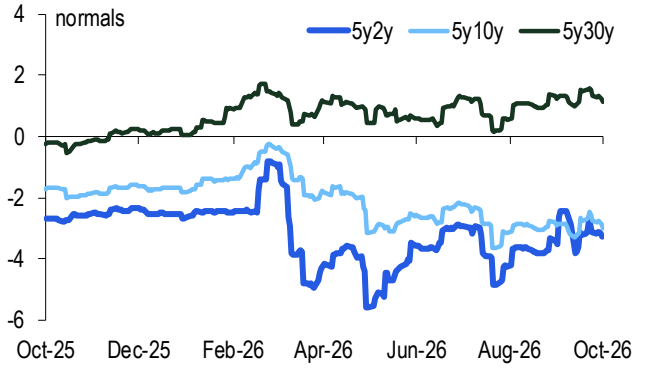
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Note: 100bp-out payer skews for 5y expiry payers. Source: Citi Research

Figure 78. 3m expiry receiver skew



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Note: 25bp-out receiver skews for 3m expiry receivers. Source: Citi Research

Figure 79. 5y expiry receiver skew



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Note: 100bp-out receiver skews for 5y expiry receivers. Source: Citi Research

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Figure 80. Skews (1y z-scores) of risk reversals, OTM payer and OTM receivers

	Risk Reversals						Payer skew						Receiver skew					
expr\tail	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y
3m	13.0 (0.8)	12.2 (0.9)	11.4 (0.9)	10.9 (0.9)	10.2 (1.0)	2.5 (0.6)	7.5 (0.2)	6.7 (0.3)	6.5 (0.4)	6.3 (0.3)	5.9 (0.2)	2.3 (-1.0)	-4.4 (-0.7)	-4.2 (-0.7)	-3.9 (-0.8)	-3.4 (-0.8)	-2.7 (-0.7)	2.0 (-0.2)
6m	18.9 (0.7)	17.5 (0.8)	16.1 (0.9)	15.2 (0.9)	14.1 (1.0)	3.2 (0.4)	13.2 (0.0)	11.7 (0.1)	11.1 (0.2)	10.7 (0.2)	10.2 (0.1)	5.3 (-1.1)	-5.2 (-0.8)	-4.9 (-0.9)	-3.8 (-0.9)	-3.3 (-1.0)	-2.6 (-1.1)	4.3 (-0.5)
1y	15.4 (0.8)	13.5 (0.8)	12.8 (0.9)	12.2 (1.0)	11.4 (1.2)	3.1 (0.3)	9.8 (0.2)	8.5 (0.2)	8.3 (0.4)	8.0 (0.3)	7.4 (0.2)	3.5 (-1.2)	-4.6 (-0.7)	-3.9 (-0.8)	-3.4 (-0.8)	-3.0 (-0.9)	-2.5 (-0.9)	2.0 (0.1)
2y	11.9 (0.5)	9.9 (0.5)	9.9 (0.8)	9.6 (1.0)	9.3 (1.2)	3.3 (0.7)	7.4 (0.1)	6.4 (0.1)	6.1 (0.2)	5.9 (0.2)	5.6 (0.1)	2.7 (-0.9)	-3.9 (-0.5)	-3.2 (-0.5)	-2.8 (-0.5)	-2.6 (-0.6)	-2.3 (-0.7)	1.2 (1.0)
3y	16.9 (0.2)	14.5 (0.1)	14.4 (0.5)	14.3 (0.8)	14.3 (1.2)	5.0 (0.0)	14.1 (0.2)	12.4 (0.1)	11.7 (0.2)	11.5 (0.3)	11.2 (0.3)	7.2 (-0.4)	-3.5 (-0.4)	-2.1 (0.0)	-2.4 (-0.5)	-2.2 (-0.7)	-1.9 (-0.9)	3.6 (0.8)
5y	13.6 (0.4)	11.3 (0.0)	11.3 (0.6)	11.5 (1.0)	11.9 (1.6)	4.2 (-0.2)	9.9 (0.5)	8.5 (0.1)	8.1 (0.4)	8.0 (0.5)	8.0 (0.7)	4.5 (-0.8)	-3.1 (0.3)	-2.5 (0.7)	-2.5 (0.2)	-2.6 (-0.4)	-2.9 (-1.2)	1.3 (1.3)
10y	9.2 (1.0)	7.7 (0.2)	7.8 (0.8)	8.1 (1.2)	8.4 (1.8)	3.8 (-0.2)	6.1 (1.1)	5.3 (0.7)	5.1 (0.7)	5.1 (0.8)	5.0 (0.9)	3.0 (-1.2)	-3.1 (-0.3)	-2.5 (0.5)	-2.6 (-0.3)	-2.7 (-0.9)	-2.8 (-1.6)	-0.4 (1.1)

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Note: Close 10/2/26. The table reports skews and 1y z-scores in parentheses for risk reversals, payer and receiver skews, defined in normal vol terms. The following OTM conventions are used: 25bp out for 3m expiries, 50bp out for 6m-2y expiries, 100bp out for 3y+ expiries. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

EUR midcurve swaptions

We report levels and z-scores of at-the-money vols, implied correlations, and implied/realized ratios of correlations. We also compute the relative richness/cheapness of midcurve vols to principal components of the plain vanilla surface.

- Implied midcurve vols of 10y10y tails continue to look cheap to swaptions when evaluated on their implied/realized correlation ratio (Figure 83).

Figure 81. Implied vols, normals (3m z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	157.3 (2.8)	119.8 (2.9)	112.4 (3.0)	140.2 (2.9)	116.5 (3.0)	113.0 (2.9)	93.3 (3.8)	79.4 (3.7)
3m	134.3 (2.5)	110.4 (2.7)	104.0 (2.8)	124.0 (2.6)	108.1 (2.8)	105.3 (2.9)	94.8 (3.7)	81.9 (3.7)
6m	124.7 (2.6)	107.8 (2.7)	101.4 (2.8)	117.3 (2.7)	105.8 (2.8)	102.9 (2.9)	95.5 (3.5)	82.7 (3.6)
1y	114.2 (2.5)	103.6 (2.9)	96.4 (2.8)	109.4 (2.7)	101.5 (2.8)	98.0 (2.9)	92.2 (3.0)	82.9 (3.2)
2y	104.3 (2.5)	99.8 (2.7)	95.2 (2.9)	102.5 (2.6)	98.9 (2.8)	97.5 (3.0)	92.8 (3.0)	82.5 (3.2)

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Close 10/2/26. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 82. Implied correlations, % (3y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	95.0 (-0.2)	99.8 (1.3)	99.9 (1.7)	91.2 (0.4)	99.3 (1.1)	98.8 (1.2)	96.9 (0.5)	98.2 (2.0)
3m	97.6 (0.7)	99.8 (1.4)	99.9 (2.0)	94.3 (0.8)	99.2 (1.1)	98.9 (1.2)	96.4 (0.1)	98.0 (2.1)
6m	98.8 (1.0)	99.8 (1.5)	100.0 (2.0)	96.5 (0.9)	99.2 (1.1)	99.0 (1.2)	96.3 (0.1)	98.0 (2.2)
1y	99.7 (1.6)	99.8 (1.8)	100.0 (2.2)	98.4 (1.4)	99.2 (1.4)	99.1 (1.5)	96.2 (0.0)	97.9 (2.2)
2y	99.7 (2.0)	99.8 (2.0)	100.0 (2.6)	98.3 (1.7)	99.1 (1.5)	99.1 (1.7)	96.4 (-0.1)	97.8 (2.4)

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Close 10/2/26. The table shows implied correlations (3y z-scores in parentheses). Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 83. Implied/realized correlation ratios (1y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	0.97 (0.2)	1.00 (1.2)	1.01 (1.3)	0.95 (1.1)	1.02 (1.3)	1.01 (1.7)	1.03 (1.3)	1.12 (2.7)
3m	0.99 (1.2)	1.00 (1.3)	1.01 (1.4)	0.97 (1.4)	1.02 (1.4)	1.01 (1.9)	1.03 (1.1)	1.11 (2.6)
6m	1.00 (1.5)	1.00 (1.4)	1.01 (1.7)	0.99 (1.5)	1.02 (1.6)	1.01 (2.5)	1.02 (1.1)	1.11 (2.6)
1y	1.00 (1.3)	1.01 (1.4)	1.01 (2.2)	1.01 (1.4)	1.02 (2.1)	1.01 (3.6)	1.02 (1.1)	1.11 (2.5)
2y	1.02 (0.9)	1.01 (2.1)	1.00 (2.7)	1.03 (2.0)	1.02 (2.8)	1.01 (3.2)	1.01 (1.3)	1.10 (2.4)

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Close 10/2/26. The table shows the ratios of implied to 3m realized correlations (in daily changes). The 1y z-scores of ratios are shown in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 84. Rich/cheap to plain vanilla swaptions (z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	12.0 (2.3)	4.5 (1.0)	1.3 (0.3)	9.8 (2.4)	5.9 (1.4)	6.4 (1.3)	-0.1 (0.0)	-5.8 (-2.1)
3m	-0.7 (-0.2)	-2.4 (-0.8)	-3.8 (-1.3)	-0.3 (-0.1)	0.3 (0.1)	1.7 (0.5)	2.7 (0.7)	-3.0 (-1.3)
6m	-2.9 (-1.1)	-3.0 (-1.5)	-3.5 (-1.5)	-1.7 (-0.8)	-0.2 (-0.1)	2.4 (1.0)	2.2 (0.7)	-3.0 (-1.5)
1y	-3.6 (-2.1)	-2.2 (-1.5)	-5.5 (-3.2)	-1.8 (-1.5)	-1.3 (-1.0)	-1.0 (-0.5)	-1.8 (-0.6)	-2.9 (-1.4)
2y	-4.1 (-2.6)	-1.8 (-1.6)	-3.8 (-2.3)	-2.0 (-2.0)	-0.2 (-0.2)	0.5 (0.3)	-0.5 (-0.2)	-1.2 (-0.7)

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Close 10/2/26. We regress midcurve vols on two corresponding plain vanilla vols using last 3y of data. For example, 1y2y1y is regressed against 1y3y and 1y2y vols. We calculate rich/cheap and z-scores in brackets. Source: Citi Research

EUR curve (spread) CMS options

We report levels and z-scores of at-the-money vols, rich/cheap on PCA, implied correlations, and implied/realized vol ratios for curve options.

- 3m 30s/5s implied correlation continued moving sharply lower, suggesting richening relative to swaption vol (Figure 87).
- Our PCA's three-year z-scores are flagging 3m through 1y expiries of implied vols across 2s5s through 2s30s and 5s10s as cheap, and in contrast short expiries of 10s30s look rich (Figure 86).

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Figure 85. Implied vols, normals (3m z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	49.6 (2.4)	84.3 (2.4)	132.9 (2.6)	43.7 (2.4)	109.6 (2.7)	79.5 (2.9)
3m	44.6 (2.2)	72.2 (2.2)	112.9 (2.5)	36.5 (2.3)	96.8 (2.6)	70.4 (2.8)
6m	43.7 (2.4)	71.7 (2.5)	110.3 (2.6)	36.4 (2.6)	92.1 (2.7)	67.5 (2.8)
1y	41.8 (2.5)	70.0 (2.6)	104.4 (2.7)	33.4 (2.8)	82.5 (2.9)	57.3 (2.9)
2y	39.5 (2.8)	64.9 (2.7)	91.5 (2.8)	31.0 (3.2)	72.1 (2.8)	49.8 (3.0)
3y	36.5 (2.8)	58.8 (3.1)	83.9 (2.8)	28.5 (3.6)	67.0 (3.1)	46.7 (3.2)
5y	33.2 (2.9)	52.2 (3.5)	75.5 (3.3)	26.9 (4.0)	58.1 (3.5)	39.2 (3.4)

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Close 10/2/26. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

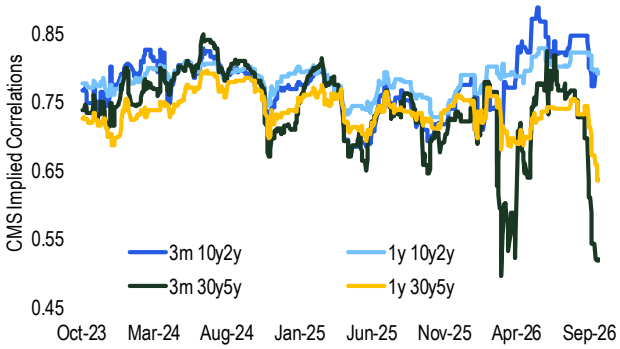
Figure 86. Rich/cheap to PCA, normals (z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	-1.3 (-0.5)	-5.5 (-1.6)	-0.5 (-0.2)	0.4 (0.2)	7.1 (4.0)	8.5 (3.3)
3m	-4.2 (-2.1)	-14.2 (-6.0)	-9.8 (-4.4)	-4.7 (-3.1)	0.4 (0.3)	5.9 (2.9)
6m	-3.8 (-2.4)	-8.9 (-4.6)	-4.4 (-2.2)	-2.5 (-2.1)	3.8 (2.5)	7.2 (3.4)
1y	-3.4 (-2.1)	-4.9 (-3.0)	-0.6 (-0.3)	-2.2 (-2.2)	2.6 (1.7)	3.5 (1.6)
2y	0.3 (0.3)	-0.1 (-0.1)	-0.5 (-0.4)	-0.2 (-0.2)	1.3 (1.1)	2.6 (1.2)
3y	1.1 (1.7)	1.4 (1.4)	0.0 (0.0)	-0.1 (-0.1)	2.4 (2.2)	3.0 (1.6)
5y	2.4 (2.8)	2.9 (2.4)	3.6 (2.8)	1.3 (1.5)	2.9 (2.0)	0.9 (0.5)

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Close 10/2/26. We calculate rich/cheap and z-scores in parentheses from PCA on curve vols using two principal components and 3y of data. By design, this framework only yields relative value. Source: Citi Research

Figure 87. Implied correlations of 2s10s and 5s30s CMS options



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Note: The chart shows implied correlation of options on the 2s10s and 5s30s curve with different expirations. Source: Citi Research

Figure 88. Implied/realized ratios

EUR Implied/1m Realized Vol Ratio (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.12 (-0.1)	1.04 (-0.6)	0.97 (-0.6)	1.07 (-0.6)	1.08 (-0.5)	1.19 (-0.2)
3m	0.93 (-0.8)	0.84 (-1.1)	0.80 (-0.9)	0.90 (-1.5)	0.95 (-0.8)	1.05 (-0.6)
6m	0.85 (-0.9)	0.80 (-1.1)	0.75 (-0.9)	0.90 (-1.5)	0.90 (-0.9)	1.01 (-0.7)
1y	0.81 (-1.2)	0.80 (-1.3)	0.73 (-1.1)	0.89 (-1.6)	0.85 (-1.1)	0.88 (-1.0)

implied/3m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.66 (1.7)	1.55 (0.9)	1.44 (0.8)	1.61 (1.3)	1.60 (1.4)	1.77 (1.6)
3m	1.36 (0.7)	1.25 (0.1)	1.17 (0.1)	1.34 (0.1)	1.40 (0.7)	1.56 (0.9)
6m	1.24 (0.2)	1.18 (-0.1)	1.10 (0.0)	1.34 (0.0)	1.33 (0.5)	1.50 (0.7)
1y	1.18 (-0.2)	1.19 (-0.3)	1.07 (-0.2)	1.33 (-0.4)	1.25 (0.0)	1.31 (0.0)

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Close 10/2/26. Close-to-close realized vol is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

EUR conditional curves

We analyze conditional trades based on the entry point, implied/realized betas, and carry.

Figure 89. Conditional curves: bear trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bear steepener				bear flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.7	31.1	18.8	1.8	23.8	4.6	2.0	65	0.10	0.13	0.03	12.3	-4.9	7.4	0.0	-12.3	4.9	-7.4	0.0
3m 1y3y	100/34.4	35.6	20.8	2.3	24.2	5.4	2.3	46	0.07	0.10	0.04	14.7	-3.4	11.3	0.0	-14.7	3.4	-11.3	0.0
3m 1y5y	100/21.3	38.4	21.7	1.7	22.3	2.8	1.5	29	0.01	-0.04	-0.05	16.7	-0.6	16.1	0.0	-16.7	0.6	-16.1	0.0
3m 2y5y	100/42.1	7.3	2.9	0.5	-1.4	-1.8	0.3	16	-0.08	-0.17	-0.08	4.4	4.3	8.7	0.0	-4.4	-4.3	-8.7	0.0
3m 2y10y	100/22.8	15.9	10.7	-0.3	0.6	-10.4	-0.4	13	-0.22	-0.37	-0.15	5.1	10.1	15.3	0.0	-5.1	-10.1	-15.3	0.0
3m 2y30y	100/10.3	1.0	-6.3	-1.1	-25.3	-35.7	-1.1	11	-0.59	-0.90	-0.31	7.3	19.1	26.4	0.0	-7.3	-19.1	-26.4	0.0
3m 3y5y	100/62.0	2.9	0.9	-0.1	-1.9	-2.5	-0.2	13	-0.06	-0.14	-0.09	2.0	2.8	4.7	0.0	-2.0	-2.8	-4.7	0.0
3m 3y10y	100/33.6	11.4	8.7	-0.6	0.2	-11.1	-0.6	13	-0.18	-0.33	-0.15	2.7	8.5	11.2	0.0	-2.7	-8.5	-11.2	0.0
3m 3y30y	100/15.2	-3.4	-8.3	-1.2	-25.4	-36.3	-1.2	10	-0.52	-0.82	-0.30	4.8	17.2	22.0	0.0	-4.8	-17.2	-22.0	0.0
3m 5y10y	100/54.2	8.6	7.8	-0.8	2.4	-8.4	-0.8	12	-0.12	-0.18	-0.06	0.8	5.4	6.2	0.0	-0.8	-5.4	-6.2	0.0
3m 5y30y	100/24.5	-6.3	-9.1	-1.4	-22.8	-33.4	-1.4	8	-0.39	-0.57	-0.18	2.9	13.7	16.6	0.0	-2.9	-13.7	-16.6	0.0
3m 10y30y	100/45.3	-14.8	-17.0	-1.7	-24.3	-24.5	-1.7	6	-0.21	-0.30	-0.09	2.1	7.3	9.5	0.0	-2.1	-7.3	-9.5	0.0
6m 1y2y	100/50.8	31.1	12.7	1.9	16.1	3.2	2.1	50	0.05	0.05	0.00	18.4	-3.4	15.0	3.2	-18.4	3.4	-15.0	-3.2
6m 1y3y	100/34.4	35.6	13.1	1.5	14.4	3.0	1.3	33	0.02	0.00	-0.02	22.5	-1.3	21.2	4.0	-22.5	1.3	-21.2	-4.0
6m 1y5y	100/21.3	38.4	12.9	0.5	10.5	-0.8	0.4	23	-0.04	-0.14	-0.11	25.5	2.4	28.0	4.7	-25.5	-2.4	-28.0	-4.7
6m 2y5y	100/42.0	7.3	0.1	0.0	-5.8	-4.0	-0.1	14	-0.08	-0.19	-0.11	7.2	5.9	13.0	1.6	-7.2	-5.9	-13.0	-1.6
6m 2y10y	100/22.7	15.9	7.5	-0.5	-5.8	-13.5	-0.6	13	-0.21	-0.40	-0.19	8.3	13.3	21.7	2.1	-8.3	-13.3	-21.7	-2.1
6m 2y30y	100/10.3	1.0	-11.3	-1.1	-35.8	-40.4	-1.1	10	-0.54	-0.92	-0.38	12.3	24.5	36.8	3.1	-12.3	-24.5	-36.8	-3.1
6m 3y5y	100/62.0	2.9	-0.2	-0.4	-3.9	-3.8	-0.4	12	-0.05	-0.14	-0.09	3.1	3.7	6.8	0.8	-3.1	-3.7	-6.8	-0.8
6m 3y10y	100/33.6	11.4	7.1	-0.7	-3.8	-13.2	-0.7	12	-0.17	-0.33	-0.17	4.3	10.9	15.2	1.4	-4.3	-10.9	-15.2	-1.4
6m 3y30y	100/15.2	-3.4	-11.7	-1.3	-33.4	-39.9	-1.3	9	-0.47	-0.82	-0.34	8.3	21.7	30.0	2.5	-8.3	-21.7	-30.0	-2.5
6m 5y10y	100/54.2	8.6	7.4	-0.8	0.6	-9.2	-0.9	11	-0.11	-0.18	-0.07	1.2	6.8	8.0	0.6	-1.2	-6.8	-8.0	-0.6
6m 5y30y	100/24.5	-6.3	-11.4	-1.4	-28.4	-35.6	-1.4	7	-0.35	-0.56	-0.21	5.2	17.0	22.1	1.9	-5.2	-17.0	-22.1	-1.9
6m 10y30y	100/45.3	-14.8	-18.8	-1.7	-27.9	-25.8	-1.7	4	-0.18	-0.29	-0.11	4.0	9.1	13.1	1.4	-4.0	-9.1	-13.1	-1.4
1y 1y2y	100/50.7	31.1	10.4	1.0	9.6	-0.6	0.3	33	-0.01	0.01	0.02	20.8	0.7	21.5	0.9	-20.8	-0.7	-21.5	-0.9
1y 1y3y	100/34.4	35.6	10.0	0.6	5.6	-2.4	0.0	23	-0.04	-0.06	-0.02	25.6	4.4	29.9	1.4	-25.6	-4.4	-29.9	-1.4
1y 1y5y	100/21.3	38.4	9.4	0.0	-1.3	-9.3	-0.5	18	-0.11	-0.19	-0.08	29.0	10.6	39.7	1.7	-29.0	-10.6	-39.7	-1.7
1y 2y5y	100/42.1	7.3	-1.0	-0.4	-10.6	-8.4	-0.8	12	-0.10	-0.19	-0.09	8.3	9.6	17.9	0.8	-8.3	-9.6	-17.9	-0.8
1y 2y10y	100/22.8	15.9	7.2	-0.7	-13.2	-20.1	-1.0	11	-0.22	-0.39	-0.16	8.6	20.4	29.0	1.1	-8.6	-20.4	-29.0	-1.1
1y 2y30y	100/10.3	1.0	-14.5	-1.3	-47.1	-47.7	-1.4	9	-0.51	-0.88	-0.37	15.5	32.6	48.2	1.5	-15.5	-32.6	-48.2	-1.5
1y 3y5y	100/62.0	2.9	-0.6	-0.8	-6.5	-6.6	-1.0	11	-0.06	-0.12	-0.06	3.5	5.9	9.3	0.3	-3.5	-5.9	-9.3	-0.3
1y 3y10y	100/33.6	11.4	7.6	-0.9	-8.7	-18.1	-1.1	11	-0.18	-0.30	-0.12	3.8	16.3	20.1	0.7	-3.8	-16.3	-20.1	-0.7
1y 3y30y	100/15.2	-3.4	-14.1	-1.4	-42.1	-45.4	-1.5	8	-0.43	-0.74	-0.31	10.7	28.0	38.7	1.1	-10.7	-28.0	-38.7	-1.1
1y 5y10y	100/54.3	8.6	8.2	-0.9	-1.5	-11.0	-1.1	11	-0.11	-0.17	-0.06	0.4	9.7	10.1	0.4	-0.4	-9.7	-10.1	-0.4
1y 5y30y	100/24.5	-6.3	-13.5	-1.5	-34.2	-37.9	-1.6	6	-0.31	-0.52	-0.22	7.3	20.7	28.0	0.8	-7.3	-20.7	-28.0	-0.8
1y 10y30y	100/45.2	-14.8	-21.7	-1.8	-31.4	-26.1	-1.8	3	-0.14	-0.27	-0.13	6.9	9.7	16.6	0.5	-6.9	-9.7	-16.6	-0.5
2y 1y2y	100/51.0	31.1	9.6	-0.3	6.3	-3.3	-0.7	27	-0.03	0.01	0.04	21.6	3.2	24.8	0.4	-21.6	-3.2	-24.8	-0.4
2y 1y3y	100/34.6	35.6	8.6	-0.8	2.2	-7.2	-0.9	20	-0.05	0.01	0.06	27.0	6.4	33.4	0.4	-27.0	-6.4	-33.4	-0.4
2y 1y5y	100/21.4	38.4	10.4	-0.9	-1.8	-13.3	-0.9	18	-0.10	-0.09	0.01	28.0	12.2	40.2	-0.1	-28.0	-12.2	-40.2	0.1
2y 2y5y	100/42.0	7.3	0.8	-1.1	-7.7	-9.7	-1.1	13	-0.07	-0.10	-0.03	6.5	8.5	15.0	-0.5	-6.5	-8.5	-15.0	0.5
2y 2y10y	100/22.8	15.9	10.8	-1.1	-8.8	-21.2	-1.2	12	-0.16	-0.26	-0.10	5.1	19.6	24.7	-0.7	-5.1	-19.6	-24.7	0.7
2y 2y30y	100/10.3	1.0	-16.4	-1.5	-50.6	-49.9	-1.6	9	-0.39	-0.63	-0.24	17.5	34.2	51.7	0.1	-17.5	-34.2	-51.7	-0.1
2y 3y5y	100/62.0	2.9	1.8	-1.0	-3.5	-5.8	-1.0	13	-0.04	-0.10	-0.06	1.0	5.3	6.4	-0.5	-1.0	-5.3	-6.4	0.5
2y 3y10y	100/33.7	11.4	11.8	-1.0	-4.3	-17.1	-1.2	12	-0.13	-0.25	-0.12	-0.4	16.1	15.8	-0.7	0.4	-16.1	-15.8	0.7
2y 3y30y	100/15.2	-3.4	-15.4	-1.6	-45.6	-45.5	-1.6	7	-0.33	-0.59	-0.26	12.0	30.2	42.2	0.1	-12.0	-30.2	-42.2	-0.1
2y 5y10y	100/54.3	8.6	10.0	-1.1	-0.3	-10.9	-1.3	11	-0.08	-0.14	-0.06	-1.4	10.3	8.9	-0.2	1.4	-10.3	-8.9	0.2
2y 5y30y	100/24.5	-6.3	-17.2	-1.7	-40.9	-38.9	-1.7	5	-0.25	-0.42	-0.17	11.0	23.7	34.7	0.6	-11.0	-23.7	-34.7	-0.6
2y 10y30y	100/45.2	-14.8	-27.2	-1.9	-39.3	-27.2	-1.9	2	-0.12	-0.21	-0.09	12.4	12.1	24.5	0.8	-12.4	-12.1	-24.5	-0.8
3y 1y2y	100/50.8	31.1	6.8	-1.9	4.6	-6.8	-2.0	25	-0.02	0.12	0.14	24.3	2.2	26.5	-0.1	-24.3	-2.2	-26.5	0.1
3y 1y3y	100/34.4	35.6	7.7	-1.6	1.0	-10.5	-1.7	19	-0.04	0.13	0.17	27.9	6.6	34.5	0.0	-27.9	-6.6	-34.5	0.0
3y 1y5y	100/21.3	38.4	11.6	-1.4	-0.4	-15.7	-1.4	18	-0.08	0.05	0.12	26.8	12.0	38.8	-0.4	-26.8	-12.0	-38.8	0.4
3y 2y5y	100/42.0	7.3	4.8	-1.1	-4.5	-8.6	-1.1	14	-0.06	-0.08	-0.02	2.4	9.4	11.8	-0.3	-2.4	-9.4	-11.8	0.3
3y 2y10y	100/22.9	15.9	15.2	-1.1	-4.0	-18.5	-1.2	13	-0.13	-0.22	-0.09	0.7	19.2	19.9	-0.4	-0.7	-19.2	-19.9	0.4
3y 2y30y	100/10.3	1.0	-17.8	-1.6	-53.7	-47.6	-1.6	8	-0.33	-0.51	-0.18	18.8	36.0	54.8	0.4	-18.8	-36.0	-54.8	-0.4
3y 3y5y	100/62.0	2.9	4.0	-1.1	-1.0	-5.0	-1.0	14	-0.03	-0.08	-0.05	-1.1	5.0	3.9	-0.3	1.1	-5.0	-3.9	0.3
3y 3y10y	100/33.7	11.4	14.3	-1.1	-0.2	-14.7	-1.2	13	-0.10	-0.22	-0.12	-2.9	14.5	11.6	-0.5	2.9	-14.5	-11.6	0.5
3y 3y30y	100/15.2	-3.4	-18.7	-1.7	-49.4	-43.5	-1.7	7	-0.28	-0.50	-0.22	15.3	30.7	46.0	0.3	-15.3	-30.7	-46.0	-0.3
3y 5y10y	100/54.4	8.6	10.3	-1.1	1.1	-9.6	-1.2	12	-0.06	-0.13	-0.06	-1.8	9.2	7.4	-0.2	1.8	-9.2	-7.4	0.2
3y 5y30y	100/24.5	-6.3	-22.6	-1.8	-47.3	-38.0	-1.8	3	-0.22	-0.35	-0.14	16.4	24.7	41.1	0.6	-16.4	-24.7	-41.1	-0.6
3y 10y30y	100/45.1	-14.8	-32.9	-2.1	-47.4	-27.7	-2.1	2	-0.12	-0.17	-0.05	18.1	14.4	32.5	0.8	-18.1	-14.4	-32.5	-0.8

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Note: Close 10/2/26. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m carry is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bear steepeners in red, and three best bear flatteners in green. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

Figure 90. Conditional curves: bull trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bull steepener				bull flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.7	31.1	18.8	1.8	14.8	2.6	1.6	47	0.10	0.13	0.03	12.3	4.1	16.4	16.7	-12.3	-4.1	-16.4	-16.7
3m 1y3y	100/34.4	35.6	20.8	2.3	18.0	4.4	2.2	37	0.07	0.10	0.04	14.7	2.8	17.6	17.9	-14.7	-2.8	-17.6	-17.9
3m 1y5y	100/21.3	38.4	21.7	1.7	21.2	3.6	1.8	28	0.01	-0.04	-0.05	16.7	0.5	17.2	17.5	-16.7	-0.5	-17.2	-17.5
3m 2y5y	100/42.1	7.3	2.9	0.5	6.4	1.0	0.9	23	-0.08	-0.17	-0.08	4.4	-3.5	0.9	0.9	-4.4	3.5	-0.9	-0.9
3m 2y10y	100/22.8	15.9	10.7	-0.3	19.0	-2.0	-0.2	20	-0.22	-0.37	-0.15	5.1	-8.2	-3.1	-2.3	-5.1	8.2	3.1	2.3
3m 2y30y	100/10.3	1.0	-6.3	-1.1	8.9	-20.5	-1.1	15	-0.59	-0.90	-0.31	7.3	-15.2	-7.9	-0.1	-7.3	15.2	7.9	0.1
3m 3y5y	100/62.0	2.9	0.9	-0.1	3.2	-0.8	0.0	20	-0.06	-0.14	-0.09	2.0	-2.3	-0.4	-0.4	-2.0	2.3	0.4	0.4
3m 3y10y	100/33.6	11.4	8.7	-0.6	15.7	-3.8	-0.5	18	-0.18	-0.33	-0.15	2.7	-6.9	-4.2	-2.3	-2.7	6.9	4.2	2.3
3m 3y30y	100/15.2	-3.4	-8.3	-1.2	5.5	-22.3	-1.2	15	-0.52	-0.82	-0.30	4.8	-13.8	-8.9	-0.1	-4.8	13.8	8.9	0.1
3m 5y10y	100/54.2	8.6	7.8	-0.8	12.3	-3.2	-0.7	17	-0.12	-0.18	-0.06	0.8	-4.5	-3.7	-2.3	-0.8	4.5	3.7	2.3
3m 5y30y	100/24.5	-6.3	-9.1	-1.4	1.9	-21.7	-1.4	15	-0.39	-0.57	-0.18	2.9	-11.1	-8.2	-0.1	-2.9	11.1	8.2	0.1
3m 10y30y	100/45.3	-14.8	-17.0	-1.7	-10.7	-18.8	-1.6	10	-0.21	-0.30	-0.09	2.1	-6.2	-4.1	-0.1	-2.1	6.2	4.1	0.1
6m 1y2y	100/50.8	31.1	12.7	1.9	10.0	2.1	1.6	34	0.05	0.05	0.00	18.4	2.8	21.2	3.2	-18.4	-2.8	-21.2	-3.2
6m 1y3y	100/34.4	35.6	13.1	1.5	12.0	3.1	1.7	30	0.02	0.00	-0.02	22.5	1.1	23.5	3.9	-22.5	-1.1	-23.5	-3.9
6m 1y5y	100/21.3	38.4	12.9	0.5	14.8	1.8	0.7	24	-0.04	-0.14	-0.11	25.5	-1.9	23.6	3.9	-25.5	1.9	-23.6	-3.9
6m 2y5y	100/42.0	7.3	0.1	0.0	4.9	-0.3	0.3	21	-0.08	-0.19	-0.11	7.2	-4.8	2.4	0.8	-7.2	4.8	-2.4	-0.8
6m 2y10y	100/22.7	15.9	7.5	-0.5	18.1	-3.1	-0.3	20	-0.21	-0.40	-0.19	8.3	-10.6	-2.2	0.4	-8.3	10.6	2.2	-0.4
6m 2y30y	100/10.3	1.0	-11.3	-1.1	7.7	-21.0	-1.1	15	-0.54	-0.92	-0.38	12.3	-19.0	-6.7	0.6	-12.3	19.0	6.7	-0.6
6m 3y5y	100/62.0	2.9	-0.2	-0.4	2.8	-1.3	-0.2	19	-0.05	-0.14	-0.09	3.1	-3.0	0.1	0.2	-3.1	3.0	-0.1	-0.2
6m 3y10y	100/33.6	11.4	7.1	-0.7	15.9	-4.2	-0.6	18	-0.17	-0.33	-0.17	4.3	-8.8	-4.5	-0.2	-4.3	8.8	4.5	0.2
6m 3y30y	100/15.2	-3.4	-11.7	-1.3	5.4	-22.1	-1.2	15	-0.47	-0.82	-0.34	8.3	-17.0	-8.8	0.0	-8.3	17.0	8.8	0.0
6m 5y10y	100/54.2	8.6	7.4	-0.8	13.0	-3.0	-0.7	18	-0.11	-0.18	-0.07	1.2	-5.6	-4.4	-0.4	-1.2	5.6	4.4	0.4
6m 5y30y	100/24.5	-6.3	-11.4	-1.4	2.2	-21.0	-1.4	15	-0.35	-0.56	-0.21	5.2	-13.6	-8.5	-0.1	-5.2	13.6	8.5	0.1
6m 10y30y	100/45.3	-14.8	-18.8	-1.7	-11.2	-18.3	-1.6	10	-0.18	-0.29	-0.11	4.0	-7.6	-3.6	0.2	-4.0	7.6	3.6	-0.2
1y 1y2y	100/50.7	31.1	10.4	1.0	10.9	2.4	1.7	36	-0.01	0.01	0.02	20.8	-0.6	20.2	-0.5	-20.8	0.6	-20.2	0.5
1y 1y3y	100/34.4	35.6	10.0	0.6	13.4	2.8	1.5	31	-0.04	-0.06	-0.02	25.6	-3.4	22.2	-0.8	-25.6	3.4	-22.2	0.8
1y 1y5y	100/21.3	38.4	9.4	0.0	17.5	2.0	0.7	25	-0.11	-0.19	-0.08	29.0	-8.2	20.9	-1.2	-29.0	8.2	-20.9	1.2
1y 2y5y	100/42.1	7.3	-1.0	-0.4	6.7	-0.4	0.2	23	-0.10	-0.19	-0.09	8.3	-7.7	0.6	-0.8	-8.3	7.7	-0.6	0.8
1y 2y10y	100/22.8	15.9	7.2	-0.7	23.2	-2.1	-0.2	22	-0.22	-0.39	-0.16	8.6	-16.0	-7.3	-1.6	-8.6	16.0	7.3	1.6
1y 2y30y	100/10.3	1.0	-14.5	-1.3	10.6	-19.9	-1.1	15	-0.51	-0.88	-0.37	15.5	-25.1	-9.5	-0.7	-15.5	25.1	9.5	0.7
1y 3y5y	100/62.0	2.9	-0.6	-0.8	4.1	-0.9	-0.3	21	-0.06	-0.12	-0.06	3.5	-4.7	-1.3	-0.4	-3.5	4.7	1.3	0.4
1y 3y10y	100/33.6	11.4	7.6	-0.9	20.5	-2.7	-0.4	20	-0.18	-0.30	-0.12	3.8	-12.9	-9.0	-1.3	-3.8	12.9	9.0	1.3
1y 3y30y	100/15.2	-3.4	-14.1	-1.4	7.6	-20.6	-1.2	15	-0.43	-0.74	-0.31	10.7	-21.7	-11.0	-0.4	-10.7	21.7	11.0	0.4
1y 5y10y	100/54.3	8.6	8.2	-0.9	16.0	-2.0	-0.5	19	-0.11	-0.17	-0.06	0.4	-7.9	-7.5	-0.9	-0.4	7.9	7.5	0.9
1y 5y30y	100/24.5	-6.3	-13.5	-1.5	2.9	-20.2	-1.3	15	-0.31	-0.52	-0.22	7.3	-16.4	-9.1	0.0	-7.3	16.4	9.1	0.0
1y 10y30y	100/45.2	-14.8	-21.7	-1.8	-13.7	-18.5	-1.7	9	-0.14	-0.27	-0.13	6.9	-8.1	-1.2	0.8	-6.9	8.1	1.2	-0.8
2y 1y2y	100/51.0	31.1	9.6	-0.3	12.1	-0.7	0.3	39	-0.03	0.01	0.04	21.6	-2.5	19.0	0.0	-21.6	2.5	-19.0	0.0
2y 1y3y	100/34.6	35.6	8.6	-0.8	13.4	-2.6	-0.5	32	-0.05	0.01	0.06	27.0	-4.9	22.1	0.1	-27.0	4.9	-22.1	-0.1
2y 1y5y	100/21.4	38.4	10.4	-0.9	19.5	-4.0	-0.7	27	-0.10	-0.09	0.01	28.0	-9.2	18.9	0.0	-28.0	9.2	-18.9	0.0
2y 2y5y	100/42.0	7.3	0.8	-1.1	7.5	-3.4	-1.1	24	-0.07	-0.10	-0.03	6.5	-6.7	-0.2	0.0	-6.5	6.7	0.2	0.0
2y 2y10y	100/22.8	15.9	10.8	-1.1	26.0	-4.9	-0.8	23	-0.16	-0.26	-0.10	5.1	-15.2	-10.1	-0.1	-5.1	15.2	10.1	0.1
2y 2y30y	100/10.3	1.0	-16.4	-1.5	9.6	-22.2	-1.4	15	-0.39	-0.63	-0.24	17.5	-26.0	-8.6	0.5	-17.5	26.0	8.6	-0.5
2y 3y5y	100/62.0	2.9	1.8	-1.0	6.0	-1.5	-0.9	24	-0.04	-0.10	-0.06	1.0	-4.2	-3.2	-0.1	-1.0	4.2	3.2	0.1
2y 3y10y	100/33.7	11.4	11.8	-1.0	24.4	-3.0	-0.7	22	-0.13	-0.25	-0.12	-0.4	-12.6	-13.0	-0.2	0.4	12.6	13.0	0.2
2y 3y30y	100/15.2	-3.4	-15.4	-1.6	7.9	-20.4	-1.4	15	-0.33	-0.59	-0.26	12.0	-23.3	-11.3	0.4	-12.0	23.3	11.3	-0.4
2y 5y10y	100/54.3	8.6	10.0	-1.1	18.2	-1.7	-0.5	21	-0.08	-0.14	-0.06	-1.4	-8.2	-9.6	-0.1	1.4	8.2	9.6	0.1
2y 5y30y	100/24.5	-6.3	-17.2	-1.7	1.4	-19.2	-1.5	15	-0.25	-0.42	-0.17	11.0	-18.7	-7.7	0.5	-11.0	18.7	7.7	-0.5
2y 10y30y	100/45.2	-14.8	-27.2	-1.9	-17.3	-17.8	-1.9	8	-0.12	-0.21	-0.09	12.4	-9.9	2.4	0.6	-12.4	9.9	-2.4	-0.6
3y 1y2y	100/50.8	31.1	6.8	-1.9	8.5	-2.9	-1.6	31	-0.02	0.12	0.14	24.3	-1.7	22.7	0.7	-24.3	1.7	-22.7	-0.7
3y 1y3y	100/34.4	35.6	7.7	-1.6	12.6	-3.2	-1.3	30	-0.04	0.13	0.17	27.9	-5.0	22.9	0.1	-27.9	5.0	-22.9	-0.1
3y 1y5y	100/21.3	38.4	11.6	-1.4	20.5	-4.5	-1.2	27	-0.08	0.05	0.12	26.8	-8.9	17.9	-0.1	-26.8	8.9	-17.9	0.1
3y 2y5y	100/42.0	7.3	4.8	-1.1	12.2	-1.6	-1.0	27	-0.06	-0.08	-0.02	2.4	-7.3	-4.9	-0.8	-2.4	7.3	4.9	0.8
3y 2y10y	100/22.9	15.9	15.2	-1.1	30.1	-3.3	-0.9	24	-0.13	-0.22	-0.09	0.7	-14.9	-14.2	-0.6	-0.7	14.9	14.2	0.6
3y 2y30y	100/10.3	1.0	-17.8	-1.6	9.7	-19.6	-1.6	15	-0.33	-0.51	-0.18	18.8	-27.5	-8.7	0.1	-18.8	27.5	8.7	-0.1
3y 3y5y	100/62.0	2.9	4.0	-1.1	7.9	-1.3	-1.0	26	-0.03	-0.08	-0.05	-1.1	-3.9	-5.0	-0.2	1.1	3.9	5.0	0.2
3y 3y10y	100/33.7	11.4	14.3	-1.1	25.6	-3.0	-0.9	23	-0.10	-0.22	-0.12	-2.9	-11.3	-14.2	0.0	2.9	11.3	14.2	0.0
3y 3y30y	100/15.2	-3.4	-18.7	-1.7	5.1	-19.4	-1.6	15	-0.28	-0.50	-0.22	15.3	-23.7	-8.5	0.6	-15.3	23.7	8.5	-0.6
3y 5y10y	100/54.4	8.6	10.3	-1.1	17.6	-1.8	-0.8	20	-0.06	-0.13	-0.06	-1.8	-7.3	-9.1	0.2	1.8	7.3	9.1	-0.2
3y 5y30y	100/24.5	-6.3	-22.6	-1.8	-3.2	-18.4	-1.7	14	-0.22	-0.35	-0.14	16.4	-19.4	-3.0	0.8	-16.4	19.4	3.0	-0.8
3y 10y30y	100/45.1	-14.8	-32.9	-2.1	-21.3	-16.8	-2.0	6	-0.12	-0.17	-0.05	18.1	-11.7	6.5	0.6	-18.1	11.7	-6.5	-0.6

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Note: Close 10/2/26. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **bull steepeners in red**, and three **best bull flatteners in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

EUR conditional flies

We analyze conditional fly trades based on the entry point, directionality with rates and carry.

Figure 91. Conditional payer flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly							costless strike		long payer in the belly				short payer in the belly			
curve	notionals, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, ZC	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	145.4/100/31.0	16.4	10.0	1.5	75	44	3.3	1.7	3.1	13.1	3.9	1.6	84	6.4	-3.1	3.3	0.0	-6.4	3.1	-3.3	0.0
3m 1y3y7y	145.4/100/22.9	15.0	8.8	1.2	81	90	2.6	1.7	4.3	13.1	5.5	1.3	95	6.2	-4.3	1.9	0.0	-6.2	4.3	-1.9	0.0
3m 2y3y5y	73.7/100/31.0	0.8	0.6	1.1	73	95	0.3	0.7	0.7	1.2	1.5	1.0	94	0.2	-0.7	-0.4	0.0	-0.2	0.7	0.4	0.0
3m 2y5y7y	118.9/100/36.9	2.3	0.3	2.2	-30	-73	4.9	1.9	-1.0	-0.6	0.5	1.7	25	2.0	1.0	2.9	0.0	-2.0	-1.0	-2.9	0.0
3m 2y5y10y	118.9/100/27.1	-0.6	-2.5	2.0	17	-32	4.4	2.1	0.5	-1.9	2.9	1.8	63	1.8	-0.5	1.3	0.0	-1.8	0.5	-1.3	0.0
3m 2y10y30y	219.4/100/22.6	15.3	13.8	3.1	42	-58	10.2	2.4	-1.2	12.7	6.9	2.8	36	1.5	1.2	2.7	0.0	-1.5	-1.2	-2.7	0.0
3m 3y5y7y	80.7/100/36.9	0.1	-0.7	2.1	-30	-70	1.9	1.9	-0.3	-0.9	0.3	1.7	32	0.8	0.3	1.0	0.0	-0.8	-0.3	-1.0	0.0
3m 3y5y10y	80.7/100/27.1	-2.9	-3.5	1.3	44	81	1.4	1.9	1.2	-2.2	2.7	1.2	91	0.6	-1.2	-0.6	0.0	-0.6	1.2	0.6	0.0
3m 3y7y10y	109.4/100/36.7	-0.2	-1.2	0.3	-29	-81	3.2	1.5	-1.0	-2.2	-0.6	0.2	14	1.0	1.0	2.0	0.0	-1.0	-1.0	-2.0	0.0
3m 3y7y15y	109.4/100/26.6	-3.4	-4.0	2.1	34	-46	3.8	2.1	0.1	-3.9	2.5	2.2	45	0.6	-0.1	0.5	0.0	-0.6	0.1	-0.5	0.0
3m 3y10y30y	148.9/100/22.6	13.1	12.8	3.0	52	-47	7.5	2.8	-0.5	12.4	6.7	2.9	42	0.3	0.5	0.8	0.0	-0.3	-0.5	-0.8	0.0
3m 5y10y15y	92.3/100/36.2	1.1	1.2	0.3	9	-80	1.8	1.3	-1.5	-0.3	-0.8	0.0	14	0.0	1.5	1.4	0.0	0.0	-1.5	-1.4	0.0
3m 5y10y30y	92.3/100/22.6	11.7	12.4	2.4	74	62	5.0	2.8	0.9	13.3	8.0	2.5	74	-0.7	-0.9	-1.6	0.0	0.7	0.9	1.6	0.0
3m 10y20y30y	85.0/100/38.5	10.2	9.7	-1.0	-8	-65	-1.5	-1.3	-0.7	8.9	-1.7	-1.0	19	0.5	0.7	1.2	0.0	-0.5	-0.7	-1.2	0.0
6m 1y3y5y	145.6/100/31.0	16.4	6.7	2.7	30	-35	4.5	2.0	2.5	9.2	3.1	3.1	76	9.7	-2.5	7.2	1.6	-9.7	2.5	-7.2	-1.6
6m 1y3y7y	145.6/100/22.9	15.0	5.7	1.7	62	56	3.9	2.0	3.9	9.6	5.0	1.8	92	9.3	-3.9	5.4	1.4	-9.3	3.9	-5.4	-1.4
6m 2y3y5y	73.9/100/31.0	0.8	0.3	1.3	60	95	0.5	1.4	0.8	1.1	1.7	1.3	93	0.5	-0.8	-0.3	0.0	-0.5	0.8	0.3	0.0
6m 2y5y7y	119.2/100/36.9	2.3	-0.9	1.3	-42	-78	4.6	1.8	-1.4	-2.3	-0.3	0.8	18	3.2	1.4	4.7	0.7	-3.2	-1.4	-4.7	-0.7
6m 2y5y10y	119.2/100/27.1	-0.6	-3.6	2.2	-4	-48	4.2	2.0	0.4	-3.2	2.2	2.0	59	3.0	-0.4	2.5	0.5	-3.0	0.4	-2.5	-0.5
6m 2y10y30y	219.9/100/22.6	15.3	13.2	3.3	32	-65	9.6	2.4	-1.7	11.5	6.1	2.7	34	2.2	1.7	3.9	0.5	-2.2	-1.7	-3.9	-0.5
6m 3y5y7y	80.6/100/36.9	0.1	-1.1	1.6	-32	-74	1.7	1.8	-0.4	-1.5	-0.1	1.2	23	1.2	0.4	1.6	0.3	-1.2	-0.4	-1.6	-0.3
6m 3y5y10y	80.6/100/27.1	-2.9	-3.8	1.2	48	78	1.3	1.8	1.5	-2.3	2.5	1.2	91	0.9	-1.5	-0.5	0.1	-0.9	1.5	0.5	-0.1
6m 3y7y10y	109.3/100/36.7	-0.2	-1.9	-0.2	-31	-83	2.7	1.3	-1.3	-3.1	-1.2	-0.2	13	1.6	1.3	2.9	0.4	-1.6	-1.3	-2.9	-0.4
6m 3y7y15y	109.3/100/26.6	-3.4	-4.2	2.3	34	-54	3.5	2.1	0.1	-4.1	2.2	2.2	44	0.8	-0.1	0.8	0.1	-0.8	0.1	-0.8	-0.1
6m 3y10y30y	148.7/100/22.6	13.1	13.0	3.2	50	-52	7.1	2.8	-0.7	12.3	6.3	2.8	42	0.1	0.7	0.9	0.0	-0.1	-0.7	-0.9	0.0
6m 5y10y15y	92.3/100/36.2	1.1	1.3	0.1	8	-82	1.7	1.3	-1.8	-0.5	-1.0	-0.1	13	-0.2	1.8	1.6	0.1	0.2	-1.8	-1.6	-0.1
6m 5y10y30y	92.3/100/22.6	11.7	13.1	2.4	72	60	4.9	2.7	1.1	14.1	8.2	2.5	77	-1.4	-1.1	-2.5	-0.4	1.4	1.1	2.5	0.4
6m 10y20y30y	85.0/100/38.5	10.2	9.2	-1.1	-8	-62	-1.8	-1.5	-1.2	8.1	-2.3	-1.2	16	0.9	1.2	2.1	0.4	-0.9	-1.2	-2.1	-0.4
1y 1y3y5y	145.4/100/31.0	16.4	5.3	3.7	-47	-61	4.8	2.3	0.8	6.1	1.8	3.7	68	11.0	-0.8	10.3	0.5	-11.0	0.8	-10.3	-0.5
1y 1y3y7y	145.4/100/22.9	15.0	4.0	2.5	0	12	4.4	2.5	2.7	6.8	4.1	2.5	89	11.0	-2.7	8.2	0.5	-11.0	2.7	-8.2	-0.5
1y 2y3y5y	73.7/100/31.0	0.8	0.1	1.8	-9	88	0.9	2.4	1.1	1.3	2.2	1.9	94	0.7	-1.1	-0.5	0.1	-0.7	1.1	0.5	-0.1
1y 2y5y7y	118.9/100/36.9	2.3	-1.8	0.1	-41	-82	3.1	1.6	-2.6	-4.4	-1.9	-0.5	14	4.1	2.6	6.7	0.4	-4.1	-2.6	-6.7	-0.4
1y 2y5y10y	118.9/100/27.1	-0.6	-4.6	2.2	-12	-60	2.9	1.9	0.1	-4.5	1.1	2.2	52	3.9	-0.1	3.9	0.2	-3.9	0.1	-3.9	-0.2
1y 2y10y30y	219.1/100/22.6	15.3	14.5	3.2	32	-68	7.7	2.5	-4.3	10.1	3.5	1.7	32	0.9	4.3	5.2	0.3	-0.9	-4.3	-5.2	-0.3
1y 3y5y7y	80.7/100/36.9	0.1	-1.6	-0.3	-11	-80	0.9	1.3	-0.9	-2.5	-0.9	-0.9	16	1.7	0.9	2.6	0.1	-1.7	-0.9	-2.6	-0.1
1y 3y5y10y	80.7/100/27.1	-2.9	-4.4	1.0	69	81	0.7	1.2	1.8	-2.6	2.1	1.2	90	1.5	-1.8	-0.3	0.0	-1.5	1.8	0.3	0.0
1y 3y7y10y	109.3/100/36.8	-0.2	-1.9	-0.7	-20	-84	1.4	0.8	-2.0	-3.9	-2.5	-0.9	12	1.6	2.0	3.7	0.1	-1.6	-2.0	-3.7	-0.1
1y 3y7y15y	109.3/100/26.6	-3.4	-3.7	2.1	52	-55	2.5	1.9	-0.2	-3.9	1.4	1.9	45	0.3	0.2	0.5	-0.2	-0.3	-0.2	-0.5	0.2
1y 3y10y30y	148.7/100/22.6	13.1	14.6	3.2	57	-46	5.8	2.8	-2.7	11.9	4.5	2.3	41	-1.5	2.7	1.2	0.1	1.5	-2.7	-1.2	-0.1
1y 5y10y15y	92.2/100/36.2	1.1	2.3	0.3	13	-82	1.5	1.3	-2.7	-0.4	-1.3	-0.5	13	-1.2	2.7	1.5	-0.1	1.2	-2.7	-1.5	0.1
1y 5y10y30y	92.2/100/22.6	11.7	15.0	2.5	70	60	4.8	2.6	0.0	15.0	7.6	2.4	79	-3.3	0.0	-3.3	-0.1	3.3	0.0	3.3	0.1
1y 10y20y30y	85.0/100/38.4	10.2	8.5	-1.3	-5	-50	-2.2	-1.9	-1.8	6.7	-2.9	-1.4	11	1.7	1.8	3.5	0.4	-1.7	-1.8	-3.5	-0.4

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Note: Close 10/2/26. Conditional flies are structured by buying (selling) payers in the belly vs selling (buying) payers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best long belly payer trades in red, and three best short belly payer trades in green. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

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Figure 92. Conditional receiver flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long receiver in the belly				short receiver in the belly			
curve	ntl, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, zs	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	145.4/100/31.0	16.4	10.0	1.5	75	44	3.3	1.7	2.6	7.4	2.5	1.4	72	-6.4	-2.6	-9.0	-9.1	6.4	2.6	9.0	9.1
3m 1y3y7y	145.4/100/22.9	15.0	8.8	1.2	81	90	2.6	1.7	3.5	5.3	3.1	1.2	87	-6.2	-3.5	-9.7	-9.9	6.2	3.5	9.7	9.9
3m 2y3y5y	73.7/100/31.0	0.8	0.6	1.1	73	95	0.3	0.7	0.5	0.0	1.2	1.2	90	-0.2	-0.5	-0.8	-0.8	0.2	0.5	0.8	0.8
3m 2y5y7y	118.9/100/36.9	2.3	0.3	2.2	-30	-73	4.9	1.9	-0.8	1.1	0.9	2.6	33	-2.0	0.8	-1.2	-1.2	2.0	-0.8	1.2	1.2
3m 2y5y10y	118.9/100/27.1	-0.6	-2.5	2.0	17	-32	4.4	2.1	0.4	-2.9	1.7	2.0	60	-1.8	-0.4	-2.3	-2.3	1.8	0.4	2.3	2.3
3m 2y10y30y	219.4/100/22.6	15.3	13.8	3.1	42	-58	10.2	2.4	-1.0	14.8	8.0	3.0	39	-1.5	1.0	-0.5	-0.5	1.5	-1.0	0.5	0.5
3m 3y5y7y	80.7/100/36.9	0.1	-0.7	2.1	-30	-70	1.9	1.9	-0.2	-0.5	0.2	2.0	38	-0.8	0.2	-0.6	-0.6	0.8	-0.2	0.6	0.6
3m 3y5y10y	80.7/100/27.1	-2.9	-3.5	1.3	44	81	1.4	1.9	1.0	-4.5	1.0	1.2	82	-0.6	-1.0	-1.6	-1.7	0.6	1.0	1.6	1.7
3m 3y7y10y	109.4/100/36.7	-0.2	-1.2	0.3	-29	-81	3.2	1.5	-0.8	-0.4	-0.3	0.3	21	-1.0	0.8	-0.2	-0.2	1.0	-0.8	0.2	0.2
3m 3y7y15y	109.4/100/26.6	-3.4	-4.0	2.1	34	-46	3.8	2.1	0.1	-4.0	2.2	1.9	44	-0.6	-0.1	-0.6	-0.6	0.6	0.1	0.6	0.6
3m 3y10y30y	148.9/100/22.6	13.1	12.8	3.0	52	-47	7.5	2.8	-0.4	13.2	7.3	2.8	44	-0.3	0.4	0.1	0.2	0.3	-0.4	-0.1	-0.2
3m 5y10y15y	92.3/100/36.2	1.1	1.2	0.3	9	-80	1.8	1.3	-1.3	2.4	0.9	0.9	23	0.0	1.3	1.3	1.4	0.0	-1.3	-1.3	-1.4
3m 5y10y30y	92.3/100/22.6	11.7	12.4	2.4	74	62	5.0	2.8	0.8	11.6	7.8	2.2	70	0.7	-0.8	-0.1	0.0	-0.7	0.8	0.1	0.0
3m 10y20y30y	85.0/100/38.5	10.2	9.7	-1.0	-8	-65	-1.5	-1.3	-0.7	10.3	-1.9	-0.9	25	-0.5	0.7	0.2	0.2	0.5	-0.7	-0.2	-0.2
6m 1y3y5y	145.6/100/31.0	16.4	6.7	2.7	30	-35	4.5	2.0	2.0	4.6	2.1	2.4	65	-9.7	-2.0	-11.7	-1.8	9.7	2.0	11.7	1.8
6m 1y3y7y	145.6/100/22.9	15.0	5.7	1.7	62	56	3.9	2.0	3.2	2.5	2.7	1.6	82	-9.3	-3.2	-12.5	-1.9	9.3	3.2	12.5	1.9
6m 2y3y5y	73.9/100/31.0	0.8	0.3	1.3	60	95	0.5	1.4	0.6	-0.3	1.0	1.4	88	-0.5	-0.6	-1.1	-0.3	0.5	0.6	1.1	0.3
6m 2y5y7y	119.2/100/36.9	2.3	-0.9	1.3	-42	-78	4.6	1.8	-1.2	0.3	0.3	1.7	30	-3.2	1.2	-2.0	-0.5	3.2	-1.2	2.0	0.5
6m 2y5y10y	119.2/100/27.1	-0.6	-3.6	2.2	-4	-48	4.2	2.0	0.4	-4.0	1.1	2.0	55	-3.0	-0.4	-3.3	-0.7	3.0	0.4	3.3	0.7
6m 2y10y30y	219.9/100/22.6	15.3	13.2	3.3	32	-65	9.6	2.4	-1.5	14.6	7.2	3.4	39	-2.2	1.5	-0.7	-0.2	2.2	-1.5	0.7	0.2
6m 3y5y7y	80.6/100/36.9	0.1	-1.1	1.6	-32	-74	1.7	1.8	-0.4	-0.7	0.0	1.5	35	-1.2	0.4	-0.8	-0.1	1.2	-0.4	0.8	0.1
6m 3y5y10y	80.6/100/27.1	-2.9	-3.8	1.2	48	78	1.3	1.8	1.2	-5.0	0.8	1.1	80	-0.9	-1.2	-2.2	-0.3	0.9	1.2	2.2	0.3
6m 3y7y10y	109.3/100/36.7	-0.2	-1.9	-0.2	-31	-83	2.7	1.3	-1.1	-0.8	-0.6	-0.1	19	-1.6	1.1	-0.6	-0.2	1.6	-1.1	0.6	0.2
6m 3y7y15y	109.3/100/26.6	-3.4	-4.2	2.3	34	-54	3.5	2.1	0.1	-4.3	1.8	2.1	43	-0.8	-0.1	-0.9	-0.1	0.8	0.1	0.9	0.1
6m 3y10y30y	148.7/100/22.6	13.1	13.0	3.2	50	-52	7.1	2.8	-0.6	13.6	6.9	3.0	44	-0.1	0.6	0.5	0.2	0.1	-0.6	-0.5	-0.2
6m 5y10y15y	92.3/100/36.2	1.1	1.3	0.1	8	-82	1.7	1.3	-1.6	2.9	1.0	0.7	25	0.2	1.6	1.8	0.3	-0.2	-1.6	-1.8	-0.3
6m 5y10y30y	92.3/100/22.6	11.7	13.1	2.4	72	60	4.9	2.7	0.9	12.2	7.6	2.2	72	1.4	-0.9	0.5	0.3	-1.4	0.9	-0.5	-0.3
6m 10y20y30y	85.0/100/38.5	10.2	9.2	-1.1	-8	-62	-1.8	-1.5	-1.1	10.3	-1.8	-0.9	25	-0.9	1.1	0.1	0.0	0.9	-1.1	-0.1	0.0
1y 1y3y5y	145.4/100/31.0	16.4	5.3	3.7	-47	-61	4.8	2.3	0.6	4.7	1.6	2.9	65	-11.0	-0.6	-11.7	0.2	11.0	0.6	11.7	-0.2
1y 1y3y7y	145.4/100/22.9	15.0	4.0	2.5	0	12	4.4	2.5	2.2	1.8	2.1	2.1	80	-11.0	-2.2	-13.2	-0.1	11.0	2.2	13.2	0.1
1y 2y3y5y	73.7/100/31.0	0.8	0.1	1.8	-9	88	0.9	2.4	0.9	-0.8	0.6	1.3	86	-0.7	-0.9	-1.6	-0.1	0.7	0.9	1.6	0.1
1y 2y5y7y	118.9/100/36.9	2.3	-1.8	0.1	-41	-82	3.1	1.6	-2.1	0.4	0.2	1.2	30	-4.1	2.1	-2.0	0.1	4.1	-2.1	2.0	-0.1
1y 2y5y10y	118.9/100/27.1	-0.6	-4.6	2.2	-12	-60	2.9	1.9	0.1	-4.6	0.6	1.7	52	-3.9	-0.1	-4.0	-0.1	3.9	0.1	4.0	0.1
1y 2y10y30y	219.1/100/22.6	15.3	14.5	3.2	32	-68	7.7	2.5	-3.7	18.2	7.8	3.6	46	-0.9	3.7	2.8	1.3	0.9	-3.7	-2.8	-1.3
1y 3y5y7y	80.7/100/36.9	0.1	-1.6	-0.3	-11	-80	0.9	1.3	-0.7	-0.8	0.0	0.9	33	-1.7	0.7	-0.9	0.0	1.7	-0.7	0.9	0.0
1y 3y5y10y	80.7/100/27.1	-2.9	-4.4	1.0	69	81	0.7	1.2	1.5	-5.9	0.5	0.7	75	-1.5	-1.5	-3.0	-0.2	1.5	1.5	3.0	0.2
1y 3y7y10y	109.3/100/36.8	-0.2	-1.9	-0.7	-20	-84	1.4	0.8	-1.7	-0.2	-0.6	-0.4	22	-1.6	1.7	0.1	0.3	1.6	-1.7	-0.1	-0.3
1y 3y7y15y	109.3/100/26.6	-3.4	-3.7	2.1	52	-55	2.5	1.9	-0.2	-3.5	1.9	2.1	47	-0.3	0.2	-0.1	0.3	0.3	-0.2	0.1	-0.3
1y 3y10y30y	148.7/100/22.6	13.1	14.6	3.2	57	-46	5.8	2.8	-2.3	16.9	7.7	3.1	51	1.5	2.3	3.8	1.1	-1.5	-2.3	-3.8	-1.1
1y 5y10y15y	92.2/100/36.2	1.1	2.3	0.3	13	-82	1.5	1.3	-2.3	4.6	1.5	2.0	28	1.2	2.3	3.5	0.5	-1.2	-2.3	-3.5	-0.5
1y 5y10y30y	92.2/100/22.6	11.7	15.0	2.5	70	60	4.8	2.6	0.0	14.9	8.2	2.4	79	3.3	0.0	3.2	0.9	-3.3	0.0	-3.2	-0.9
1y 10y20y30y	85.0/100/38.4	10.2	8.5	-1.3	-5	-50	-2.2	-1.9	-1.7	10.2	-1.9	-1.0	24	-1.7	1.7	0.0	0.0	1.7	-1.7	0.0	0.0

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Note: Close 10/2/26. Conditional flies are structured by buying (selling) receivers in the belly vs selling (buying) receivers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best long belly receiver trades in red, and three best short belly receiver trades in green. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

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Appendix A-1

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